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Deflagration to Detonation Transition

Update: XTIV (eXplosive Thermal Ignition to Violence)

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ABSTRACT

This report extends the recent SAND reports on the XDDT deflagration-to-detonation transition code [1, 2] with the development of XTIV (eXplosive Thermal Ignition to Violence), a pipeline that couples the XCHEM prethermal-ignition code [3, 4] to the XDDT solver, enabling end-to-end simulation from first exposure to an abnormal thermal environment such as a fire, through thermal ignition, to the final reactive event—a benign pressure burst or a catastrophic detonation. The pipeline transfers the XCHEM state at thermal runaway through a single-timestep transfer file with two schema options based on different gas EOS assumptions, augmented by diagnostic and smoothing redesigns required to interpret coupled trajectories without ambiguity. The methodology is demonstrated on four HMX configurations spanning the regime spectrum from fizzle through transition propagation to a CJ-class detonation for an imposed-boost initial state: a single-fragment slab in cold bulk that fails to engage surface-burn kinetics, a NATO AOP-4382 [5, 6] slow cook-off cylinder producing transition propagation through preheating of the HMX bed, a natural-cookoff twin-fragment laminate configuration also landing in the transition band, and a twin-fragment-impact configuration with an imposed twin hot-spot boost that produces a CJ-class wave for the artificially elevated, partially compacted initial state the boost establishes (benchmarked against a TIGER [7] hot-CJ calculation at the matching internal-energy condition). The progression establishes bulk-preheating extent as the regime parameter governing deflagration-to-detonation outcomes in cookoff-coupled BN [8, 9] modeling. Several limitations are identified for future work.

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Use of AI-assisted tools. This report was prepared with substantive use of large language model tools. See the postlog (§C) for a full description of the human–AI workflow, the division of labor between the project lead and the AI tools, the reproducibility and code-protection considerations that apply, and the export-control assessment for unlimited distribution.

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EXECUTIVE SUMMARY

This report extends the recent SAND reports titled “Deflagration to Detonation Transition Update: XDDT Code Modularization” [1] and “Deflagration to Detonation Transition XDDT Update: SI Units, Adaptive Gridding, TIGER EOS” [2] with the development of XTIV (eXplosive Thermal Ignition to Violence), a pipeline that couples the XCHEM prethermal-ignition code [3,4] to the XDDT deflagration-to-detonation solver, enabling end-to-end simulation from (1) first exposure to an abnormal thermal environment such as a fire, through (2) thermal ignition, to (3) the final event—whether a benign pressure burst or a catastrophic detonation.

The pipeline transfers the XCHEM state field at thermal runaway into XDDT through a single-timestep transfer file generated by a dedicated extractor. Two transfer schemas were developed. Schema 1 passes temperature, gas density, solid volume fraction, pressure, and solid mass fraction $(T, \rho_g, \phi_s, P, S_f)$ directly from XCHEM. Schema 2 drops P from the file and adds a mass-balance diagnostic $(T, \rho_g, \phi_s, S_f, m_f)$, letting XDDT recompute pressure from (T, ρ_g) via its own TIGER-based gas equation of state. The two schemas thus differ in which code’s gas EOS owns the handoff pressure. Because XCHEM and XDDT use different gas equations of state, the same physical state can produce different pressure values in each code; schema 1 keeps XCHEM’s value, schema 2 lets XDDT recompute.

Two diagnostic redesigns were required to interpret the resulting coupled trajectories. First, the burn-front tracker was rebuilt to follow the active reaction band rather than its rearmost edge, eliminating three previously-undetected failure modes; the new tracker reproduces the SAND-1 and SAND-2 validation results on the historical piston-driven cases, so earlier calibrations remain valid. Second, the gauge-history output was extended with new diagnostics that distinguish kinetic from compaction-driven reaction at probe locations.

XCHEM solves only the conductive form of the energy equation, with no convective transport. At thermal runaway, the local temperature rises very steeply: in part because the reaction kinetics accelerate exponentially, and in part because conduction alone cannot redistribute the released energy on the timescale of the runaway. In a real system, gas-phase convection would smooth the stiff energy equation; in the conduction-only handoff state delivered by XCHEM, the spike is narrower than a single grid cell and would cause unphysical integrator failure at XDDT’s sub-microsecond timescales. A handoff-state smoothing knob applies a tanh ramp to the four state fields (T, P, ϕ_s, ρ_g) over a configurable inner-edge length to mitigate this stiffness. The need is most acute in spherical and cylindrical geometries, where wave convergence further steepens gradients, but it also arises in slab geometries with sharp material-interface contrasts.

The pipeline is demonstrated on four configurations spanning the regime spectrum from fizzle through transition propagation to a CJ-class detonation for an imposed-boost initial state.

A single-fragment hot-steel slab in cold HMX bulk produces only compaction-driven activity pinned at the steel/HMX interface; the BN surface-burn kinetic switch never engages, the cold bulk beyond the contact face never sees the reaction, and the integration terminates after $\sim 4 \mu\text{s}$ with a fitted velocity $|D| \approx 2 \text{ km/s}$ and a peak pressure of $\sim 143 \text{ MPa}$ (two orders of magnitude below CJ). The case establishes the lower-bound fizzle regime: contact-face ignition in cold bulk cannot sustain a propagating wave.

A NATO AOP-4382 slow cook-off cylinder [5, 6] (18-inch HMX bed with 1-inch steel jacket, 15°C/hr ramp through ignition) exhibits inward-propagating reactive wave amplification consistent with classical converging-wave detonation transition: peak gas temperatures exceeding 4800 K , peak pressures exceeding 15 GPa , and a fitted steady wave speed of $|D| = 5.39 \text{ km/s}$ (roughly 85% of HMX’s ideal detonation velocity $D_{\text{CJ}} = 6.36 \text{ km/s}$ computed from TIGER thermochemistry). The preheating of the HMX bed (outer wall at $\sim 500 \text{ K}$ under the AOP-4382 ramp at the moment of ignition) supplies the bulk preheating extent that the surface-burn kinetics require to engage.

A natural-cookoff twin-fragment configuration — laminate steel-HMX-steel sandwich with 540 K steel and a 4 cm HMX bed, the same geometry as the headline twin-fragment-impact case but without any imposed boost — produces transition propagation at $|D| = 1.43 \text{ km/s}$ (roughly 22% of D_{CJ}), similar to the cylinder case. A fragment-spacing sensitivity sweep across fragment spacings of $1, 2, \text{ and } 4 \text{ cm}$ establishes that two-sided natural cookoff produces fizzle or transition outcomes depending on the two-sided preheating overlap but never reaches the detonation regime.

A twin-fragment-impact configuration with an asymmetric twin hot-spot boost layered on the laminate steel-HMX-steel handoff produces front-on-front collision of twin reaction bands and fires the BN-DDT criterion at the collision frame ($t_{\text{DDT}} = 2.20 \mu\text{s}$ at $x_{\text{DDT}} = 5.98 \text{ cm}$) with a steady wave speed of $|D| = 8.05 \text{ km/s}$. Benchmarking against a TIGER hot-CJ calculation at the matching internal-energy condition ($e_{\text{initial}} = 1.38 \text{ kJ/g}$, corresponding to HMX at $\sim 1000 \text{ K}$) places this velocity on the hot-CJ trend at $\rho \approx 1.54 \text{ g/cc}$ ($\phi_s \approx 0.81$ of TMD) — a CJ-class wave running at its hot-bed ceiling for the boost-imposed initial state, not an overdriven detonation in the classical sense. The cold-granular CJ value (6.36 km/s) is the wrong baseline for this boosted initial state; the hot-CJ benchmark establishes thermodynamic consistency between the BN result and the CJ steady state at the same elevated initial-energy condition. Both the BN integration and the TIGER hot-CJ calculation are idealized model probes against an artificially elevated initial state, not predictions of a physically realizable detonation following a realistic cookoff handoff.

The progression establishes bulk-preheating extent as the regime parameter governing deflagration-to-detonation outcomes in cookoff-coupled BN modeling. Single isolated hot fragments lack the thermal mass to preheat cold bulk and produce only fizzle-class outcomes. Two-sided or multi-source thermal loading provides the centimeter-scale preheated extent that the BN compaction-burn coupling requires for sustained transition propagation. Reaching CJ-class detonation requires preheating intensity beyond what natural cookoff produces in any of the demonstrated geometries; the boosted twin-fragment-impact case probes that upper bound against an artificially elevated initial state.

Several limitations are identified for future work, including a true rigid-wall outer-boundary condition (the present implementation lets reaction products vent freely outward instead of reflecting off

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1. INTRODUCTION

The XDDT (eXplosive Deflagration to Detonation Transition) code, descended from the two-phase mixture-theory framework of Baer and Nunziato [8, 9], models the post-ignition compressible reactive flow that carries a granular explosive from a localized hot spot through deflagration and—under sufficient confinement—to a steady Chapman–Jouguet detonation. Confinement in the physical problem is a multi-dimensional phenomenon, set by case strength, lateral inertia, and the geometry of any pressure-relief path perpendicular to the wave direction; its representation in the present one-dimensional treatment is necessarily idealized, and the implications of that idealization are discussed in §10.2 and §10.1. XDDT alone presumes that an ignition state already exists. In real-world safety analysis, the ignition state is itself the result of a thermal cook-off process that takes place over minutes to hours of slow heating, comprising heat conduction through case material, thermal decomposition kinetics, and gas–solid coupling at the prethermal-ignition timescale. The XCHEM (eXplosive CHEMistry) prethermal-ignition code [3, 4] models this slow phase but not the post-ignition compressible dynamics.

The disparate timescales (XCHEM: 1 s–10 h; XDDT: $0.1\ \mu\text{s}$ – $100\ \mu\text{s}$) and the disparate physics (XCHEM: thermal-conduction-dominated, slow chemistry; XDDT: compressible two-phase flow, fast chemistry) make a single end-to-end solver impractical. The natural division of labor is to let each code do what it does best: XCHEM resolves the cook-off through thermal runaway, and XDDT picks up the post-ignition state and integrates the deflagration–to–detonation transition.

This report describes XTIV (eXplosive Thermal Ignition to Violence), the pipeline that connects them. XTIV consists of: (a) a transfer-file format that captures the XCHEM state at handoff time in a form XDDT can read; (b) an extractor that converts XCHEM’s plot output into the transfer-file format, with optional preconditioning of the handoff state to manage numerical stiffness at material interfaces; (c) modifications to XDDT to accept the transfer-file as initial condition (IBC=3 mode); and (d) a set of diagnostic and reporting redesigns required to interpret the resulting coupled trajectories without ambiguity.

Symbols and acronyms used in this report are listed with units in the Nomenclature chapter.

1.1. Prior reports in the DDT series

The XDDT code was modernized from a legacy FORTRAN-77 implementation in the first report of this series [1], which restructured the solver into Fortran 90 with namelist-driven input, externalized material-specific parameters, and corrected two long-standing defects (an inconsistency in nodal solid-pressure evaluation and a non-physical burn-front tracking criterion). Validation against piston-driven DDT experiments of McAfee et al. [11] (Shot B-9036, 65% TMD granular HMX)

reproduced the experimental ionization-pin trajectory within 0.3% in transition time and 3.8% in steady detonation velocity.

The second report [2] extended the modernization with three developments—SI-units modernization, repair of the adaptive mesh refinement, and adoption of a TIGER thermochemical-equilibrium gas-phase EOS—and added a primary validation against the Price–Bernecker thermal-ignition experiments [12]. The TIGER tabular EOS removed the constant- c_v and polynomial-isentrope approximations of the legacy Jones–Wilkins–Lee (JWL) form. A companion sensitivity study asked whether a single generic CHNO TIGER table could substitute for explosive-specific tables in BN-model calibration: the reactant identity used to generate the EOS was varied while all other BN parameters were held fixed at HMX-tuned values. The study found a $\sim 70\%$ spread in induction timing across reactants—a sensitivity-study result rather than a full DDT calibration, since only the gas EOS was varied—but large enough to settle the question: an explosive-specific TIGER table is required for each calibration target rather than a single generic table serving all CHNO materials.

1.2. Why coupling matters for IM safety analysis

The dominant accident scenarios in insensitive-munitions (IM) design are slow heating events: container fires, fuel pool exposure, and prolonged solar heating in storage. Standard tests for IM compliance, including the NATO AOP-4382 slow cook-off test [5] (governed in the United States by MIL-STD-2105 [6]), require thermal exposures at 15°C/hr through ignition. These events develop the thermal initial condition that determines whether a subsequent reaction propagates as a benign deflagration or escalates to a violent detonation. The bulk-temperature profile at ignition—in particular, the radial extent of preheating in cylinder or sphere geometries representative of munition cases—is a first-order parameter in setting the wave dynamics that follow. Without the coupling, the post-ignition simulation has to assume an ignition state, which loses the physical link between the test conditions and the predicted response.

XTIV restores that link. The pipeline transfers the XCHEM state field (temperature, gas density, solid volume fraction, pressure, and solid mass fraction) at the moment of thermal runaway into XDDT, where it becomes the initial condition for the compressible reactive integration. This report demonstrates the pipeline on four configurations spanning the regime spectrum: a single-fragment slab in cold HMX bulk that fizzles (Chapter 6), a NATO AOP-4382 slow-cook-off cylinder that produces transition propagation through preheating of the HMX bed (Chapter 7), a natural-cookoff twin-fragment laminate configuration that also lands in the transition band, and a twin-fragment-impact configuration with an imposed twin hot-spot boost that produces a CJ-class wave for the elevated initial state (Chapter 8).

1.3. Frequently asked methodology questions

The XDDT code implements the two-phase Baer–Nunziato (BN) framework as extended in subsequent literature, with several auxiliary closure terms inherited from the legacy implementation.

Several questions about the model’s form and applicability recur often enough in review and discussion to be worth addressing here.

1.3.1. *What is the difference between DDT and SDT, and why does this report target DDT?*

Insensitive-munitions analysis cares about two distinct paths that take a granular explosive from a benign state to a violent response. The two paths have qualitatively different energy sources, initial conditions, characteristic timescales, and wave structures, and modeling them well requires different tools.

Deflagration-to-detonation transition (DDT). The explosive is first driven to ignition by slow thermal processes — external heating, internal heat generation, or contact with hot fragments — on a timescale of hours to days. The reaction begins as a low-velocity deflagration in the preheated bulk. If the bed’s porosity allows pressure to accumulate behind the reaction front, the deflagration mechanically compacts the unreacted material ahead of it, and the combination of compaction-driven hot-spot formation and the pressure-dependent surface-burn rate accelerates the wave from deflagration through a transition regime and ultimately to a steady detonation. The bed’s initial porosity is essential to the transition mechanism, and the slow thermal precursor is what produces the preheated bed configuration that DDT requires.

Shock-to-detonation transition (SDT). A mechanical shock is imposed directly on cold, unreacted explosive — by a planar shock, a donor charge, or a high-velocity fragment impact in the absence of thermal preconditioning. Ignition emerges at the shock front from compressive heating of the bulk, not from any preconditioned state. There is no extended thermal precursor; the ignition-to-steady-detonation timescale is microseconds, and the bulk ahead of the shock is at its initial cold density. SDT in fully dense pressed charges is governed by mechanical hot-spot formation under shock loading rather than by porosity-mediated deflagration acceleration.

Why XTIV targets DDT. The XTIV pipeline (XCHEM cookoff handed off to XDDT integration) is specifically built for the DDT regime: XCHEM resolves the slow thermal precursor, the handoff delivers the bed at its post-cookoff porous state, and XDDT integrates the post-handoff wave dynamics through the BN compaction-burn coupling. The underlying XDDT solver also supports purely mechanical initialization (piston-ramp boundary conditions, IBC=1) and can be used standalone for SDT studies, but the SDT regime is better served by reactive-flow codes built around mechanical hot-spot ignition models (Lee–Tarver ignition-and-growth [13] or Forest Fire [14]) than by XTIV’s BN-with-cookoff-handoff construction. The four demonstrations in this report (Chapters 6, 7, and 8) span the DDT spectrum from natural-cookoff fizzle through hot-fragment-boosted DDT through twin-fragment hybrid initiation; none are pure-SDT configurations.

1.3.2. Where does the compaction-burn term come from?

The original BN 1986 paper [8] contains only the pressure-dependent surface-burn rate as the source for the compaction equation $\dot{\phi}_s$. The extended form used in this work—with an additional contribution proportional to a compaction-burn coefficient α^* —is from the 1989 follow-on paper [9] on compaction-induced combustion in ball propellants, which was explicitly motivated by phenomena (friction heating at grain contacts, void collapse, plastic deformation under compaction) that the surface-burn term cannot capture on its own. In the XDDT compaction-equation source the two contributions combine to give the local rate of mass conversion from solid to gas:

$$C = -\gamma_s a_2 \left(\frac{P_g}{P_{\text{ref}}} \right)^{b_n} \frac{3 \phi_s}{a_p} H (1 - f_{dv})^{2/3} - \gamma_s \alpha_0 \left(\frac{P_{\text{mix}}}{P_{\text{ref}}} \right)^2 f_{dv} (1 - \phi_{s,0}), \quad (1.1)$$

where the first term is the BN 1986 pressure-dependent surface-burn rate, turned on by the Heaviside step function H , and the second is the Baer 1989 compaction-burn term, with the $(P_{\text{mix}}/P_{\text{ref}})^2$ pressure dependence inherited from the Lee–Tarver hot-spot reaction-time scaling $\tau_H^{-1} = b p_m^2$ [9, 13]. This P^2 factor on the compaction-burn term was *not* displayed in the prose equations of SAND-1 [1] or SAND-2 [2] but is present in the XDDT source; the present report restores it after a side-by-side comparison of the BN 1989 paper against the XDDT implementation. C is the local mass conversion rate from solid to gas (units kg/m³/s); the volume-fraction time derivative used elsewhere in the report is $\dot{\phi}_s = C/\gamma_s$. The variables in equation (1.1) are:

- a_2 : surface-burn coefficient [m/s at $P = P_{\text{ref}}$]; $a_2 = 0.1227$ m/s for HMX (Boggs strand-burn calibration [15]).
- a_p : granular particle radius [m] (input variable `ap_in`).
- b_n : pressure exponent [dimensionless], $b_n \approx 0.9$ for HMX.
- $f_{dv} = \max(0, (\phi_s - \phi_{s,0})/(1 - \phi_{s,0}))$: the compaction (densification) fraction, vanishing in undisturbed bed at initial packing and approaching unity as the bed compacts toward fully dense.
- H : ignition switch; $H = 1$ when the granular surface temperature T_{surface} exceeds the ignition threshold T_{ign} , $H = 0$ otherwise. Called `eti` in the XDDT source.
- P_g : gas pressure (used in the surface-burn term). Code variable `pg_nod`; the dimensionless ratio P_g/P_{ref} is discussed below.
- P_{mix} : mixture pressure $\phi_s P_s + \phi_g P_g - P_{\text{intergranular}}$ (used in the compaction-burn term); code variable `pmix`. Distinct from P_g when solid and gas pressures differ.
- P_{ref} : reference pressure for the surface-burn calibration, $P_{\text{ref}} = 100$ MPa.
- T_{ign} : ignition temperature threshold for the Heaviside switch, $T_{\text{ign}} = 800$ K in the demonstrations of this report. Pegged just above the slow-cockoff XCHEM handoff plateau (~ 807 K) so the surface-burn switch fires on compacted, post-handoff cells rather than on the cookoff-preheated handoff state itself; see §6.3 for the sensitivity analysis.

- T_{surface} : granular surface temperature, computed from the interface heat-transfer model derived by Baer [8]. Distinct from the gas temperature T_g and the bulk solid temperature T_s . Called `tinf` in the XDDT source.
- α_0 : compaction-burn rate coefficient [1/s]; $\alpha_0 = 100$ in the present XDDT deck (input variable `alstar_coeff_in`). SAND-1 [1] reported $\alpha^* = 75$ from a calibration against the McAfee Shot B-9036 piston-driven DDT experiment [11]; that calibration value can be recovered by setting `alstar_coeff_in` = 75. The full pressure-dependent coefficient that multiplies $f_{dv}\gamma_s(1 - \phi_{s,0})$ in equation (1.1) is $\alpha^* \equiv \alpha_0(P_{\text{mix}}/P_{\text{ref}})^2$.
- γ_s : solid intrinsic (crystal) mass density [kg/m³].
- ϕ_s : current solid volume fraction at the node.
- $\phi_{s,0}$: initial bed solid volume fraction (= TMD fraction at packing).

The factor $3\phi_s/a_p$ in the surface-burn term is the surface-to-volume ratio of an idealized spherical-particle bed at solid volume fraction ϕ_s with particle radius a_p . The $(1 - f_{dv})^{2/3}$ exponent captures the geometric reduction in available reactive surface area as the bed compacts past initial packing: $1 - f_{dv} = \phi_g/\phi_{g,0}$ is the ratio of current gas volume fraction to initial gas volume fraction (i.e., the fraction of original porosity that remains), and the $2/3$ power is the surface-to-volume scaling for spherical particles. Together, $3\phi_s/a_p$ and $(1 - f_{dv})^{2/3}$ form the shape factor ϵ defined in Baer and Nunziato [8] (their eq. 65).

In equation (1.1), P is a physical pressure quantity and the ratio P/P_{ref} is dimensionless: P may be expressed in any consistent units, with P_{ref} interpreted in the same units. The reference pressure $P_{\text{ref}} = 100$ MPa is the calibration reference for the HMX strand-burn coefficient a_2 , inherited from the legacy strand-burn measurements with $a_2 = 0.1227$ m/(s·100 MPa) in the SI input deck. The compute engine, which runs in CGS internally, evaluates the dimensionless ratio P/P_{ref} via `pres_loc` $\times 10^{-9}$ where `pres_loc` is in dyn/cm² and the implicit denominator is 10^9 dyn/cm² = P_{ref} . Readers comparing against SAND-1 [1] eq. 2.24, which renders the reference as 1 GPa, should note the factor-of-ten difference in the rendered reference but no difference in the numerical calibration: a_2 would scale by 10^{b_n} if SAND-1's labelling were taken at face value with the same numerical a_2 . See the Nomenclature chapter for the unit-equivalence list of P_{ref} .

The compaction fraction f_{dv} vanishes in undisturbed bed, so the compaction-burn term contributes only when the bed has compacted past its initial porosity. $\alpha^* = 75$ from SAND-1 [1] is the dominant driver of DDT in piston-driven cases. In pure thermal-ignition cases (as in this report's slow cook-off cylinder of §7), α^* has reduced sensitivity for the run-to-detonation distance; primary calibration in those scenarios is available through the surface-burn coefficient a_2 and the particle radius a_p .

On the term “Arrhenius.” The pressure-dependent surface-burn rate $a_2(P/P_{\text{ref}})^{b_n} H$ is sometimes called the “Arrhenius rate” in BN literature, but it is not an Arrhenius rate in the chemical-kinetics sense. A true Arrhenius rate has the form $A \exp(-E_a/RT)$ with explicit pre-exponential factor A and activation energy E_a , neither of which appears in equation (1.1). The actual form

is the empirical pressure-dependent surface-regression law $r = aP^n$ commonly used to fit solid-propellant strand-burn data, with the temperature dependence reduced to a Heaviside switch at a single ignition threshold T_{ign} . The “Arrhenius” label persisted as community shorthand for the kinetics-driven branch (as distinguished from the compaction-driven α^* branch); the present report drops the misnomer and uses “pressure-dependent surface-burn rate” for accuracy. Beyond the absence of explicit activation-energy and pre-exponential-factor structure, the BN form has effectively no temperature dependence above T_{ign} : once the Heaviside switch fires, the rate is constant in T (modulated only by pressure). The threshold T_{ign} itself is, however, a first-order tuning parameter in cookoff-coupled scenarios—it must be selected to sit just above the XCHEM handoff temperature plateau and just below the gas temperatures reachable by post-handoff compaction work, a narrow scenario-dependent window documented in §6.3. The BN form’s single thermal degree of freedom is therefore not just inert above threshold but *brittle* at the threshold itself, requiring hand-tuned per-scenario calibration where a smoothly-varying Arrhenius/DAE rate would adapt physically without intervention. Calibration is consequently overloaded: the surface-burn parameters (a_2 , b_n , T_{ign}) and the compaction-burn coefficient α_0 must absorb physics they do not structurally represent, and parameters tuned to one scenario rarely transfer cleanly to another. This is a substantive limitation of the BN form, and a primary motivation for the distributed-activation-energy adoption proposed in §10.5.

1.3.3. **How is the granular surface temperature T_{surface} computed?**

The Heaviside switch H in equation (1.1) turns on the surface-burn term once the granular surface temperature T_{surface} — the temperature at the gas–grain interface — crosses the ignition threshold. T_{surface} is not directly equal to either the gas temperature T_g or the bulk solid temperature T_s : the grain has finite thermal conductivity, so its surface heats up over time as gas–grain convection delivers heat to the interface, but the rate of heating depends on how quickly that heat conducts inward into the grain.

The Baer–Nunziato model handles this by tracking a quantity ζ — the *surface-temperature integral function* of Baer–Nunziato 1989 — that accumulates the cumulative effect of past gas–grain heating on each grain’s surface. A useful analogy: ζ behaves like a savings account, with the gas–grain temperature difference as the deposit rate. Whenever the gas is hotter than the grain surface, the account grows; when the two equilibrate, growth pauses. Formally, $\zeta = Q_{\text{cumulative}} / (3 m c_{p,s})$, where $Q_{\text{cumulative}}$ is the total heat the grain has absorbed from gas–grain convection and $m c_{p,s}$ is its heat capacity. The factor of 3 is the sphere’s surface-to-volume ratio ($3/R_p$); a different grain geometry (slab, rod) would give a different numerical factor, as flagged in the caveats below. Pristine bed at gas–solid equilibrium has $\zeta = 0$. The grain surface temperature at any moment depends on this accumulated record together with the current gas and solid temperatures. The model uses two equations to close the picture: an evolution equation for ζ , and an algebraic relation that gives T_{surface} in terms of ζ , T_g , and T_s .

Evolution of ζ . The evolution follows the SAND88-1073 / BN 1989 derivation [9, 16]. ζ advects with the solid velocity v_s and is driven by the gas–grain temperature gradient:

$$\frac{\partial \zeta}{\partial t} + v_s \frac{\partial \zeta}{\partial x} = \frac{\alpha_s \text{Bi}}{R_p^2} (T_g - T_{\text{surface}}), \quad (1.2)$$

where $\alpha_s = k_s/(\rho_s c_{p,s})$ is the solid thermal diffusivity, Bi is the gas–grain Biot number, and R_p is the grain radius.

Closure for T_{surface} . The closure is Baer–Nunziato 1989 [9], Eq. 14, derived in Baer 1988 [16]:

$$T_{\text{surface}} = \frac{1}{\text{Bi} + 1} \left[\frac{5(\zeta + T_s) + \text{Bi} T_g}{5 + \text{Bi}} + \text{Bi} \left(T_s + \sqrt{\psi^2 + 2\psi |T_g - T_s|} - \psi \right) \right], \quad (1.3)$$

with the variables (alphabetical):

- $\alpha_s = k_s/(\rho_s c_{p,s})$ [m²/s]: solid thermal diffusivity.
- Bi: a Biot number that sets how much the gas versus the solid drives surface heating. Namelist-tunable via `htfact_in` and `akm_in`.
- R_p [m]: grain radius.
- T_g, T_s [K]: gas and bulk-solid temperatures.
- v_s [m/s]: solid velocity (used for advection of ζ).
- ζ [K]: surface-temperature integral function — a running total of past gas–grain heating delivered to the grain surface (see the savings-account analogy above).
- $\psi = \max(0, \zeta \text{Bi}/2)$ [K]: a coupling parameter between ζ and the gas–solid temperature gradient inside the closure.

Equations (1.2) and (1.3) form a coupled implicit closure: ζ depends on T_{surface} through its evolution, and T_{surface} depends on ζ through equation (1.3).

How equations (1.1), (1.2), and (1.3) couple. The three equations together close the kinetic–heat-transfer loop:

- **Compaction work heats the gas.** The compaction-burn term in equation (1.1), proportional to $\alpha_0(P_{\text{mix}}/P_{\text{ref}})^2$, converts solid mass to gas, raising T_g and opening a gas–grain temperature gradient $T_g - T_{\text{surface}}$.
- **The temperature gradient grows ζ .** Equation (1.2) integrates the gradient over time, producing a running total ζ of the cumulative heat delivered to the grain surface.
- **ζ raises the surface temperature.** Equation (1.3) computes T_{surface} from ζ , T_g , and T_s . As ζ accumulates, T_{surface} rises above the simple gas–solid blend.

- **The surface temperature turns on the surface-burn kinetics.** The Heaviside switch H in equation (1.1) changes from 0 to 1 when T_{surface} crosses the ignition temperature threshold T_{ign} , turning on the pressure-dependent surface-burn term.

The full reactive cycle is therefore: compaction work $\rightarrow$ gas heating (Eq. 1.1) $\rightarrow T_g - T_{\text{surface}}$ gradient $\rightarrow \zeta$ accumulation (Eq. 1.2) $\rightarrow T_{\text{surface}}$ rises (Eq. 1.3) $\rightarrow$ Heaviside switch fires $\rightarrow$ surface-burn engages in equation (1.1). Equations (1.2) and (1.3) are the heat-transfer link without which the loop would not close.

Where these equations come from. The derivation lives in Baer 1988 [16] and is purely a heat-transfer analysis — no reaction kinetics involved. The setup is a representative spherical grain heated externally by gas convection, characterized by the Biot number Bi defined above. Rather than solving the heat equation inside the grain exactly, Baer assumes a plausible shape for the temperature profile and requires it to satisfy the heat balance only in an averaged sense — a standard approximate technique for problems of this kind. Equation (1.2) is the time-evolution of the integrated profile, and equation (1.3) is the closed-form algebraic relation it produces for the grain surface temperature. The two terms inside the square brackets of equation (1.3) come from this analysis in two limits:

- The **low- Bi limit** (first term inside the brackets) assumes a parabolic temperature profile inside the grain. Averaging this profile over the spherical grain volume produces the constant 5 that appears in the formula. The constant is geometry-dependent: a slab or rod would give a different number.
- The **high- Bi limit** (second term inside the brackets) assumes a thin heated layer just inside the grain surface, with a linear profile across that layer. Matching the surface heat flux to the heat carried by the layer gives a quadratic equation; its positive root produces the square root in equation (1.3). The square root is purely an algebraic feature of that quadratic, not a runaway-kinetics signature.

The two limits are blended by the overall $1/(Bi + 1)$ factor in equation (1.3).

Caveats acknowledged in the derivation. Baer 1988 [16] explicitly flags several approximations that the XDDT implementation inherits: (a) interparticle conduction is neglected; (b) the analysis assumes spherical grains — slabs or rods would require different constants; (c) the assumed temperature profiles are plausible fits, not exact solutions; (d) the formula that blends the low- Bi and high- Bi limits is a chosen weighting rather than one derived from the equations; (e) the method is expected to fail for non-monotonic or periodic thermal forcing such as cyclic heating; (f) mass transfer, phase change, and radiation are not represented. Readers wanting the technical names of the profile-fitting methods and the supporting derivation should consult [16] directly.

Is the form adequate? Equations (1.2) and (1.3) form a closed-form heat-transfer model: they capture the finite-conductivity equilibration of the grain surface temperature with the gas. The equations contain no reaction kinetics — those live upstream in the compaction-burn equation (1.1) of §1.3.2. Within the cookoff-driven DDT regime this report addresses, the Baer 1988 closure has been adequate: it provides a tractable surface-temperature relationship driven by a small number of physically interpretable inputs (gas–grain heat transfer, grain radius, solid conductivity) that can be measured or estimated from independent experiments rather than fit to the cookoff response data themselves. The caveats listed above bound where the form should not be applied — cyclic forcing, non-spherical grains, regimes where the assumed temperature profiles break down — but inside that scope the form has supported reproducible calibration across BN-model studies of granular explosives. The distributed-activation-energy upgrade proposed in §10.5 acts on the compaction-burn *kinetics* of equation (1.1), not on this heat-transfer closure.

Terminology. The Baer–Nunziato model carries multiple state variables that are sometimes loosely lumped together as “hot-spot” quantities. Three are distinct and worth keeping straight:

- The **surface-temperature integral** ζ introduced here is a running total of past gas–grain heating, evolved by equation (1.2). It is not an energy — it is a temperature-like quantity that records how much heat the grain surface has accumulated from gas–grain convection so far.
- A separate **hot-spot energy field** carries energy released by compaction-driven hot-spot mechanisms. It is a different physical quantity from ζ , governed by its own evolution equation, and does not enter equation (1.3). This field is not analyzed further in the present report.
- The **imposed hot-spot patch** of §8.3 (the asymmetric twin hot-spot boost in the boosted Chapter 8 deck) is a methodology-level initial condition for XDDT integration, applied through the `&xtiv_boost` namelist. It is not a state variable.

1.3.4. *How are radiative effects treated in the model?*

Radiation is included in the gas-phase energy transport via the Rosseland diffusion approximation. The radiative contribution is added to the gas thermal conductivity:

$$k_{\text{gas,eff}} = k_{\text{gas,mol}} + \frac{16 \sigma_{\text{SB}} T_g^3}{3 \beta_R}, \quad (1.4)$$

where σ_{SB} is the Stefan–Boltzmann constant, T_g is the local gas temperature, and β_R is the Rosseland mean absorption coefficient (a namelist parameter, $\beta_R = 500 \text{ m}^{-1}$ in this work). At pre-ignition gas temperatures ($\sim 300\text{--}600 \text{ K}$) the radiative term is negligibly small relative to molecular conduction; at post-ignition reaction-zone temperatures ($\sim 4000\text{--}5000 \text{ K}$, observed in the cylinder slow cook-off demonstration of §7) it is comparable in magnitude to the molecular gas conductivity. Particle heating from radiation enters indirectly: the radiation-enhanced gas conductivity drives the gas–solid interface heat transfer in the granular surface-temperature model, and that interface-temperature mechanism heats the solid. The solid-phase energy equation has no separate radiation

term, and the model does not represent solid-to-solid radiative exchange between hot and cool grains. This treatment is appropriate for optically-thick reactive gas in dense granular beds (typical of energetic materials at TMD fractions $\gtrsim 60\%$); in optically-thin regions it would underpredict radiative transport.

1.3.5. *Can XDDT model high-density energetic materials with little or no initial porosity?*

XDDT is built on the Baer–Nunziato two-phase mixture-theory framework, which presumes both gas and solid phases coexist throughout the integration domain. For a fully dense single crystal or pressed charge with $\phi_g = 0$ exactly, the BN gas-phase equations have no domain and XDDT cannot integrate from such a state.

In the cookoff-coupled use case treated in this report, this limitation is moot: porosity is generated naturally during the XCHEM thermal-runaway phase as the solid decomposes into gas-phase products that pressurize the void space. The handoff state delivered to XDDT therefore always has finite gas content, regardless of the original pristine porosity — XCHEM provides the porosity that XDDT requires.

For materials with very small but nonzero initial porosity (e.g., $\phi_g = 10^{-4}$), XDDT in principle remains applicable, but two practical considerations apply. First, numerical stiffness: the gas-phase partial density per cell is small, gradients at material interfaces are steep, and the integrator strains. The handoff-state smoothing knob (§5) was designed for exactly this regime. Second, calibration scope: the BN constitutive parameters used here (a_2 , b_n , α^* , T_{ign}) are tuned against mid-porosity granular-HMX experiments [11, 12, 17], where the void-collapse mechanics encoded in the compaction-burn term are the dominant ignition-amplification pathway. At nearly-fully-dense loadings void collapse is no longer the limiting mechanism, and the calibration does not extrapolate cleanly — DDT could in principle still be reached, but the time, position, and velocity of transition would not be reliable extrapolations from the mid-porosity calibration.

For nearly-fully-dense pressed charges under shock loading — the SDT regime described in §1.3.1 — the relevant physics is mechanical hot-spot formation, which is the natural domain of reactive-flow or ignition-and-growth models [18] rather than BN’s compaction-burn coupling. Analysts attempting to apply the XTIV methodology to that regime should expect the model to fail.

1.3.6. *What is the steady detonation-wave speed D , and why no subscript?*

Following the convention introduced in SAND-1 [1] of this series, the steady detonation-wave speed reached after the deflagration-to-detonation transition is denoted D_2 , distinguishing it from a transient D_1 that can appear earlier — a faster wave propagating through an inert plug ahead of the reaction zone, observed in some experiments (e.g. McAfee’s ionization-pin data on heated HMX cylinders [11]). None of the four demonstration cases in this report exhibit a D_1 stage; in every case the burn front transitions directly from compaction-driven propagation to a single steady wave. To keep the body chapters readable for non-specialist audiences we therefore drop the subscript and

write D for the steady detonation-wave speed throughout this report. Readers comparing values to SAND-1 or SAND-2 [2] should equate D here with D_2 there.

1.4. Roadmap

Chapter 2 formulates the BN equations in the unified one-dimensional form covering slab, cylindrical, and spherical geometries, and lays out the boundary-condition treatment used throughout the report. Chapter 3 describes the architecture of the coupling pipeline: the transfer-file format, the extractor, and XDDT's IBC=3 initialization mode. Chapter 4 presents the burn-front and gauge-history diagnostic redesigns required to interpret the coupled trajectories. Chapter 5 documents the handoff-state smoothing knob that addresses sharp temperature and density jumps at material interfaces, which would otherwise destabilize the integrator. Chapter 6 reports a series of slab demonstration runs, including parameter sweeps that establish how a hot-spot ignition patch controls when and where ignition occurs but does not change the speed of the resulting detonation wave once it stabilizes. Chapter 7 applies the pipeline to the NATO AOP-4382 cylinder configuration and reports inward-propagating reactive wave amplification consistent with classical converging-wave detonation transition. Chapter 8 extends the demonstration to a twin-fragment-impact configuration. A natural-cockoff fragment-spacing sensitivity sweep establishes the natural-cockoff baseline (fizzle to transition, depending on fragment spacing); an asymmetric twin hot-spot boost layered on the natural-cockoff handoff then produces a CJ-class wave through front-on-front collision of two converging reactive bands, benchmarked against a TIGER hot-CJ calculation at the boost-imposed initial-energy condition. Chapter 9 synthesizes the demonstration results into a single organizing finding: the outcome — fizzle, transition, or CJ-class detonation for an imposed-boost initial state — depends on how much of the explosive has been preheated before ignition. Chapter 10 describes open issues including the outer-boundary-condition handling, the sphere-geometry / ODTX cockoff handoff window, and the path forward to a distributed activation-energy kinetic upgrade in the next report of the DDT series. Chapter 11 summarizes contributions and identifies next steps. Appendix A reproduces the full specification of the XTIV transfer-file format for readers who want to write their own extractor or read the file from another program. The postlog (Appendix C) closes the report with the workflow disclosure for the AI-assisted preparation of this document, including the human–AI division of labor, reproducibility considerations, and the export-control assessment.

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2. GEOMETRY GENERALIZATION AND BOUNDARY CONDITIONS

XDDT supports three 1D geometric configurations: planar slab, cylindrical, and spherical. The cylindrical and spherical configurations each accommodate both *solid* beds extending to the axis or geometric center and *hollow* beds with a finite inner radius $R_i > 0$ (annulus or shell); see §2.3. All three share the Baer–Nunziato two-phase mixture description summarized below; they differ only in the form of the divergence operator that appears in each conservation equation and in the boundary conditions imposed at the inner edge of the bed. The full Baer–Nunziato model with all source terms, equation-of-state closures, and constitutive relations is documented in the first report of this series [1] and originates with Baer and Nunziato’s 1986 mixture-theory paper [8], with the compaction equation extended in their 1989 follow-on [9]. This chapter records only the small subset of that material that the geometry generalization changes.

2.1. Unified 1D form

Let r denote the single spatial coordinate, with n the geometric index: $n = 1$ for the planar slab, $n = 2$ for the cylindrical geometry, $n = 3$ for the spherical geometry. Each conservation equation in the BN model then takes the unified form

$$\frac{\partial Q}{\partial t} + \frac{1}{r^{n-1}} \frac{\partial}{\partial r} (r^{n-1} \mathcal{F}(Q)) = \mathcal{S}(Q), \quad (2.1)$$

where Q is the conserved variable for the equation in question ($\phi_\alpha \rho_\alpha$ for continuity, $\phi_\alpha \rho_\alpha u_\alpha$ for momentum, $\phi_\alpha \rho_\alpha E_\alpha$ for energy, $\alpha \in \{s, g\}$), $\mathcal{F}(Q)$ is the convective flux, and $\mathcal{S}(Q)$ is the source term collecting the inter-phase exchange, the non-conservative pressure-gradient contribution, and (for the compaction equation) the intergranular-stress-difference closure. The pressure-gradient and compaction terms retain their planar form; only the convective-flux divergence picks up the geometric weighting.

Expanding the divergence in (2.1) gives the equivalent quasi-conservative form

$$\frac{\partial Q}{\partial t} + \frac{\partial \mathcal{F}(Q)}{\partial r} + \frac{n-1}{r} \mathcal{F}(Q) = \mathcal{S}(Q), \quad (2.2)$$

in which the geometric source $(n-1) \mathcal{F}(Q)/r$ appears explicitly and is the term responsible for the inward pressure focusing observed in the cylinder demonstration of Chapter 7.

Table 2-1 itemizes, equation by equation, what plays the role of Q , $\mathcal{F}(Q)$, and the main contributions to $\mathcal{S}(Q)$ across the BN conservation laws. The geometric weighting acts on $\mathcal{F}(Q)$ uniformly — every convective flux picks it up, regardless of the equation. The terms in $\mathcal{S}(Q)$ retain their planar form. The momentum equation is the case where this distinction matters most: pressure

is isotropic, so its gradient stays in $\mathcal{S}$ rather than being folded into $\mathcal{F}$; if it were, the geometric expansion would generate a spurious $(n-1)\phi_\alpha P_\alpha/r$ source that does not represent a real geometric effect on momentum.

Table 2-1. Geometric weighting across the BN conservation equations. The convective flux $\mathcal{F}(Q)$ acquires the $1/r^{n-1}$ weighting (equivalently the $(n-1)\mathcal{F}/r$ geometric source); the contributions to $\mathcal{S}(Q)$ retain their planar form. Phase index $\alpha \in \{s, g\}$.

Equation	Conserved Q	Convective flux $\mathcal{F}(Q)$	Main contributions to $\mathcal{S}(Q)$
Continuity ($\times 2$)	$\phi_\alpha \rho_\alpha$	$\phi_\alpha \rho_\alpha u_\alpha$	inter-phase mass exchange
Momentum ($\times 2$)	$\phi_\alpha \rho_\alpha u_\alpha$	$\phi_\alpha \rho_\alpha u_\alpha^2$	$-\phi_\alpha \partial P_\alpha / \partial r$, drag
Energy ($\times 2$)	$\phi_\alpha \rho_\alpha E_\alpha$	$(\phi_\alpha \rho_\alpha E_\alpha + \phi_\alpha P_\alpha) u_\alpha$	inter-phase heat exchange, body forces
Compaction	ϕ_s	$\phi_s u_s$	intergranular stress difference
Scalars ($\zeta, f_{dv}, \text{ehs}$)	$\phi_s \rho_s \chi$	$\phi_s \rho_s \chi u_s$	scalar-specific source

2.2. Side-by-side comparison

Table 2-2 collects the geometry-specific pieces of (2.1) and (2.2) along with the boundary-condition conventions adopted by XDDT.

Table 2-2. 1D geometric configurations supported by XDDT. The geometric index n enters every BN conservation equation through the divergence operator. Each geometry has two physical edges with boundary conditions selected from the transfer file (see §2.3); cylindrical and spherical additionally accept `centerline` as the inner-edge BC for the solid case $R_i = 0$.

	Slab	Cylindrical	Spherical
Geometric index n	1	2	3
Coordinate	$x \in [0, L]$	$r \in [R_i, R_o]$	$r \in [R_i, R_o]$
Divergence operator	$\frac{\partial \mathcal{F}}{\partial x}$	$\frac{1}{r} \frac{\partial(r \mathcal{F})}{\partial r}$	$\frac{1}{r^2} \frac{\partial(r^2 \mathcal{F})}{\partial r}$
Geometric source	0	$\mathcal{F}/r$	$2\mathcal{F}/r$
Inner edge	physical BC	physical BC [†]	physical BC [†]
Outer edge	physical BC	physical BC	physical BC

[†]Solid bed ($R_i = 0$): `centerline` BC pins $v_g = v_m = 0$ on the symmetry axis or center. Hollow bed ($R_i > 0$): the inner edge takes `free` or `layer:X` just like any other physical surface.

2.3. Boundary conditions

The slab geometry is two-sided in the BC sense: both endpoints are physical surfaces and each independently selects an XDDT boundary-condition mode—piston-driven (IBC=1), specified-state (IBC=2), or transfer-file handoff (IBC=3, see §3.3). For the IBC=3 path used by all demonstrations in this report, the transfer file carries an `inner_side` and `outer_side` confinement designation that resolves to a free-surface, wall, or layer-impedance treatment. The twin-fragment-impact

case of Chapter 8 exercises this two-sidedness directly: distinct hot-spot patches and smoothing parameters are applied at each edge of the HMX bed.

The cylindrical and spherical geometries accommodate two distinct inner-edge configurations through the same `inner_side` dispatcher used by the slab path. A *solid* bed extends to the symmetry axis ($n = 2$) or geometric center ($n = 3$), with $R_i = 0$ and `inner_side = 'centerline'`; the dispatcher then pins $v_g = v_m = 0$ on the gas- and solid-momentum components at the first node (Dirichlet), while the scalar fields $\phi_s, \rho_\alpha, T_\alpha, P_\alpha$ retain Neumann. The geometric source $(n-1)\mathcal{F}/r$ in (2.2) is singular at $r = 0$; the discretization regularizes it by placing the first node at $r = \Delta r/2$ (cell-centered treatment) together with the centerline momentum Dirichlet. The slow cook-off cylinder of Chapter 7 is the solid-bed example in this report; the AOP-4382 thermal ramp on the outer wall is the matching outer-edge physical BC.

A *hollow* bed — an annular cylinder or spherical shell — has a finite inner radius $R_i > 0$ and takes a physical BC on the inner surface through `inner_side = 'free'` (Neumann) or `inner_side = 'layer:X'` (impedance match; currently a Neumann fallback pending the work documented in §10.1). The geometric source $(n-1)\mathcal{F}/r$ is regular throughout the domain in the hollow case, so no centerline regularization is required. The inner and outer edges of a hollow bed are treated symmetrically by the dispatcher — functionally indistinguishable from the two-sided slab path apart from the geometric source carried by $n > 1$. Hollow configurations are supported by the schema (`x0_exp`, `x1_exp`, `inner_side`) and the solver directly; no hollow demonstration is included in the present report, but reviewers interested in annular or shell cookoff geometries (e.g., HMX surrounding a central void or inert core) should expect the pipeline to run them through the same IBC=3 path as the solid cases.

The current pipeline implementation supports all three geometries on the IBC=3 path, but the demonstration matrix in this report exercises the slab path (Chapters 6 and 8) and the cylindrical path (Chapter 7). An attempt to extend the matrix to spherical geometry through the standard One-Dimensional Time-to-eXplosion (ODTX) test of McGuire and Tarver [10]—a small-scale HMX cookoff configuration consisting of a 0.25-inch-radius HMX core inside an aluminum jacket held at a fixed boundary temperature—was attempted during the current development. The XDDT integrator failed within a small fraction of a microsecond of simulation time. DDEBDF — the backward-differentiation-formula integrator that XDDT uses for the stiff BN system — reported its convergence-failure code (IDID= -6) at $t \approx 60$ ns: the integrator’s implicit corrector iteration could not converge, which in practice means that the local system had become too stiff for the integrator to advance with any usable step size. No meaningful reactive-wave development was observed before the failure. The BN equations admit the spherical case in principle, but the additional geometric source factor of two relative to the cylindrical $1/r$ source, combined with the $1/r^2$ singularity scaling at the centerline, has so far defeated extended integrations on the coupled spherical configurations attempted to date. A full diagnosis of the failure modes—including the interaction between the handoff-state smoothing convention, the multi-cell hot-shell-on-cold-bulk character of the ODTX handoff, and the gradient stiffness at the reactive face—is laid out in §10.3.

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3. COUPLING PIPELINE ARCHITECTURE

This chapter describes the three components of the XTIV coupling pipeline: the transfer-file format that captures the XCHEMA state at handoff (§3.1), the extractor that converts XCHEMA’s `ofield.plt` output into the transfer-file format (§3.2), and the IBC=3 initialization mode added to XDDT to consume the transfer file (§3.3). Figure 3-1 shows the data flow.

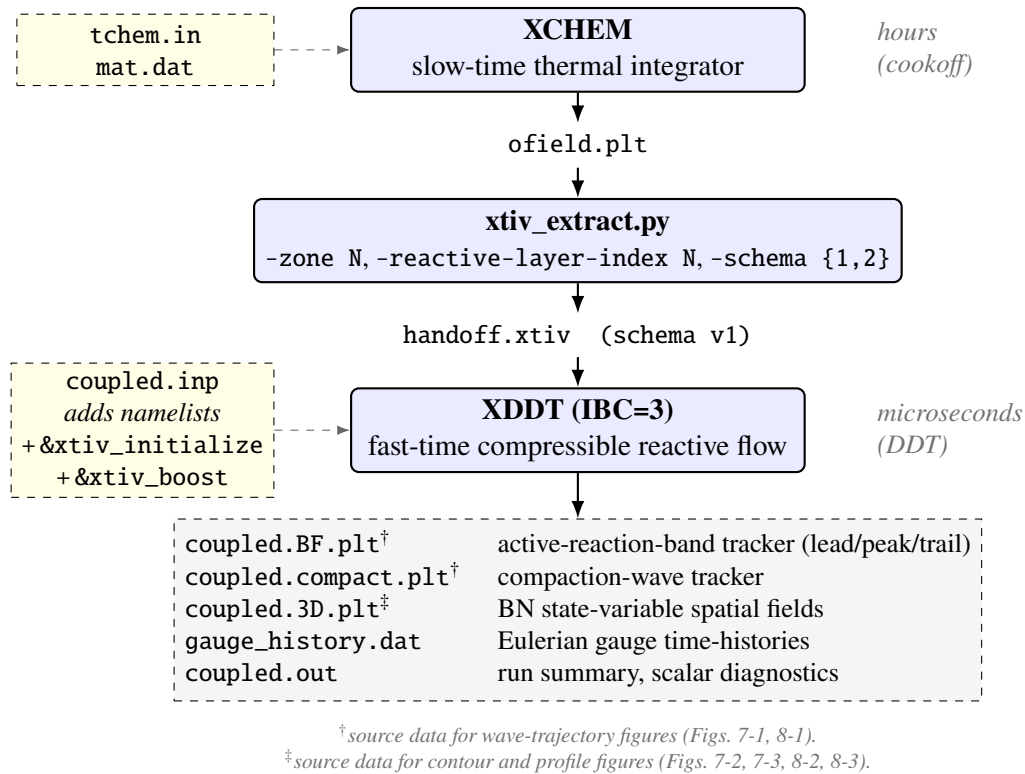


Figure 3-1. XTIV coupling pipeline data-flow schematic. The pipeline runs in three stages. **XCHEMA** (top) integrates the slow-time thermal evolution from external heating through cookoff over hours of simulated time, taking the deck file `tchem.in` and the shared material database `mat.dat` as input. **xtiv_extract.py** (§3.2) reads XCHEMA’s `ofield.plt` field-state output at the chosen pre-runaway zone and writes a `handoff.xtiv` transfer file in schema-v1 format (§3.1). **XDDT** (bottom) consumes the transfer file via the IBC=3 initialization mode (§3.3) and integrates the fast-time compressible reactive flow over microseconds of simulated time, with user-controlled preconditioning via the `&xtiv_initialize` and `&xtiv_boost` namelists in the XDDT deck `coupled.inp`. The pipeline assumes material consistency between stages (§3.4); the `xtiv_expected_material_in` parameter enforces this at runtime. Solid arrows trace the data flow between stages; dashed arrows show user-supplied deck files entering each program.

3.1. Transfer-file format

A single file is used to pass information from the XCHEM code to the XDDT code. It holds a snapshot of the XCHEM state at the chosen handoff moment, written as plain ASCII text with named sections in a fixed order. The file extension is `.xtiv`. All values are SI; no unit conversion happens at the interface (XDDT applies SI→CGS conversions internally after reading, mirroring the pattern in the rest of the source tree).

Two schema versions exist. Schema v1 (the default format) carries the XCHEM-reported gas pressure p alongside the rest of the state, and XDDT cross-checks p against its own equation-of-state evaluation as a soft consistency check. Schema v2 drops p from the file — XDDT owns gas pressure via its own EOS — and adds a `mass_fraction` diagnostic that lets XDDT audit local mass conservation between the two codes. The extractor’s `--schema` option (set when the user runs the extractor) selects which version to write; XDDT keys on the `schema_version` field in the file’s `[meta]` block and accepts either. The demonstrations in this report use the default schema v1; schema v2 is recommended for new work because the `mass_fraction` field it adds enables an automated mass-balance audit of the handoff state at read time.

The file’s seven sections, in fixed order, are:

- `[meta]` — schema version, problem identity, handoff timing (`t_src`, `zone_index`, `ignition_flag`, `ignition_cause`), code-build provenance.
- `[geometry]` — 1D coordinate system (slab/cylinder/sphere), reactive-layer extent, run-derived geometry scalars.
- `[material]` — reactive-material constants needed for the XDDT-side reconstruction of ρ_c at each node (ρ_{co} , thermal-expansion coefficients, phase-change ramps).
- `[confinement]` — inner- and outer-edge boundary-condition selectors for the XDDT integration.
- `[grid]` — per-node coordinates and refinement-rank metadata.
- `[fields]` — per-node primitive state. The columns differ between schema versions (Table 3-1).
- `[end]` — end-of-file marker; XDDT checks this is the last section before declaring the file valid.

Table 3-1. Per-node [fields] columns by schema version. *Primary* columns are consumed to populate XDDT’s state vector; *cross-check* columns are consumed only as safety-belt consistency checks.

Schema	Column	Unit	Quantity	Role
v1, v2	T_K	K	temperature	primary
v1, v2	rho_g	kg/m ³	gas partial density	primary
v1, v2	alpha_s	–	solid volume fraction	primary
v1	p_Pa	Pa	gas pressure	cross-check (v1)
v1, v2	S_f	–	unreacted-mass fraction	primary
v2	mass_fraction	–	mass-balance regime indicator	cross-check (v2)
v1, v2	rho_m_check	kg/m ³	extractor’s ρ_c recomputation	cross-check

Example fragment. A schema-v2 file (fields shown for two nodes, with the bulk of the [meta] and [material] sections elided):

```
[meta]
schema_version = 2
problem_title  = SITI-HMX
xchem_input    = tchem.in
write_time     = 2026-04-28T10:34:55Z    ! UTC clock time (ISO-8601)
t_src          = 2.92000e+03    ! seconds; XCHEM time at this zone
zone_index     = 145
ignition_flag  = T
tmax_reached   = 5.04503e+02    ! K

[geometry]
geom           = 1                ! 0 = slab, 1 = cylinder, 2 = sphere
x0_exp         = 0.0000e+00    ! m, inner edge of reactive layer
x1_exp         = 1.2700e-02    ! m, outer edge of reactive layer

[grid]
n_nodes        = 45
columns = i  r_m  ilev
1  1.36853e-05  1
2  2.73707e-05  1
... (43 more rows) ...

[fields]
columns = T_K  rho_g  alpha_s  S_f  mass_fraction  rho_m_check
3.7349e+03  1.2657e+03  3.9105e-02  6.480e-02  0.998  9.85e+02
3.7100e+03  1.2540e+03  3.8800e-02  6.510e-02  0.997  9.87e+02
... (43 more rows) ...

[end]
```

Built-in cross-checks. Two safety-belt consistency checks are built into the file format:

1. ρ_c *reconstruction*. Both the extractor and XDDT independently recompute the condensed-phase density ρ_c from T and the [material] constants using the same thermal-expansion formula (with optional phase-change cosine ramps). The extractor writes its result as `rho_m_check`; XDDT recomputes its own value at read time and aborts on a mismatch above 1%. This catches the entire class of “ ρ_c formula skew between the two codes” bugs — wrong expansion coefficient, mismatched phase-change parameters, sign error in either reconstruction.
2. *Mass-balance closure (schema v2 only)*. The extractor writes the dimensionless ratio

$$\text{mass_fraction} = \frac{\rho_g (1 - \alpha_s)}{\rho_{co} (1 - \phi_o) (1 - S_f)}, \quad (3.1)$$

which equals unity for a sealed, mass-conserved cell, drops below unity in cells losing gas through permeable transport, and exceeds unity in cells gaining gas. XDDT independently recomputes the same ratio from the transferred fields and aborts on a mismatch above 1% (a placeholder value of -1 marks pristine cells where $1 - S_f$ approaches zero and the denominator is degenerate).

The full schema — including the complete [meta], [geometry], [material], and [confinement] key list, the ρ_c -reconstruction formula, and XDDT’s full list of error conditions on the IBC=3 path — is reproduced in Appendix A.

3.2. Extractor: `xtiv_extract.py`

The extractor is a Python tool that reads XCHEM’s text output and writes the transfer file described in §3.1. It is a stand-alone script with no run-time dependencies beyond the Python standard library, so it can be invoked from the same shell environment that runs XCHEM and XDDT, with no separate installation step. The source lives at `tools/xtiv_extract.py` in the source tree.

Usage modes. The tool runs in three mutually-exclusive modes (one of `-list`, `-zone`, or `-time` must be given on each invocation):

- `-list <ofield.plt>` — print one line per zone (zone number, time, max temperature, max pressure, minimum unreacted-mass fraction). Zones are numbered starting from 1, matching XCHEM’s own diagnostic output: zone 1 is the first zone written, zone 2 the second, and so on. No transfer file is written. Useful for picking a zone or time to extract. If `-tchem-in` and `-mat-dat` are also supplied, a per-layer table is printed first showing which layers are reactive — the table needed when the input deck contains multiple reactive layers.
- `-zone N <ofield.plt>` — extract zone N into a transfer file ($N = 1$ selects the first zone listed by `-list`).

- `-time T <ofield.plt>` — extract the zone whose stored time `t_src` is closest to T (seconds); the actual `t_src` used is recorded in the output file's `[meta]` block.

`-zone` and `-time` also require `-tchem-in <path>`, `-mat-dat <path>`, and `-o <out.xtiv>`.

Optional command-line options.

- `-schema {1,2}` — transfer-file schema version to write (default 1). Schema v2 omits gas pressure and adds the `mass_fraction` diagnostic per §3.1.
- `-reactive-layer-index N` — index of the reactive layer to extract, counting from 0: $N = 0$ selects the first reactive layer in the per-layer table, $N = 1$ selects the second, and so on. (The offset relative to `-zone`, which counts from 1, follows the extractor's internal Python list convention for layers; zone numbering follows XCHEM's own 1-based diagnostic output.) Required when `tchem.in` contains multiple reactive layers (e.g., the twin-fragment-impact configuration of §8 uses a 3-layer steel–HMX–steel deck and selects the HMX layer); ignored when only one reactive layer is present. Run `-list` with `-tchem-in` and `-mat-dat` to see the layer table.
- `-override-ndv N` — force the number of phase-change ramps used in the ρ_c reconstruction (debugging only; $N=0$ skips the cosine-ramp summation entirely).
- `-write-time <ISO-8601>` — explicit write-time stamp for the `[meta]` block; the default is the current UTC time at run time. Setting an explicit value is useful when the goal is exact reproduction of an archived transfer file.
- `-quiet` — suppress the per-zone summary line that is otherwise printed to the terminal after a successful extraction.

Input files read.

- `ofield.plt` — per-node field history written by XCHEM. Each zone in the file is one timestep of the cookoff integration, with one row per spatial node. Column 21 carries the refinement rank `ilev`; the extractor copies this verbatim to the transfer file's `[grid]` block.
- `tchem.in` — the XCHEM input deck. The extractor reads the geometry block (`geom`, `pgeom`, layer interfaces), the per-layer table (layer name, reference temperature `tlay`, permeability flag), the pressure-unit and initial/maximum-pressure entries, and the excess volume `evol` used to compute `maxstrain` for the `[geometry]` block.
- `mat.dat` — the shared material database. The extractor pulls the reactive layer's constants (condensed-density at reference temperature `rho_co`, thermal-expansion coefficients `bvol` and `bslope`, number of phase-change ramps `ndv`, and per-ramp triplets) for the `[material]` section and for the per-node ρ_c recomputation.

The extractor does not read any in-memory XCHEM state; it works exclusively from text files XCHEM has already written. This separation keeps the pipeline reproducible from the same archived inputs and outputs without requiring a re-run of XCHEM.

Behavior. Given a zone selection and the three input files, the extractor:

1. Identifies the unique reactive layer in the `tchem.in` layer list (by cross-referencing `mat.dat`); aborts when multiple reactive layers are present and `-reactive-layer-index` was not supplied.
2. Filters the selected zone's rows to spatial coordinates inside the chosen reactive layer.
3. Computes the run-derived `maxstrain` from the layer's geometric volume, the excess volume `evol`, and the initial gas volume fraction (formula in Appendix A).
4. Derives the inner- and outer-edge confinement labels for the `[confinement]` block from the flanking layers ('`centerline`' for the inner edge of a centered cylinder or sphere, '`free`' for an exposed edge, '`layer:<name>`' when a non-reactive layer flanks the reactive one).
5. Sets the `ignition_flag` and `ignition_cause` fields in `[meta]` based on whether the maximum temperature in the selected zone reached XCHEM's runaway threshold.
6. Replicates XCHEM's ρ_c formula (Appendix A, eqs. A.1–A.4) at each node and writes the result as the `rho_m_check` column in `[fields]`. When `-schema 2` is selected, the extractor additionally computes the `mass_fraction` diagnostic (eq. 3.1 of §3.1) at each node, with a placeholder value of `-1` marking pristine cells.
7. Writes the assembled file to the path given by `-o/-output`. All numeric values are written in fixed-precision scientific notation (five significant digits for primary state fields, six for `rho_m_check`); these formats match the reference-fixture corpus used to validate the extractor against the schema specification.

The default schema version is 1, and the demonstrations in this report use that default. Schema v2 is recommended for new work: the `mass_fraction` field it carries lets XDDT cross-check mass conservation against XCHEM at read time, an audit that schema v1 omits.

3.3. IBC=3 initialization mode

XDDT's initial-boundary-condition selector IBC chooses how the integration is initialized. The legacy options `IBC=0, 1, 2` (used in SAND-1 and SAND-2 piston-driven and specified-state setups) remain unchanged. This work adds `IBC=3` as a fourth option that initializes the integration from a transfer file rather than from a procedural initial condition.

Triggering and entry point. Setting `IBC=3` in the `&numerical` namelist of the XDDT input deck routes the run through the `IBC=3` path on entry to the initialization routine `nvalue0`. The transfer-file path is given by the namelist parameter `xtiv_file_in` (no default; the run aborts with a clear diagnostic if the file is missing or unreadable). The legacy `IBC=1` namelist parameters (`pg_ign`, `tg_ign`, `l1_ign`) are still read by the namelist machinery for backward compatibility but are unused on the `IBC=3` path.

Behavior on entry. Given a valid transfer-file path, XDDT performs the following steps in order, aborting on any cross-check failure:

1. Opens the transfer file and parses [meta]. Verifies that `schema_version` is in the accepted set $\{1, 2\}$ and uses the appropriate schema-specific parsing for the per-node fields.
2. Parses [geometry]. The `geom` value (slab/cylinder/sphere) is inherited from the file unless the user has set `xtiv_override_geom_in` ≥ 0 in the XDDT deck, in which case the deck value wins (an escape hatch for sensitivity studies).
3. Parses [material]. If the user has set `xtiv_expected_material_in` in the XDDT deck, the file's `name` field is cross-checked against that value, and a mismatch (e.g. HMX-tuned XDDT parameters with a non-HMX XCHEM handoff) aborts the run. With this parameter empty (the default), this check is skipped; see §3.4 for the full material-consistency model.
4. Parses [confinement]. Resolves the `inner_side` and `outer_side` strings to XDDT's internal boundary-condition selectors. An optional user override is available via `xtiv_outer_bc_in` in the XDDT deck.
5. Parses [grid]. Sets the per-node coordinate arrays and the refinement-rank metadata that XDDT's adaptive-mesh subsystem consumes.
6. Parses [fields]. For each node:
 - Reconstructs ρ_c from the transferred temperature and the [material] constants (Appendix A). Compares against `rho_m_check`; aborts on a mismatch above 1%.
 - For schema v2, recomputes the `mass_fraction` diagnostic and aborts on a mismatch above 1% at any non-pristine node.
 - Applies the `&xtiv_boost` preconditioning if it is present in the deck (see below).
 - Evaluates XDDT's gas equation of state at (T, ρ_g) and its solid equation of state at (T, ρ_c) to populate the per-node energy fields.
 - Populates XDDT's per-node BN state from the loaded primitives. The BN model's hot-spot state variables, including the surface-temperature integral ζ of §1.3.3, start at zero — pristine bed is assumed at the moment XDDT integration begins. The handoff of the related pyrolysis-progress variable π_y versus the XCHEM solid-mass fraction S_f is discussed in §3.3.1.
7. Verifies that the [end] marker was reached. A truncated file aborts the run.
8. Logs the handoff to `xddt.log`: file path, `schema_version`, `t_src`, the per-node minimum and maximum of the primary state fields, and a summary of any `&xtiv_boost` modifications applied.

After step 8 the IBC=3 path returns control to the regular XDDT integrator before reaching the legacy burst-pressure logic that IBC=0/1/2 use — the burst-pressure path is bypassed because the boundary state is already specified by the transfer file. The complete list of fatal-error conditions XDDT enforces on the IBC=3 path is reproduced in Appendix A.

The &xtiv_boost preconditioning namelist. When IBC=3 is active, the XDDT deck may optionally include the &xtiv_boost namelist to apply controlled modifications to the loaded handoff state before the integration starts. Defaults preserve the transferred state. Each modified parameter is echoed to `xddt.log` at startup so the modified state is auditable after the run.

Table 3-2. Parameters in the &xtiv_boost namelist. Defaults preserve the state delivered by the transfer file.

Parameter	Default	Effect
<i>Global scaling (every node)</i>		
t_scale_in	1.0	multiplies the transferred temperature at every node by this factor
p_scale_in	1.0	multiplies the transferred gas pressure (informational only on schema v2; logged but does not affect integration)
alpha_s_scale_in	1.0	multiplies the solid volume fraction at every node by this factor (clamped to $[10^{-4}, 1 - 10^{-4}]$)
<i>Inner-edge hot-spot patch</i>		
hot_spot_l_in	0	length (m) of the inner-edge patch; zero disables the patch
hot_spot_T_in	0	temperature (K) override inside the inner-edge patch
hot_spot_P_in	0	pressure (Pa) override inside the inner-edge patch
hot_spot_alpha_s_in	0	solid-volume-fraction override inside the inner-edge patch
<i>Outer-edge hot-spot patch (twin-source configurations)</i>		
hot_spot_outer_l_in	0	length (m) of the outer-edge patch; zero disables it
hot_spot_outer_T_in	0	temperature (K) override inside the outer-edge patch
hot_spot_outer_P_in	0	pressure (Pa) override inside the outer-edge patch
hot_spot_outer_alpha_s_in	0	solid-volume-fraction override inside the outer-edge patch

The two groups of `hot_spot_*_in` parameters together implement the imposed hot-spot patch initial condition used by the slab demonstration of §6.1 (inner edge only) and the twin-fragment-impact configuration of §8 (both edges, applied asymmetrically to capture the LHS-of-mid collision frame discussed in that chapter). The inner-edge patch is applied as a uniform override over the first `hot_spot_l_in` meters of bed adjacent to the inner boundary; the outer-edge patch covers the last `hot_spot_outer_l_in` meters adjacent to the outer boundary. Each non-zero override field takes precedence over the loaded handoff state inside its patch. The imposed hot-spot patch is the methodology-level surrogate for preheated-bed extent that the slab and fragment-impact chapters discuss; it is distinct from the BN model’s internal hot-spot state variables described in §1.3.3.

3.3.1. Pyrolysis-progress versus solid-mass-burnup

The pyrolysis-progress variable π_y (slot 8 in XDDT’s per-node solution vector) tracks the mass density of pyrolysed gas under XDDT’s P_{mix}^2 -driven dynamic-burn kinetics; the MMP solid mass fraction S_f transferred from XCHEM tracks a different reaction — the Arrhenius solid-phase decomposition that drives the cookoff up to runaway. At the handoff frame, $\pi_y = 0$ is the physically correct initial condition: XDDT’s dynamic-burn mechanism has not yet acted, and the gas density

ρ_g delivered by XCHEM already reflects whatever solid has decomposed during cookoff. The mass-balance information carried by S_f is therefore preserved through the handoff via (ρ_g, ϕ_s) rather than through the `F_sol(8)` slot, and the schema-2 mass-fraction cross-check (§4) audits this consistency at read time. π_y does not participate in either the burn-front tracker or the BN-DDT criterion, so this packaging choice does not affect the report’s principal diagnostics.

3.4. Material consistency between XCHEM and XDDT

XCHEM and XDDT describe disjoint physical regimes and consequently need disjoint sets of material parameters. XCHEM’s `mat.dat` holds the parameters for the slow-time cookoff phase: condensed density ρ_{co} , TMD, thermal-expansion coefficients, phase-change ramps, and the pre-thermal-ignition kinetics. The XDDT deck holds the BN-closure parameters that describe the post-handoff dynamics: a_2 , b_n , α_0 , T_{ign} , grain radius, gas-grain heat-transfer factor, and solid-matrix conductivity (namelist parameters `a2_in`, `bn_in`, `alstar_coeff_in`, `tign_in`, `ap_in`, `htfact_in`, `akm_in`). These two parameter sets do not overlap, so they cannot “disagree” against each other in any meaningful sense.

The thermal and density constants XDDT needs at the handoff frame — principally ρ_{co} and TMD, used in the `rho_m_check` cross-check of §3.1 — ride along in the transfer file’s `[material]` block. XDDT consumes them directly without an XDDT-side copy of `mat.dat`, so the values used by both codes at the handoff frame are guaranteed equal by construction.

On entry to IBC=3 the pipeline supports an optional sanity check. When the namelist variable `xtiv_expected_material_in` is set in the XDDT deck, the transfer file’s `[material]` name field must match that user-asserted value, and a mismatch aborts the run with a diagnostic naming both values. The default is empty, in which case the check is skipped. This guards against an XDDT deck calibrated for one explosive being handed a transfer file produced from an XCHEM run on a different explosive — e.g., an HMX-tuned BN deck paired with a PETN cookoff handoff — which is the only consistency failure that the current code paths can produce.

A separate observation is that the user must currently populate XCHEM’s `mat.dat` and the XDDT deck independently when working with a new material, even though both describe the same physical explosive. A unified material library spanning both codes — pick HMX (or PETN, or another material of interest) once, and both the cookoff and the BN-closure parameters are sourced from a shared entry — would remove the duplicated bookkeeping. Whether the resulting tighter coupling between the two codes is worth the reduced flexibility (e.g., the ability to swap in a one-off BN calibration without touching the cookoff library) is an open design question and is left as future work in §10.

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4. DIAGNOSTICS REDESIGN

The coupled XCHEMA–XDDT trajectories presented in the demonstration chapters could not be interpreted with the diagnostics inherited from SAND-1 and SAND-2. Two redesigns were required: the burn-front tracker, which now follows the active reaction band rather than the rearmost satisfying node, and the gauge-history output, which now distinguishes kinetic engagement from compaction-driven motion at probe locations. This chapter documents both redesigns, illustrates them on the demonstration cases, and confirms that they preserve the SAND-1 and SAND-2 validation results exactly on the historical piston-driven test cases.

4.1. Burn-front tracker redesign

The legacy burn-front tracker used in SAND-1 and SAND-2 located the burn front by scanning the bed for the rightmost node satisfying a cumulative-state criterion (specifically, $\phi_s < 0.98 \phi_{s,0}$) and reporting that node’s index as `ibf`. This worked well on the piston-driven validation cases of those reports, but three failure modes surfaced when it was applied to coupled XCHEMA–XDDT trajectories.

Failure mode 1: boundary-noise lock-in. When the outer-edge cell carries even a small numerical perturbation — introduced by the boundary-condition discretization or by handoff-state interpolation in the IBC=3 path of §3.3 — it can satisfy the legacy tracker’s threshold by accident. The tracker latches onto that cell and reports the burn-front position pinned at the bed end while the real reaction is happening elsewhere. This was confirmed in the unsmoothed-handoff slab cases during the current development.

Failure mode 2: compaction masking. When compaction has globally raised ϕ_s above $\phi_{s,0}$ across the bed — as it does in the boost-driven configurations that compact the entire bed during the first few microseconds — no cell satisfies the cumulative criterion $\phi_s < 0.98 \phi_{s,0}$, even at locations where reaction is actively proceeding. The tracker reports no burn front, and the reaction proceeds invisibly to the diagnostic. Confirmed in the $t_{\text{scale}} = 1.5$ boost-driven case from the current development.

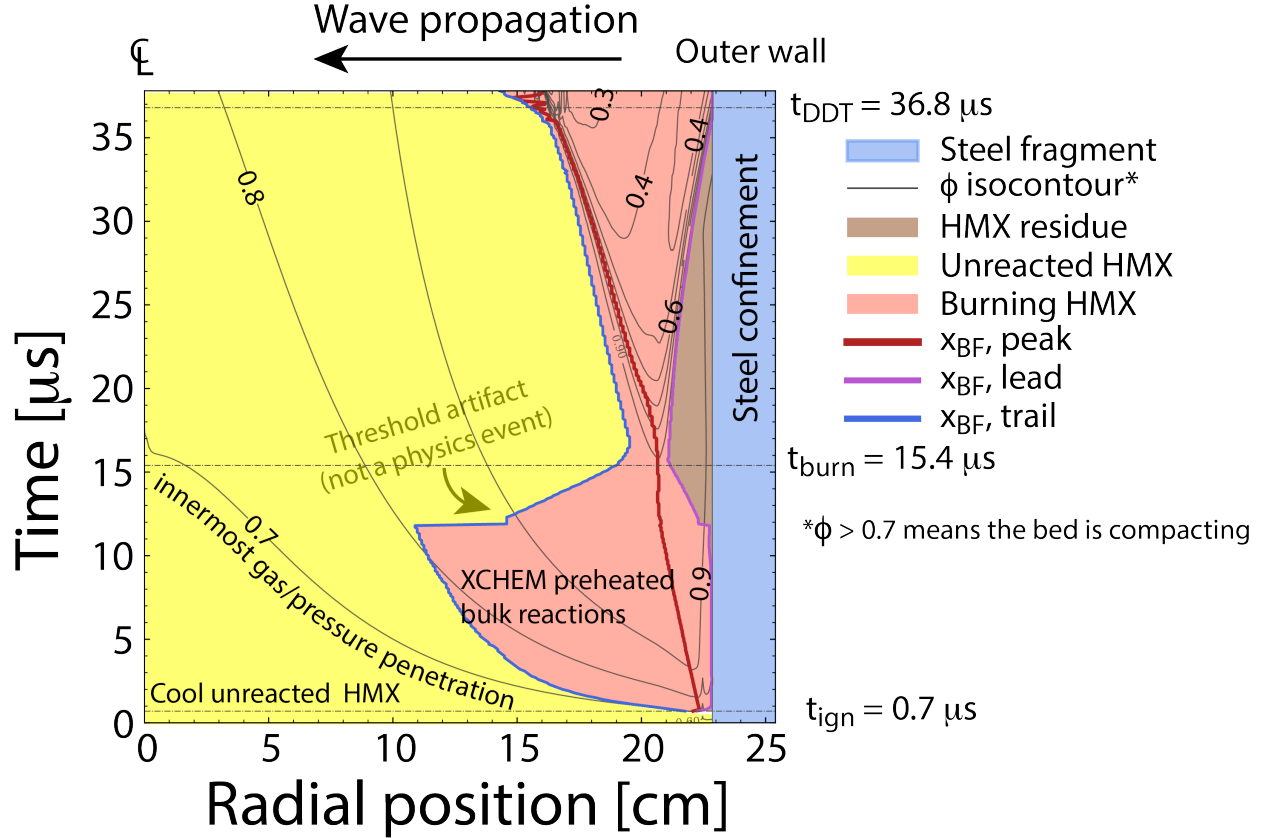


Figure 4-1. Burn-front tracker output for the post-thermal-ignition XDDT phase of the slow cook-off cylinder demonstration of Chapter 7. Color-coded regions show the bed state: yellow is unreacted HMX, pink is burning HMX, brown is HMX residue, and the right-hand strip is the steel jacket. Thin black lines are ϕ isocontours, with $\phi > 0.7$ indicating that the bed is compacting at that location and time. The active-reaction-band tracker's three positions are overlaid: a lead edge (magenta), the peak (red), and a trail edge (blue). The qualitative phases the figure exposes are: an early period with no reaction detected anywhere; a wide volumetric burn across the XCHEM-preheated HMX bed (the broad pink wedge before $t \approx 12 \mu\text{s}$); surface-burn engagement at $t_{\text{burn}} = 15.4 \mu\text{s}$, after which the band narrows to a localized inward-propagating wave; and the BN-DDT criterion firing at $t_{\text{DDT}} = 36.8 \mu\text{s}$. None of these phases was visible to the legacy single-position tracker. The annotated “threshold artifact” marks a feature of the noise-floor threshold that is not a physics event.

Failure mode 3: volumetric-vs-propagating ambiguity. A volumetric burn (uniform reaction across a preheated zone) and a propagating wave through cold bed both produce “rightmost satisfying” readings under the legacy criterion. The diagnostic cannot tell them apart. This distinction matters for the slow cook-off cylinder demonstration of §7, where the early-time reaction is volumetric across the preheated HMX bed before transitioning to a localized propagating wave converging toward the centerline.

Active-reaction-band redesign. The new tracker locates the contiguous band of cells where the local solid-fraction time derivative magnitude $|\dot{\phi}_s|$ exceeds a self-scaled noise floor, and reports three positions along that band: a lead edge (in the direction of propagation), the peak (the cell with

maximum $|\phi_s|$), and a trail edge. When no contiguous active band is detected the tracker emits a placeholder value, distinguishing “no reaction yet” from “reaction pinned at the bed end.” Two placeholder conventions appear in the report’s output files and are worth flagging together so they aren’t confused. The per-output-step trackers (`coupled.BF.plt`, `coupled.compact.plt`) emit a numeric sentinel row $-1.0 \ -1.0 \ -1.0$ for the lead/peak/trail positions when no band is currently detected. The scalar event markers in the run summary (`coupled.out`’s t_{ign} , t_{burn} , t_{DDT} , P_{max} , D_2 , etc.) instead use the textual placeholder “---” when the underlying condition is never satisfied over the integration — e.g., “ $t_{\text{burn}} = \text{---}$ ” means T_{mix} never crossed T_{ign} anywhere in the bed and the surface-burn kinetic switch never armed. The direction of propagation is computed from the sign of $dx_{\text{BF,peak}}/dt$ over a recent window of samples; this lets downstream diagnostics such as the BN-DDT criterion handle inward-propagating waves correctly without the legacy tracker’s implicit assumption that the wave moves left-to-right. Figure 4-1 (above) shows the tracker’s output on the slow cook-off cylinder run; the same view is reproduced in Fig. 7-1(b) of §7, where the cookoff physics behind the tracked phases is discussed in detail.

Hybrid retention for SAND-1 and SAND-2 markers. The legacy criterion’s `ibf` index is preserved internally and continues to drive the calibrated state-machine markers — ignition time, DDT firing, D extraction, stuck-physics detection — that were validated against SAND-1 and SAND-2 piston-driven data. Switching those markers to the new tracker would have required re-deriving the calibration thresholds against the historical regression suite; retaining `ibf` keeps both prior reports’ results unchanged while making the new cookoff trajectories in this report interpretable. The validation tests themselves are summarized in §4.3.

4.2. Gauge-history instrumentation

The legacy `gauge_history.dat` output carried per-probe time histories of mixture pressure, gas and solid temperatures, gas density, solid volume fraction, and a few derived quantities. These were sufficient for the piston-driven cases of SAND-1 and SAND-2 but missed two distinctions that this report requires: whether reaction at a probe is currently active or has stalled, and whether the active reaction is being driven by surface-burn kinetics or by compaction alone. Two columns were added.

Solid-fraction time derivative $\dot{\phi}_s$ (1/s). Signed. Negative means solid is being consumed at the probe location (reaction proceeding); positive means compaction is locally densifying the solid phase. This column reveals the time evolution of reaction at each probe independently of the global burn-front tracker — a probe deep inside the bed can show reaction beginning even when the bed-wide tracker has not yet detected a coherent band.

Surface-burn velocity r_{burn} (m/s). The pressure-dependent surface-burn velocity from the BN compaction-equation source, $r_{\text{burn}} = a_2(P_g/P_{\text{ref}})^{b_n} H$, where H is the Heaviside switch on $T_{\text{surface}} - T_{\text{ign}}$ that fires when the granular surface temperature crosses the ignition threshold (see §1.3.3).

Zero unless the surface-burn term is active at that node. This column is what distinguishes whether reaction at a probe is kinetic (surface-burn engaged, $r_{\text{burn}} > 0$) or compaction-driven only ($r_{\text{burn}} = 0$, with the compaction-burn term in equation (1.1) carrying any reaction that proceeds).

The slow cook-off cylinder demonstration relies directly on this distinction. The wave at gauge 2 ($r = 18$ cm) of §7.4 shows r_{burn} rising to 1.94 m/s at $t = 29.9$ μs , evidence of kinetic engagement that the legacy diagnostics could not have surfaced.

4.3. Validation: SAND-1 and SAND-2 calibration markers preserved

The diagnostic redesigns were validated against the regression test suite from SAND-1 and SAND-2 to verify that they did not disturb the published calibration markers. Two regression cases were exercised.

The McAfee Tier-1 piston-driven test — the principal SAND-1 reference case — passes exactly with the new tracker active. The `coupled.BF.plt` output now carries the lead/peak/trail edges in addition to the legacy `ibf` index, but the reported ignition time, DDT firing time, and steady detonation-wave speed D exactly match the SAND-1 published values.

The Fig. 3 reference case from SAND-2 produces unchanged behavior on the portion of the run that integrates successfully. The late-time integrator stiffness in that case is a property of the SAND-2 BN closure (specifically, the surface-burn rate at high mixture pressure) and is independent of the tracker redesign; the diagnostic output up to the point of stiffness exactly matches the SAND-2 reference values.

The hybrid-retention strategy of §4.1 is what makes this exact match possible: the calibrated state-machine markers in both prior reports continue to use the legacy `ibf`, while the active-reaction-band edges that this report needs are exposed as additional output columns.

5. HANDOFF-STATE SMOOTHING

Coupled XCHEM–XDDT runs initially failed within nanoseconds of XDDT entry on geometries that had succeeded for SAND-1 and SAND-2’s piston-driven cases. The cause was numerical stiffness at the handoff seam: sub-millimeter gradients in the XCHEM state that the XDDT integrator could not advance through on its microsecond timescale. This chapter documents the handoff-state smoothing knob added to address the problem — its motivation, the tanh-ramp design, the tuning trade-off, and a validation case.

5.1. Motivation: integrator stiffness at the handoff seam

The initial coupled XCHEM–XDDT runs in spherical geometry failed within roughly 30 nanoseconds of the start of XDDT integration, with the DDEBDF integrator returning the convergence-failure code IDID=-6 on the first or second integration step. The same physical handoff state integrated successfully for approximately 12 μ s in slab geometry — the difference was the geometric source. The $1/r^2$ spherical source in equation (2.1) of §2 amplifies inner-edge gradients by a factor of $(R/r)^2$ as the radial coordinate r approaches the centerline; in slab geometry the geometric source vanishes ($n = 1$) and gradients are not amplified.

A diagnostic on the failing handoff state showed the cause. In the inner-edge cells of the bed, the solid volume fraction ϕ_s jumped from approximately 0.4 to 0.9 to 0.98 across three adjacent cells, and the gas density ρ_g dropped by a factor of 27 across two adjacent cells. These are artifacts of the steel/HMX interfacial transition in the XCHEM cookoff calculation: the transition is a thin spatial layer in XCHEM’s slow-time integration, but it appears as a near-discontinuity to XDDT’s microsecond-timescale integrator, and the converging-geometry amplification was enough to push the numerical Jacobian to non-convergence on the first integration step.

5.2. Tanh-ramp design

The smoothing knob is implemented as the routine `xtiv_smooth_state` in `xtiv_glue.f90`. It applies a tanh ramp to four state fields — gas temperature T_g , mixture pressure P_{mix} , solid volume fraction ϕ_s , and gas density ρ_g — over the first cells of bed past the inner edge of the explosive layer. The smoothing extent is controlled by the namelist parameter `xtiv_smooth_l_in`, with an analogous `xtiv_smooth_outer_l_in` for the outer-edge counterpart. For each smoothed field F , the value at $x = L$ (just past the smoothing zone) serves as the inner-region target, and cells inside $[0, L]$ are blended toward it via:

$$F_{\text{smoothed}}(x) = (1 - s(x)) F_{\text{target}} + s(x) F_{\text{loaded}}(x), \quad s(x) = \frac{1}{2} \left[1 + \tanh \left(4 \left(\frac{x}{L} - \frac{1}{2} \right) \right) \right], \quad (5.1)$$

where $L = \text{xtiv_smooth_l_in}$. At $x = 0$ (the innermost cell), $s \approx 0.018$ and the smoothed value is approximately 98% of the way from the original loaded value toward the target; at $x = L$, $s \approx 0.982$ and the smoothed value sits within 2% of the original loaded value, so the bulk of the bed beyond the smoothing zone is left essentially untouched.

Why all four fields? The dominant stiffness drivers are ϕ_s and ρ_g , but two-field smoothing experiments — smoothing only those two and leaving T_g and P_{mix} untouched — achieved only a 22% improvement in integration survival time. Smoothing all four fields together gave a 12× improvement, matching the natural smoothness of a different XCHEM handoff zone (one for which the integrator already succeeded without smoothing). Two-field smoothing remains an attractive future option (§10) because it would preserve a single-cell-class temperature spike in T_g , but the present implementation is four-field and validates against the demonstrations in this report.

Why tanh and not a local diffusion operator? A local smoothing operation (running average or Gaussian convolution) would propagate effects beyond the intended smoothing region and would require iteration to a stopping criterion. The tanh-to-target ramp has a hard endpoint at $x = L$ with a known reduction factor at $x = 0$; the smoothing extent is set entirely by the namelist parameter, requires no iteration, and is independent of bed-scale or grid-resolution choices.

Effect on the mass and energy budget. Because the tanh ramp modifies ϕ_s , ρ_g , T_g , and P_{mix} over the smoothing length, the smoothed state’s integrated mass and energy in the smoothed region differ from those of the raw XCHEM handoff. In the typical case — low ϕ_s and high ρ_g at the inner edge, both pulled toward bulk-like target values — the smoothed region carries slightly more solid mass and slightly less gas mass and thermal energy than the raw handoff. The perturbation is small in absolute terms: the smoothing length is at most a few millimeters compared to the centimeter-to-decimeter bed scale, so the perturbed region is typically 1–5% of the bed by volume. More importantly, the modification is applied as a one-time operation at the handoff frame; XDDT’s compaction and energy equations then conserve mass and energy strictly from the smoothed state forward. Smoothing is therefore a discrete shift between the raw XCHEM end-state and the XDDT initial condition, not an ongoing conservation violation during the integration. The per-cell modifications are echoed in the run-time provenance log (step 8 of §3.3) so the shift is auditable case-by-case.

Order of operations. The smoothing operation is applied after the `&xtiv_boost` preconditioning of §3.3, so an imposed hot-spot patch (if present) cannot reintroduce a sharp gradient that the smoother just removed.

5.3. Tuning the smoothing length

Handoff-state smoothing of the XCHEMA runaway temperature spike
(2frag XCHEMA precursor at outer steel/HMX interface, $t = 62.20$ s)

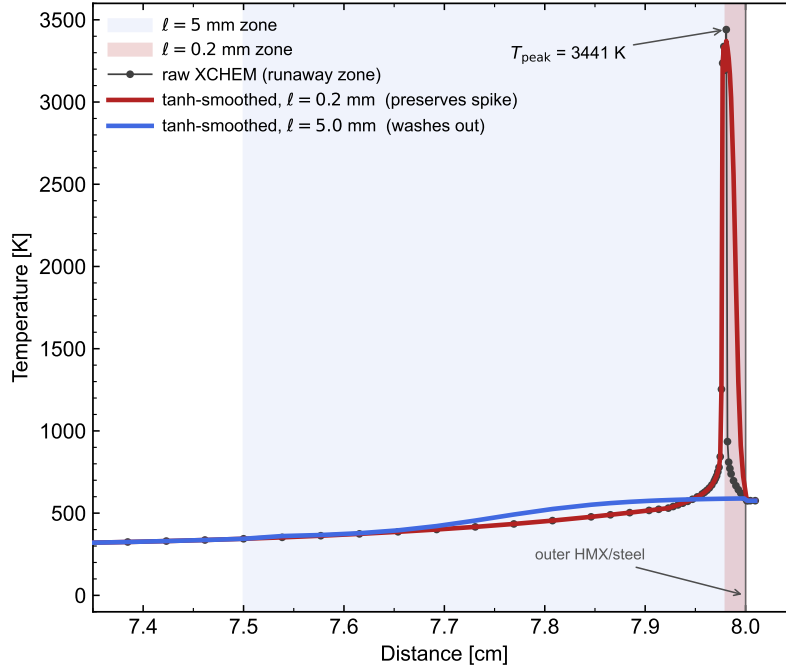


Figure 5-1. Handoff-state smoothing trade-off illustrated on the twin-fragment runaway-zone temperature profile near the outer HMX/steel interface (Chapter 8). Top curve: raw XCHEMA output, with a single-cell-class temperature spike at the interface. Middle curve: tanh-smoothed at `xtiv_smooth_outer_l_in = 0.2 mm`, preserving the spike's peak temperature while removing the steepest portion of the gradient. Bottom curve: tanh-smoothed at `xtiv_smooth_outer_l_in = 5.0 mm`, with the spike substantially washed out — the peak temperature drops below the BN ignition threshold, and surface-burn engagement is suppressed once the smoothed state is handed to XDDT.

The smoothing length must balance two competing requirements that do not always coexist on the same length scale, illustrated on the twin-fragment runaway-zone temperature profile in Fig. 5-1:

- It must be *long enough* to remove the inner-edge ϕ_s and ρ_g jumps that destabilize the integrator — typically a few cells, on the order of 0.5–1 mm in the AMR-refined XCHEMA output.
- It must be *short enough* to preserve the localized temperature feature at the runaway center. For natural cookoff ignition, this feature is single-cell-class (~ 0.1 – 0.5 mm after AMR refinement); a smoothing length of 1 mm or more washes the spike out, the spike's peak temperature drops below the BN ignition threshold T_{ign} , and the surface-burn switch in equation (1.1) never engages.

For the imposed-hot-spot-patch demonstrations of §6.1 and §8, the two requirements do not interfere: the patches are centimeter-scale, and a smoothing length of 5 mm is comfortably long enough

to remove the ϕ_s and ρ_g jumps while being too short to damage the patch. Both requirements are satisfied at the same value of L .

The natural-cockoff cases — the single-fragment slab of §6.1 and the slow cook-off cylinder of §7 — are harder. The temperature spike is only $\sim 0.1\text{--}0.5$ mm wide, so the upper bound on L (spike preservation) is comparable to or below the lower bound on L (stability). Because the current implementation uses one knob (`xtiv_smooth_l_in`) that applies the same smoothing length to all four fields, no single choice of L can both stabilize the integrator and keep the spike intact: making L long enough for stability washes the spike below T_{ign} , and making L short enough to preserve the spike fails to smooth the ϕ_s/ρ_g jumps. The choices documented in this report compromise toward one end or the other on a case-by-case basis.

A cleaner fix is to decouple the two requirements: smooth only ϕ_s and ρ_g (the fields that drive the stiffness) and leave T_g and P_{mix} untouched (so the spike survives intact regardless of L). Selective smoothing along these lines is identified as future work in §10.

5.4. Validation

The smoothing knob was validated on two test cases that had failed without it. XCHEM writes its slow-time cookoff history to `ofield.plt` as a series of per-timestep snapshots (“zones”); the extractor of §3.2 selects one zone for handoff. The two cases below are identified by their zone indices in the corresponding XCHEM output file — the integers below are simply timestep numbers, not node counts or any geometric quantity.

Sphere case: ODTX zone 72 versus zone 30. The first validation case is the One-Dimensional Time-to-eXplosion (ODTX) sphere cookoff configuration of McGuire and Tarver [10] introduced in §2 — the test that originally motivated the smoothing work. Two zones were extracted from the XCHEM run for this validation: zone 72 (XCHEM time $t_{\text{XCHEM}} = 7.05$ s, a runaway-state snapshot near the moment of thermal ignition, with the steep inner-edge ϕ_s and ρ_g gradients described in §5.1) and zone 30 (XCHEM time $t_{\text{XCHEM}} = 2.85$ s, an earlier pre-runaway snapshot whose inner-edge profile happens to be naturally smooth and integrated successfully without smoothing). Without smoothing, zone 72 fails at $t \approx 0.03$ μs in the converging spherical geometry. With `xtiv_smooth_l_in` = 0.2 mm applied, the same handoff integrates to $t = 0.36$ μs , matching the survival time of zone 30. The match between smoothed-zone-72 and unsmoothed-zone-30 on otherwise identical physics is direct evidence that the failure was geometric amplification of inner-edge stiffness rather than a deeper property of the post-handoff dynamics: smoothing recovers the same survival time that natural smoothness already gives. Smoothing is necessary but not sufficient for extended sphere integrations; the residual sphere-convergence problem is the future-work item documented in §10.3.

Slab case: single-fragment natural cookoff. The second validation case is the single-fragment slab demonstration of §6.1, with the XCHEM handoff extracted at the last zone (the runaway snapshot at which the steel/HMX interface gas-phase temperature has spiked through the kinetic chain

reaction). Without smoothing the case fails within $\sim 0.1 \mu\text{s}$ of XDDT entry with DDEBDF returning `IDID=-7`. `IDID=-7` is a more severe form of the convergence failure introduced in §2: the integrator has reduced its step size to its lower limit without recovering corrector convergence, here because the chemistry Jacobian has become ill-conditioned at the steep inner-edge ϕ_s and ρ_g jumps inherited from XCHEM. With `xtiv_smooth_l_in = 5 mm`, the same case runs through to $t = 4.28 \mu\text{s}$ before reaching the same `IDID=-7` wall (the long integration is reported in §6.1). The smoothing knob is the difference between a near-immediate halt and several microseconds of usable integration — necessary but not sufficient for indefinite propagation.

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6. SLAB DEMONSTRATIONS

This chapter reports the single-fragment slab-geometry demonstration of the XTIV pipeline. A 1-cm steel slab at hot-fragment temperature is placed in contact with a 10-cm bed of granular HMX, and the coupled XCHEMA–XDDT pipeline is exercised end-to-end with no boost imposed. The outcome (§6.1) is a fizzle: XCHEMA drives the contact face through thermal runaway over tens of seconds, but the resulting handoff state delivers a gas-phase spike confined to a single cell adjacent to the steel/HMX interface, surrounded by a bed at its initial cold temperature. Post-handoff XDDT integration produces only a brief compaction-driven response at the boundary; the BN surface-burn kinetic switch never arms, the bed beyond the contact face never sees the reaction, and the integration terminates after 4.28 μ s with no propagating wave. This case sets the lower bound of the preheat-extent parameter sweep developed in Chapter 9.

The chapter is structured around this one demonstration plus three methodology asides used elsewhere in the report. §6.2 documents the parameter guidance for the imposed hot-spot patch (&xtiv_boost), which is the device used in the twin-fragment chapter (Chapter 8) to stand in for prior thermal history of the bed; the patch itself is not exercised in §6.1. §6.3 reports the T_{ign} sensitivity sweep on the slow-cookoff cylinder (Chapter 7), the only configuration in this report whose handoff temperature sits close enough to the kinetic switch threshold for T_{ign} to be a first-order tuning knob. §6.4 provides the mechanistic reasoning behind the single-fragment fizzle observation of §6.1 and contrasts it with the multi-fragment configurations developed in Chapter 8, where two-sided preheating between coordinated fragments supplies the bed extent that a single fragment alone cannot deliver.

6.1. Single-fragment slab with natural ignition

The first demonstration is the original hot-fragment-into-explosive geometry: a 1 cm steel slab at 600 K placed in contact with a 10 cm HMX bed (initially 300 K, granular at 70% TMD, vented at the outer surface). XCHEMA determines the slow temperature evolution until thermal runaway occurs at the steel/HMX interface. The resulting handoff state is passed to XDDT through the standard IBC=3 transfer-file path (§3.3) with no boost imposed. This is the natural-ignition case for a single fragment in a vented HMX bed, and the outcome — a brief compaction-driven response followed by quench, with no surface-burn engagement and no propagating wave — sets the lower bound of the parameter sweep developed in §9.

The XCHEMA cookoff history is shown as panel (a) of Figure 6-1. Time slices from $t = 0$ to $t = 33$ s capture the slow thermal soak: the steel at $x < 1$ cm holds near 575 K, the HMX/steel interface at $x = 1$ cm conducts heat into the cold HMX bed, and a temperature gradient develops in the inner millimeter of HMX. The latest profile shows gas-phase thermal runaway at the interface —

the local gas temperature exceeds 3200 K — but only at the single cell where the kinetic chain has accelerated; the surrounding bulk remains near its initial 300 K with only modest skin-depth warming. This is the characteristic single-fragment cookoff signature: a sharp gas-phase spike at the contact face superimposed on a cold bed.

XDDT integration of the handoff state, shown as panel (b) of Figure 6-1, yields a strongly stalled response:

- Surface burn never engages. The BN kinetic switch $H(T_{\text{surface}} - T_{\text{ign}})$ evaluates on the granular mixture-surface temperature (§1.3.3), not on the gas-phase temperature. Even though the XCHEM handoff carries a localized gas-phase spike above 3000 K, the surface temperature lags the gas through Biot-limited heat exchange into the cold porous solid; with the HMX bulk at 300 K throughout and only the single interface cell carrying any meaningful preheating, the surface temperature stays below 800 K for the entire integration. The diagnostic reports $t_{\text{burn}} = \text{---}$.
- Compaction-driven activity is detected at $t_{\text{ign}} = 3.00 \mu\text{s}$, when $|\dot{\phi}_s|$ first exceeds the burn-front tracker's noise floor in a thin layer at the steel/HMX interface. Once detected, the burn-front triple (lead/peak/trail) sits within $x \in [1.0, 1.4] \text{ cm}$ for the remainder of the run; the compaction-driven activity does not propagate outward.
- Peak mixture pressure $P_{\text{max}} = 143 \text{ MPa}$, two to three orders of magnitude below detonation-class pressures.
- DDEBDF terminates the integration at $t = 4.28 \mu\text{s}$ with the wave still pinned at the interface. The fitted burn-front velocity is $D_2 = -2.10 \text{ km/s}$, *negative* (i.e., the active-band peak migrates slightly inward over the few microseconds during which it is detected, not outward into the bulk). No DDT criterion fires.

The result is qualitatively the single-fragment quench mechanism documented in §6.4: a hot fragment readily ignites the explosive at the contact face, but the surrounding cold bulk thermally suppresses the locally ignited reaction before a self-sustaining propagating wave can establish. The contact-face HMX has temperature, but the bed has no preheat extent. Without that extent, the BN compaction-burn coupling has no warm bed length to engage across, the kinetic switch never arms, and the post-handoff response collapses to a compaction-only transient at the boundary.

This configuration establishes the lower bound of the parameter sweep: a hot single fragment delivers contact-face ignition without the centimeter-scale preheated bed extent the BN compaction-burn coupling requires for sustained propagation. The wave never develops; the kinetic switch never arms; the bed beyond the contact face never sees the reaction. The sections that follow exercise the same configuration with progressively more preheating supplied — a hot-spot patch standing in for prior thermal history (§6.2), then two-sided cookoff between coordinated fragments at decreasing separations (Chapter 8) — to identify the preheat extent at which the model transitions out of the fizzle regime.

Sec. 6.1 Single-fragment slab, natural ignition, fizzle

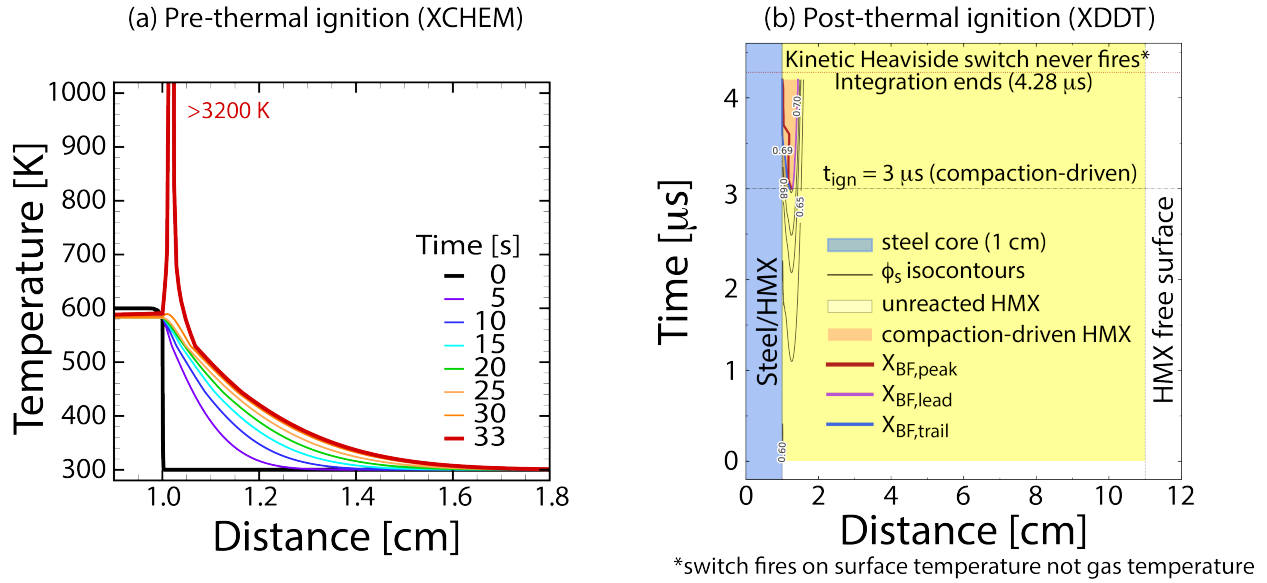


Figure 6-1. Coupled XCHEM→XDDT response for the single-fragment natural-ignition configuration — a 1 cm steel slab at 600 K against a 10 cm vented HMX bed (initially 300 K) with no boost. (a) XCHEM temperature profiles from $t = 0$ to $t = 33$ s of natural cookoff: gas-phase temperature at the steel/HMX interface ($x = 1$ cm) accelerates through the kinetic chain reaction to local runaway above 3200 K, with the surrounding HMX bulk remaining cold. (b) XDDT integration of the resulting handoff state. Position runs horizontally, time advances upward. Steel core occupies $x \in [0, 1]$ cm (cornflowerblue); unreacted HMX (yellow background) fills the bed from $x = 1$ cm out to the free surface at $x = 11$ cm. The burn-front tracker triple (lead, peak, trail) is confined to $x \approx 1.0$ to 1.4 cm throughout the run, signifying compaction-driven activity pinned at the interface with no outward propagation. The BN kinetic switch evaluates on surface temperature, not on gas-phase temperature; with the bulk cold and the gas-phase spike at a single cell, the surface temperature never crosses $T_{\text{ign}} = 800$ K, and the switch never arms. Compaction-driven activity is detected at $t_{\text{ign}} = 3.00$ μs (annotation in panel (b)); the integration terminates at $t = 4.28$ μs without a propagating wave establishing. Peak mixture pressure $P_{\text{max}} = 143$ MPa. Fitted burn-front velocity $D_2 = -2.10$ km/s (negative, indicating the brief active band migrates slightly inward rather than outward). This is the characteristic single-fragment fizzle outcome in vented HMX.

6.2. Methodology guidance for the hot-spot-patch boost

The single-fragment fizzle of §6.1 demonstrates that natural XCHEM cookoff alone does not deliver the preheated bed extent the BN compaction-burn coupling needs to drive sustained propagation. In the twin-fragment configuration of Chapter 8, this gap is bridged by applying an imposed hot-spot patch through the `xtiv_boost` namelist — a methodology device that stands in for the prior thermal history a real fragment would have imparted to the bed. This section documents the parameter guidance for that patch so the values used in Chapter 8 are traceable to a physical rationale. The patch is not exercised in §6.1 itself; the guidance applies wherever the boost is used in the report.

The boost is parameterized by three values, each set for a distinct physical reason, and is coordinated with a fourth parameter that controls the numerical handoff smoothing.

Patch temperature `hot_spot_T_in`. *Physical purpose: arm the surface-burn kinetics.* The kinetic switch in the BN compaction-burn term (the Heaviside H on $T_{\text{surface}} - T_{\text{ign}}$ in equation (1.1)) does not engage until the granular surface temperature crosses the explosive’s ignition threshold. `hot_spot_T_in` must therefore exceed T_{ign} (800 K for HMX); below the threshold the wave will not transition out of pure compaction-driven deflagration regardless of patch length or pressure. The recommended patch values (900 K in the laminate interior, 1100 K on the twin-fragment outer-edge mirror) sit above the threshold by construction and are correspondingly higher than the thermal kernel a real fragment would deposit by impact alone. The threshold itself is a tunable parameter (§6.3) and could be lowered through a coordinated choice of the XCHEM stop-at-temperature parameter; the patch values would then drop in turn, bringing the boost closer to physically realizable post-impact kernel temperatures. That coordinated tuning is not exercised in the present report.

Patch length `hot_spot_l_in`. *Physical purpose: supply enough preheated bed for the wave to sustain propagation.* A localized ignition into cold bulk quenches before a self-sustaining wave establishes (the single-fragment quench mechanism of §6.4). `hot_spot_l_in` sets the spatial extent of the preheated zone the BN compaction-burn coupling is handed, and it must be wide enough that propagation is no longer cold-bulk limited. Centimeter-scale values are the operating point for the demonstrations in this report. Large patch lengths are not realistic, and are only used as an example to emphasize the importance of a heated bed.

Patch pressure `hot_spot_P_in`. *Physical purpose: stand in for post-compaction kernel pressures.* The fragment-impact and shock-passage mechanisms that a single-fragment hot-spot represents in reality produce high-pressure ignition kernels that a purely thermal cookoff precursor underrepresents. The calibrated default of 50 MPa reflects realistic post-compaction kernel pressures.

Numerical coordination with `xtiv_smooth_l_in`. *Numerical purpose: keep the integrator from choking on the inner-edge gradients.* `xtiv_smooth_l_in` is a separate namelist parameter set in `&xtiv_glue` (not in `&xtiv_boost`) that controls the length over which the handoff-state smoothing of Chapter 5 relaxes the steep XCHEM inner-edge ϕ_s and ρ_g jumps, so that DDEBDF does not encounter a near-discontinuity at XDDT entry. Because the smoothing routine runs after the boost is applied (§5.2), `hot_spot_l_in` must exceed `xtiv_smooth_l_in` or the smoothing tanh ramp spans the entire patch and washes it out before XDDT integrates. The constraint is therefore a coordination between two independent user-set knobs — a physics knob (`hot_spot_l_in`) and a numerics knob (`xtiv_smooth_l_in`) — not a single-parameter limit.

Sensitivity behavior. A brief sensitivity study spanning $T \in [900, 1200]$ K (at $\ell = 2$ cm) and $\ell \in [10, 40]$ mm (at $T = 900$ K) confirmed that above-threshold variations affect D by approximately 30 % (range 4.5–5.9 km/s, all in transition territory) and shift x_{DDT} linearly with ℓ relative to the inner explosive edge; no above-threshold parameter combination produces a qualitatively different wave regime. The patch thus acts effectively as a threshold switch between sub-ignition (no DDT) and transition DDT, with ℓ acting as a geographic seed for the DDT crossing position rather than as a continuous tuning knob.

Table 6-1. Hot-spot-patch boost parameter guidance. Threshold below which DDT does not occur, recommended value above threshold, and dominant effect of above-threshold variation.

Parameter	Threshold	Recommended	Above-threshold effect
<code>hot_spot_T_in</code>	$T > T_{\text{ign}}$	900 K	Reduces t_{DDT} ; no trend on D
<code>hot_spot_l_in</code>	$\ell > \text{xtiv_smooth_l_in}$	0.02 m (2 cm)	Sets x_{DDT} as geographic seed
<code>hot_spot_P_in</code>	none calibrated	5×10^7 Pa	Modest effect on t_{DDT}

The hot-spot-patch boost discussed in this section is exercised in the twin-fragment configuration of Chapter 8, where the values used (`hot_spot_T_in` = 900 K, `hot_spot_l_in` = 0.02 m, `hot_spot_P_in` = 5×10^7 Pa, with the outer-edge mirror raised to 1100 K / 100 MPa for the converging collision geometry) are the recommended above-threshold defaults.

6.3. Methodology guidance for T_{ign} in cookoff-coupled scenarios

The Heaviside ignition switch in equation (1.1) turns on the BN surface-burn term when the local granular surface temperature exceeds T_{ign} ; below the threshold the term is off. In piston-driven scenarios where the ignition boundary condition deposits energy far above any plausible T_{ign} , the threshold value is effectively calibration-insensitive at the SAND-1 reference value of 525 K (HMX low-pressure decomposition onset). In cookoff-coupled scenarios where the XCHEM handoff state itself sits near the threshold, however, T_{ign} becomes a first-order parameter that must be tuned to the scenario’s handoff temperature plateau.

The methodology lesson below is reported here, in the slab chapter, even though the slab demonstration of §6.1 side-steps the T_{ign} sensitivity: the single-fragment natural-cookoff case has T_{mix}

staying so far below T_{ign} throughout the integration that the switch never engages at any plausible threshold. The twin-fragment configuration of Chapter 8 sits at the other extreme: the imposed boost (`hot_spot_T_in` = 900 K) overrides T_{ign} by construction. The slow-cockoff cylinder of Chapter 7 is the only case in this report whose handoff temperature lands close enough to the threshold for T_{ign} to be a first-order tuning knob. The guidance derived below is therefore based on a sweep on that cylinder case; it applies broadly to all cockoff-coupled scenarios where the handoff temperature sits near the kinetic switch threshold.

The slow-cockoff cylinder of Chapter 7 provides a clean illustration. The XCHEM-to-XDDT handoff at the moment of thermal runaway carries a peak temperature of approximately 807 K—the slow-cockoff thermal wave has nearly reached the BN ignition criterion at the outer-edge cells but has not yet satisfied it at every cell. A sensitivity sweep at three values of T_{ign} , with all other deck parameters held identical to the reference SCO run, spans three qualitatively distinct regimes:

- $T_{\text{ign}} = 525 \text{ K}$ (HMX decomposition onset, the XDDT deck default). Many handoff cells are already above 525 K, so the Heaviside switch fires at $t = 0.10 \text{ }\mu\text{s}$ on cells that are hot but not yet compacted. The surface-burn velocity peaks at $r_{\text{burn}} \approx 0.7 \text{ m/s}$ —far below typical deflagration rates—and the bed pressure stalls near 0.7 GPa. The run terminates at $4.1 \text{ }\mu\text{s}$ with the wave still pinned to the outer edge. Premature, non-physical ignition.
- $T_{\text{ign}} = 800 \text{ K}$ (this report’s value). The threshold sits just below the XCHEM handoff peak gas temperature, but the BN Heaviside switch evaluates on the granular surface temperature T_{surface} — a Biot-weighted blend of gas and bulk-solid temperatures, with an additional contribution from the surface-temperature integral ζ that grows as the gas–grain temperature gradient accumulates over time ([9], Eq. 14; see §1.3.3) — which lags T_g when the bulk solid is cold and consequently sits well below 800 K at handoff despite the gas peak being only 7 K above threshold. Surface-burn fires at $t = 11.9 \text{ }\mu\text{s}$ once compaction-driven heating has elevated T_{surface} past 800 K. The wave sweeps inward 7.3 cm and reaches DDT at $x_{\text{DDT}} = 15.6 \text{ cm}$: peak mixture pressure climbs to 15.6 GPa, peak gas temperature approaches 6000 K (near the EOS table ceiling), and the surface-burn velocity peaks at $\approx 29 \text{ m/s}$. Expected SCO deflagration-to-detonation behavior.
- $T_{\text{ign}} = 1100 \text{ K}$ (high-bound stress test). This case is conceptually the simplest of the three: the kinetic switch is set so high that the granular surface temperature never reaches it. The Heaviside condition in equation (1.1) stays at zero for the entire integration; no surface-burn ever fires; no chemical reaction occurs. The bed is not, however, inert. The compaction-burn term in the same equation does not depend on the kinetic switch, so the post-handoff pressure wave still drives the compaction front through the bed (sweeping from $r = 22.9$ to 7.65 cm). Surface burn never engages: the peak surface-burn velocity stays at $r_{\text{burn}} \approx 0.1 \text{ m/s}$ (negligible) and no reaction wave forms. The burn-front tracker’s least-squares velocity fit, applied to this non-smooth, reaction-free velocity field (compaction-front motion alone), cannot be reliably interpreted: the large positive D it reports in `coupled.out` is an artifact of that fit, and the wave speed is simply not cleanly characterizable for this case. This is a burn-front-tracker least-squares-fit limitation, distinct from the direction-aware diagnostics (§10.4) that the present work corrected. Mechanically the bed is in motion; chemically nothing is happening.

Figure 6-2 visualizes the three regimes through their burn-front trajectory, gas temperature, mixture pressure, and surface-burn velocity histories. The 525 K and 1100 K cases diverge from the 800 K reference trajectory in qualitatively opposite ways: 525 K shows surface burn but no compression, 1100 K shows compression but no surface burn.

SCO cylinder: T_{ign} sensitivity sweep with other parameters identicle.
The handoff temperature from XCHEM was 807 K.

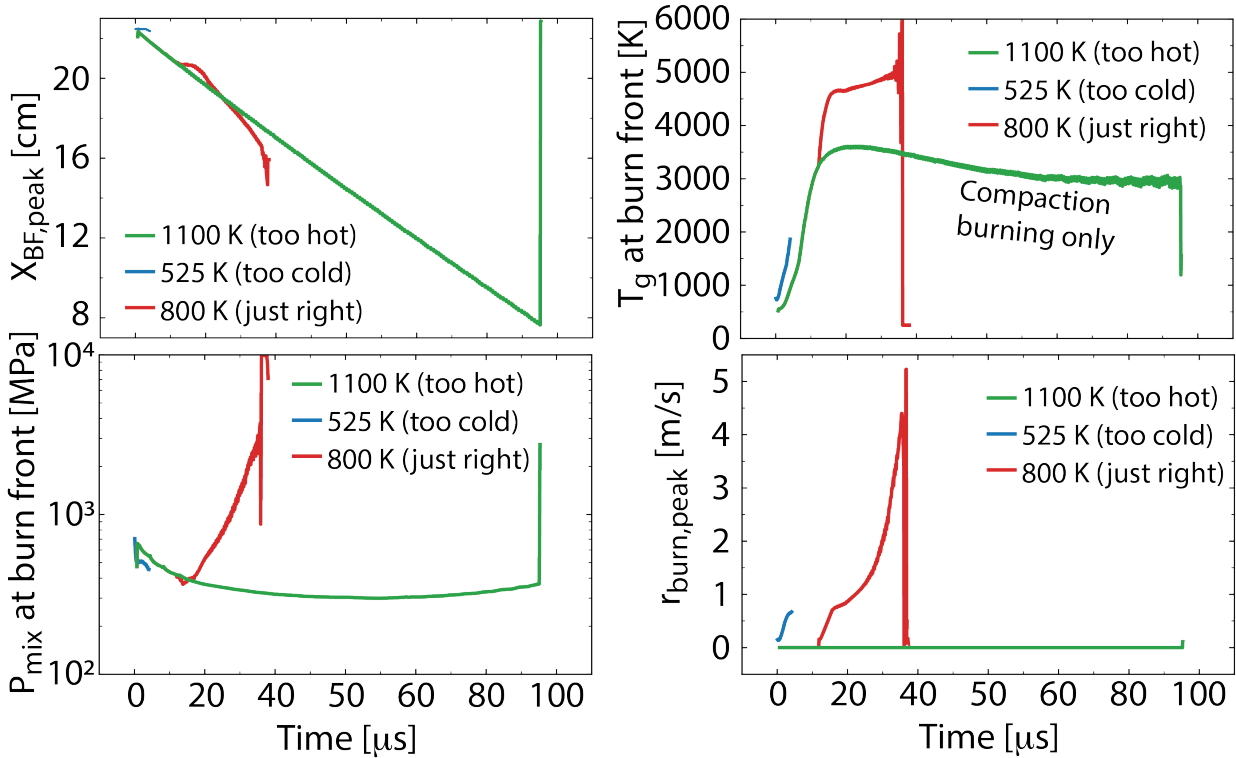


Figure 6-2. T_{ign} sensitivity sweep on the slow-cockoff cylinder demonstration. All other deck parameters held identical to the reference SCO run; the XCHEM handoff peak temperature is 807 K. Top-left: peak burn-front position $x_{\text{BF,peak}}$ vs. time. Top-right: gas temperature at burn front (log scale). Bottom-left: mixture pressure at burn front (log scale). Bottom-right: peak surface-burn velocity $r_{\text{burn,peak}}$ (symlog scale). 525 K (blue): premature ignition on uncompacted handoff state, run terminates at 4 μs with $r_{\text{burn}} \approx 0.7$ m/s and pressure stalled below 1 GPa. 800 K (red): reference SCO deflagration-to-detonation (reaching DDT), surface-burn engaging at $t = 12 \mu\text{s}$ after compaction-driven gas heating crosses the threshold. 1100 K (green): surface-burn never engages; compaction-front motion alone, demonstrating that T_{ign} has been set above any reachable gas temperature.

The methodological lesson is that T_{ign} in cookoff-coupled scenarios must be selected such that $T_{\text{handoff,peak}} \lesssim T_{\text{ign}} \ll T_{\text{gas,reachable}}$, where $T_{\text{handoff,peak}}$ is the peak gas temperature in the XCHEM handoff state and $T_{\text{gas,reachable}}$ is the gas temperature attainable through subsequent XDDT compaction-driven heating. For the present report's slow-cockoff cylinder this gives a workable range of approximately 800 K up to ≈ 5000 K; the 800 K choice sits just below the handoff gas peak (807 K) but allows the surface-burn switch to fire exclusively on cells whose temperature has been further elevated by post-handoff compaction work. This works because the BN switch responds

to a surface temperature that lags the gas temperature when the surrounding solid is cold; the practical tuning levers are the gas–solid heat-transfer coupling parameters (`htfact_in`, `akm_in`) that drive the surface-temperature integral ζ (§1.3.3), not the gas-peak temperature itself. The twin-fragment-impact scenario of this report is not affected by this constraint because the boost-driven inner-edge heating (`hot_spot_T_in` $\gg T_{\text{ign}}$) overrides the threshold by construction; the single-fragment slab of §6.1 is also unaffected because the surface-burn term never engages there at any plausible T_{ign} (the handoff T_{mix} stays well below threshold for the entire integration). A near-term refinement direction — short of the distributed-activation-energy upgrade discussed below — is to coordinate T_{ign} with the XCHEM stop-at-temperature parameter (`tmax` in `tchem.in`; see the stop-at- T workaround in §10.3). Instead of letting XCHEM run to natural runaway and choosing T_{ign} just below the resulting handoff peak (≈ 800 K in the SCO cylinder), the user could halt XCHEM at a deliberately lower temperature (e.g., `tmax` = 500 K) and set T_{ign} just above it (e.g., 525 K, the SAND-1 reference value). Because T_{ign} then sits well below the gas temperatures reachable through post-handoff compaction work, the surface-burn switch still fires exclusively on compacted, post-handoff cells — the same selectivity property that motivated the 800 K choice above — but the threshold itself is now anchored to a chosen physical temperature rather than to whichever runaway temperature XCHEM happens to produce. A second consequence is that the hot-spot patch values of §6.2 can drop in turn: with $T_{\text{ign}} \approx 525$ K, patch temperatures of 600–700 K are sufficient to arm the kinetics, bringing the boost values closer to physically realizable post-impact kernels than the 900–1100 K values used in the present report. The inequality $T_{\text{handoff,peak}} \lesssim T_{\text{ign}} \ll T_{\text{gas,reachable}}$ is preserved under either choice; the coordinated low-threshold tuning is a methodology refinement recommended for future cookoff-coupled studies where physically defensible patch values are required. It is not exercised in the present report.

The brittleness of T_{ign} in cookoff-coupled scenarios is itself an argument for the distributed-activation-energy adoption proposed in §10.5: a smooth Arrhenius/DAE rate naturally handles the same physics without requiring per-scenario threshold tuning.

6.4. Single-fragment quench versus multi-fragment propagation

The single-fragment fizzle of §6.1 does not arise from a failure to ignite the HMX. HMX is a reactive material and a hot fragment readily drives the bed through thermal runaway at the contact interface — the XCHEM cookoff history in panel (a) of Figure 6-1 shows exactly that. The fizzle arises instead from a failure to *sustain* propagation once ignition has occurred. A single hot fragment heats only a thin boundary layer of the surrounding HMX; the bulk beyond that layer remains cold, and the cold bulk thermally suppresses the reaction front before a self-sustaining wave can establish. In every coupled single-fragment configuration attempted in this work the locally-ignited reaction quenched before propagation could be recovered, regardless of the fragment temperature within plausible bounds.

Three paths to the preheated-bed condition the BN compaction-burn coupling requires for sustained propagation are demonstrated elsewhere in this report: a longer cookoff time scale that lets heat diffuse deeply into the bed (the slow cook-off cylinder of Chapter 7), *multiple* hot-fragment sources whose thermal influence overlaps in the bed interior (the twin-fragment-impact case of Chapter 8,

which produces detonation-class propagation precisely because the two-sided preheating supplies the bed extent the BN compaction-burn coupling needs), and a hotter or larger driving source able to reach further into the bed (the central steel-rod scenario considered as future work in Chapter 10). The single-fragment-fizzle versus multi-fragment-DDT contrast is developed as a regime finding in §9.2.1.

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7. SLOW COOK-OFF CYLINDER (NATO AOP-4382)

7.1. Configuration

The headline demonstration of the XTIV pipeline is a NATO AOP-4382 slow cook-off simulation [5, 6] on a demonstration cylinder of realistic geometry filled with a sensitive HMX charge — not an IM-qualified composition — so that the pipeline is exercised against a configuration that actually undergoes thermal runaway and transitions into a violent response under the AOP-4382 ramp. The geometry is an 18-inch diameter HMX charge enclosed in a 1-inch steel jacket, 60 inches in cylinder height, with the HMX layer treated as permeable to gas flow during the XCHEMA cookoff phase (player='p': gases flow uniformly through the open pore space, no spatial pressure gradients, but bulk pressure rises as decomposition proceeds), and the steel jacket as the inert outer layer (player='n'). In cylinder coordinates the HMX bed runs from $r = 0$ (centerline) to $r = 22.86$ cm (HMX/jacket interface). The outer-wall thermal boundary condition is the AOP-4382 slow cook-off ramp: $T_{\text{wall}}(t)$ rises linearly from 300 K to 1050 K over 180,000 s of simulated time, equivalent to the current NATO standard heating rate of 15°C/hr. This rate represents a recent revision [5]: as recorded in the AOP-4382 custodial-working-group history, the rate was narrowed from a candidate range of 10–25°C/hr down to 15–20°C/hr and finally established at 15°C/hr as the value that both bounds 92% of real-world incidents reviewed and remains close enough to the historical test rate to preserve continuity with the existing baseline test data. The XCHEMA cookoff phase resolves the full thermal evolution to runaway, after which the XDDT phase resolves the post-ignition compressible reactive flow at microsecond timescales.

The XCHEMA cookoff phase produced 329 output zones, ending with thermal runaway at the outer HMX/jacket interface at $t = 12$ h 53 min 30 s (i.e., $t = 46,410$ s into the cookoff), where the local temperature crossed $T_{\text{ign}} = 800$ K and the gas-phase reaction rate diverged. The handoff zone for the XDDT integration is the runaway zone (zone 329), with peak handoff temperature $T_{\text{max}} = 3100$ K concentrated in 1–2 cells at the outer wall.

Figure 7-1 previews both halves of the coupled simulation on a single page so the connection between the slow-time thermal preconditioning and the microsecond-scale post-handoff wave dynamics can be read at a glance. Panel (a) shows the pre-thermal-ignition XCHEMA phase: a stack of radial temperature profiles $T(r)$ at hourly intervals as the 15°C/hr ramp on the outer wall pushes heat inward across the bed, ending with the red runaway curve at $t = 12$ h 53 min 30 s when the outer-wall cells cross T_{ign} and the gas-phase reaction rate diverges. Panel (b) shows the post-thermal-ignition XDDT phase: ~ 38 μs of compressible reactive flow handed off from the XCHEMA runaway zone, with the compaction wave propagating inward from the outer wall (black, $x_{\text{comp,peak}}$), the active-reaction band trailing it (peak/lead/trail triple in red/light-purple/blue), and the burnt residue accumulating between the band's lead edge and the outer wall (dark-purple fill). The pre-ignition panel sets the inhomogeneous initial condition that the post-ignition panel inherits and runs forward

in time: the runaway is concentrated at the outer wall, with a smoothly decreasing preheat extending inward across the rest of the HMX bed. The XCHEM results underlying panel (a) are summarized briefly in §7.2; the XDDT wave dynamics in panel (b) are analyzed in detail in §7.3 and §7.4.

7.2. Pre-ignition thermal profile

The slow heating timescale produces a non-uniform thermal initial condition for the XDDT integration. Panel (a) of Fig. 7-1 shows the radial temperature profile $T(r)$ throughout the XCHEM cookoff: the outer-wall ramp drives a smooth, monotonically decreasing preheat into the bed, with the runaway curve (red, at $t = 12$ h 53 min 30 s) concentrated at the wall and the deep interior still essentially undisturbed. The handoff state delivered to XDDT carries this gradient directly, and the post-ignition wave dynamics described below are shaped by it.

7.3. XDDT integration of the post-ignition handoff

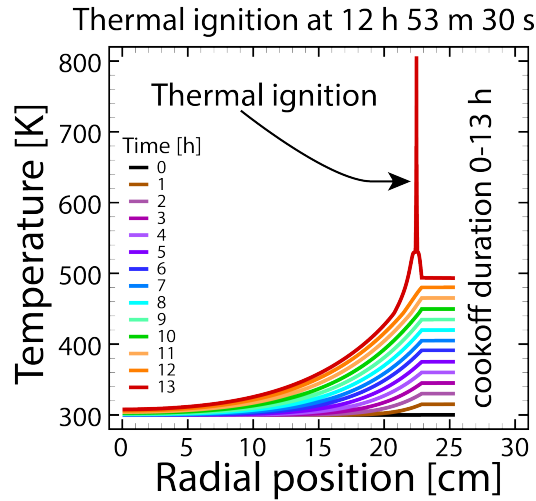
The XDDT integration of the runaway zone, with the inner-edge smoothing knob set to `xtiv_smooth_l_in` = 0.0002 m and the hot-spot patch disabled (i.e., natural cookoff ignition only), produces an inward-propagating reactive wave with several characteristic markers:

- $t_{\text{ign}} = 0.7 \mu\text{s}$: compaction-driven burn first crosses the consumption threshold at the outer wall.
- $t_{\text{burn}} = 15.4 \mu\text{s}$: $T_{\text{mix}} > T_{\text{ign}}$ somewhere in the bed for the first time — the pressure-dependent surface-burn term engages. This event is absent in the slab demonstrations (see §6), confirming that bulk preheating is what permits kinetic engagement in coupled cookoff scenarios.
- $t_{\text{DDT}} = 36.8 \mu\text{s}$ at $x_{\text{DDT}} = 15.6$ cm: burn front overtakes the compaction-front-at-ignition position; classical BN definition of DDT is satisfied.
- $D = -5.386$ km/s (the negative sign reflects inward propagation in the cylinder coordinate; magnitude is ~85% of $D_{\text{CJ}} = 6.36$ km/s).
- $P_{\text{max}} = 15619$ MPa — peak mixture pressure observed at the burn front.

The wave is in late pre-detonation transition territory: well past benign deflagration ($D \sim 0.3$ km/s would be deflagration class) and at ~85% of D_{CJ} — approaching, though not yet reaching, the ideal detonation velocity — with strong continuing pressure amplification driven by the cylinder's $1/r$ geometric source term as the wave converges toward the centerline. Figure 7-1(b) shows the wave-position structure; Fig. 7-2 shows the full BN state-variable fields underlying that wave structure.

Slow cookoff cylindrical (NATO AOP-4382, 15 K/h)

(a) Pre-thermal ignition, XCHEM



(b) Post-thermal ignition, XDDT

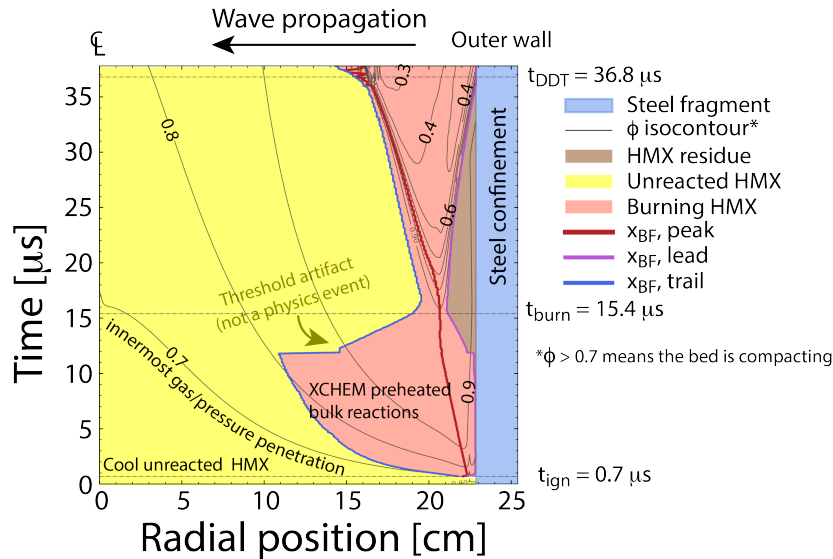


Figure 7-1. XTIV coupling pipeline applied to the NATO AOP-4382 slow cook-off scenario in an 18-inch HMX cylinder. **(a)** Pre-thermal-ignition XCHEM phase: temperature profiles $T(r)$ at hourly intervals during slow heating (300 K initial, 15°C/hr ramp on the outer wall), with thermal runaway at the outer HMX/jacket interface at $t = 12$ h 53 min 30 s (red curve). **(b)** Post-thermal-ignition XDDT phase: wave dynamics over $\sim 38 \mu s$ of compressible reactive flow following the XCHEM handoff. The compaction wave (black, $x_{comp, peak}$) propagates inward from the outer wall; the active-reaction band (red/light-purple/blue triple: peak/lead/trail edges) follows the compaction wave; burnt residue (dark purple fill) accumulates between the lead BF and the outer wall as the wave consumes the bed. Time advances upward; wave propagation direction is leftward (inward toward the centerline).

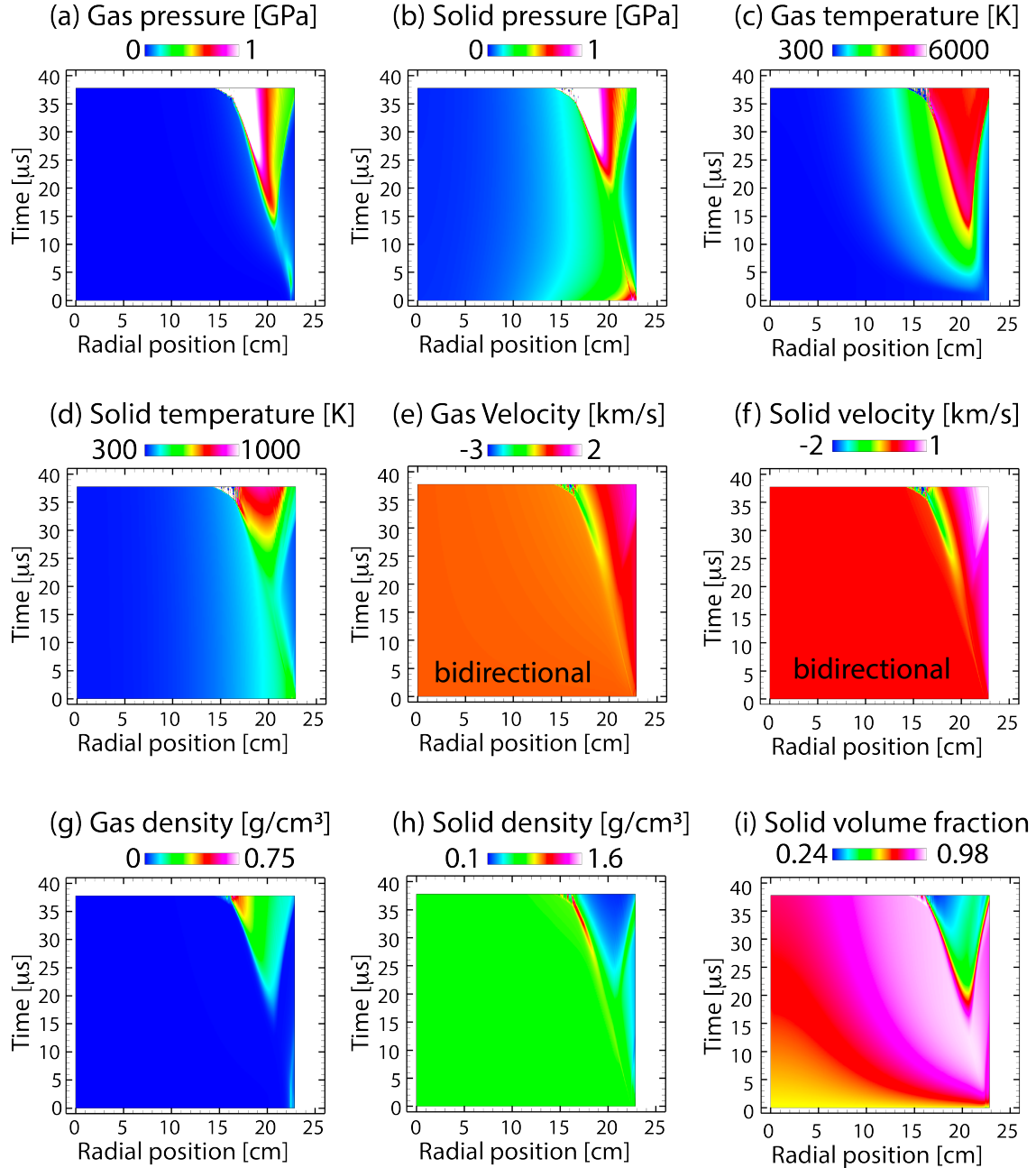


Figure 7-2. State-variable contour fields for the post-thermal-ignition XDDT phase of the cylinder slow cook-off (companion to Fig. 7-1(b)). Each panel shows one BN-model state field as a function of time (vertical) and radial position (horizontal): **(a)** gas pressure; **(b)** solid pressure; **(c)** gas temperature; **(d)** solid temperature; **(e)** gas velocity (negative values = inward flow); **(f)** solid velocity; **(g)** gas density; **(h)** solid density; **(i)** solid volume fraction. The inward-converging wave is visible in every field, with the strongest signatures in gas pressure (a), gas temperature (c), and solid-volume-fraction depletion behind the wave (i). Panel (e) shows the bidirectional gas flow: inward-converging precursor (negative) ahead of the wave and outward-venting reaction products (positive) behind it.

7.4. Wave amplification: gauge-data evidence

Three Eulerian gauges instrument the bed at radial positions $r = 22$ cm, $r = 18$ cm, and $r = 10$ cm, capturing the wave at near-ignition, mid-propagation, and deep-interior positions respectively. The mid-radial gauge (probe 2 at $r = 18$ cm) records the wave passage at $t = 29.9$ μs with several signatures of pre-detonation transition:

- Gas temperature T_g jumps from 3055 K to 4873 K.
- Mixture pressure P_{mix} jumps from 250 MPa to 1316 MPa — a factor of ~ 5 amplification over a ~ 5 μs window.
- Solid volume fraction ϕ_s drops from 0.952 to 0.732 (solid consumption).
- Surface-burn velocity r_{burn} rises to ~ 1.94 m/s, evidence of kinetic engagement at the gauge location.

The outer gauge (probe 1 at $r = 22$ cm) sees weaker amplification ($P_{\text{mix}} = 286$ MPa at the corresponding time) because the outer-edge reaction has been venting through the open Neumann outer boundary condition (a known limitation; see §10). The inner gauge (probe 3 at $r = 10$ cm) is unreached during the run: the wave was still in transit through the cool-bed boundary at the run end. The pressure amplification from probe 1 to probe 2 is direct evidence of $1/r$ geometric focusing at work.

The full radial structure of the wave at multiple time slices is shown in Fig. 7-3, which plots the nine BN state-variable profiles $r \mapsto Q(r, t_i)$ at hourly post-handoff intervals between $t = 5$ and $t = 35$ μs . The early-time wide volumetric burn driven by the XCHEM-preheated HMX bed (§7.2) and the late-time collapse to a narrower inward-propagating wave structure following surface-burn engagement at $t_{\text{burn}} = 15.4$ μs are both visible directly in the profile evolution.

7.5. Interpretation

The wave initiates at the outer HMX/jacket interface, propagates inward through the cookoff-preheated HMX bed, and amplifies via the cylinder's $1/r$ geometric source term as it converges toward the centerline. Within the simulated window it reaches DDT (at $t_{\text{DDT}} = 36.8$ μs) and climbs to $\sim 85\%$ of D_{CJ} , with transition-territory signatures (surface-burn engagement, multi-GPa peak pressures, multi-thousand-K peak temperatures, $1/r$ amplification visible in gauge data) all present, but does not attain the full ideal detonation velocity within the bed. The principal factor in the incomplete transition is that the inner core ($r < 10$ cm) was not cookoff-preheated, so the wave must drive its own preheating ahead of itself as it converges. (With the direction-aware bed-end diagnostic corrected per §10, the run is no longer truncated before DDT — the wave reaches transition within the bed.) Whether this configuration would eventually reach D_{CJ} in an extended simulation is an open question for the future work in §10.

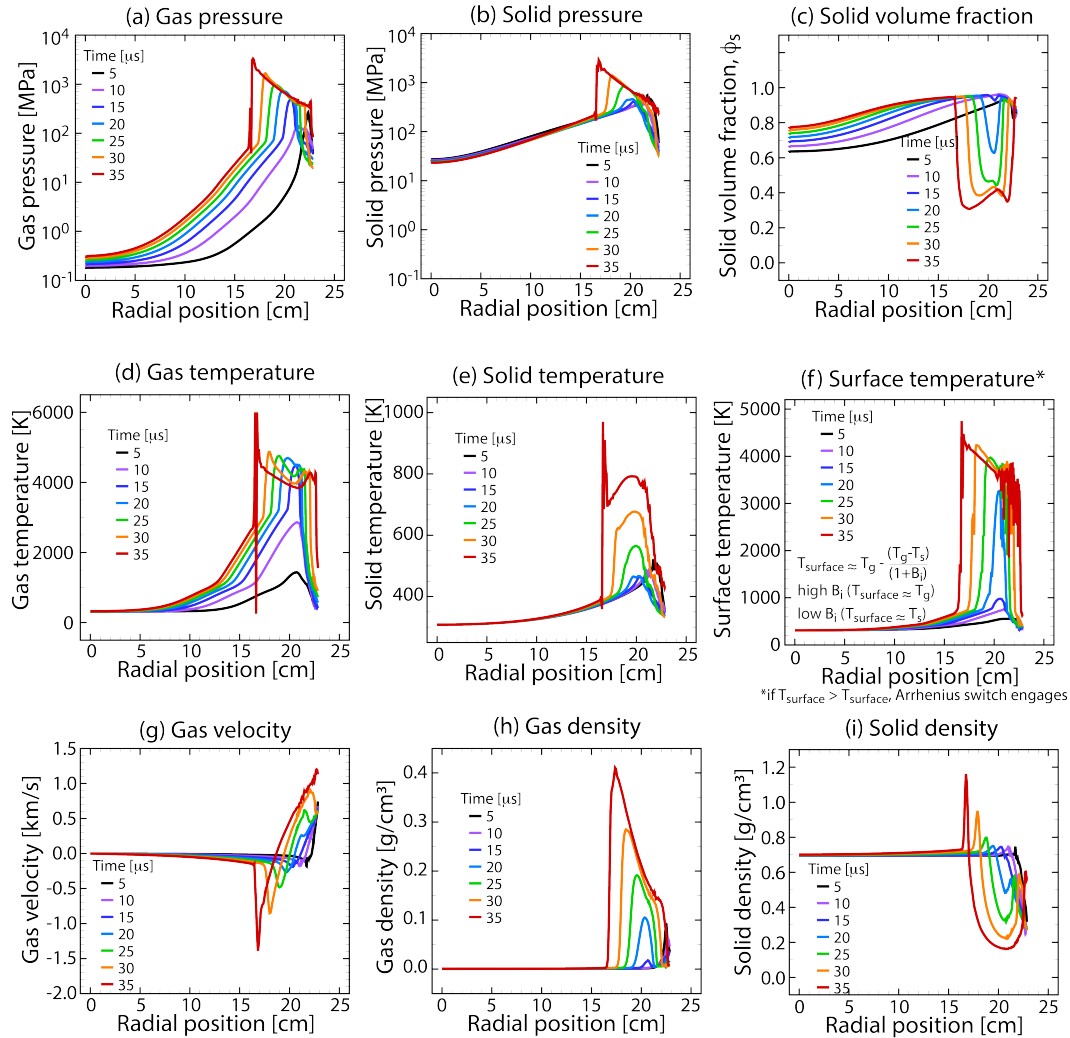


Figure 7-3. Radial profiles of the nine BN state variables at seven time slices (5, 10, 15, 20, 25, 30, 35 μs) for the slow cook-off cylinder XDDT phase. Each panel shows one field as a function of radial position, with the time slices color-coded. Panels: **(a)** gas pressure (log axis); **(b)** solid pressure (log axis); **(c)** solid volume fraction; **(d)** gas temperature; **(e)** solid temperature; **(f)** surface temperature; **(g)** gas velocity (negative = inward); **(h)** gas density; **(i)** solid density. The inward propagation of pressure and temperature peaks from the outer wall ($r = 22.86$ cm) toward the bed interior is visible across all panels, with the strongest signatures in gas pressure (a) and gas temperature (d). The early-time wide reaction zone characteristic of compaction-only burning across the XCHEM-preheated HMX bed is most evident in panel (e), which shows the solid temperature climbing across a wide radial extent in the early time slices.

8. TWIN-FRAGMENT IMPACT

This chapter reports the headline twin-fragment-impact demonstration of the XTIV pipeline. The geometry — two hot steel fragments separated by a cold HMX bed — carries both thermal and shock-like initiation features simultaneously, spanning the DDT and SDT regimes introduced in §1.3.1. Two inward-propagating reactive waves develop, collide near mid-bed, and fire the classical BN deflagration-to-detonation transition criterion at the collision. The configuration is exercised in a boost-augmented mode — an asymmetric twin hot-spot patch (900/1100 K, 50/100 MPa) layered on the XCHEM pre-thermal-runaway handoff — that exceeds natural cookoff for vented HMX; the boost stands in for sub-grid-scale shock-driven ignition physics that XCHEM’s 1D thermal model cannot resolve, and the resulting initial state is an idealized model assumption rather than a physically realizable cookoff outcome. The detonation velocity recovered from this boosted initial state (§8.3.1) is benchmarked against a TIGER hot-CJ calculation at the matching internal-energy condition in §8.5, which establishes that the observed $D = 8.05$ km/s is CJ-class for the artificially elevated initial temperature state rather than an overdriven detonation in the classical sense. This configuration turned out to be the most numerically demanding case in the validation matrix, requiring a coordinated suite of methodology fixes before the run would integrate; §8.4 retains a record of that development in case its lessons aid future coupled-pipeline work.

The boost-augmented run is one of two demonstrations this chapter reports. A companion fragment-spacing sweep (§8.6) probes what the same laminate steel-HMX-steel geometry produces under *natural* cookoff — with no imposed boost — across fragment spacings of 1, 2, and 4 cm. The natural-cookoff cases are the physically realizable companion to the boost-augmented probe, and they are illuminating in their own right: across the three fragment spacings the simulation either fails to ignite (1 cm), ignites and briefly propagates before the two fronts collide (2 cm), or sustains slow inward burning (4 cm); none of the three reaches the detonation regime the boost accesses. Together the two demonstrations bracket the regime spectrum the coupled pipeline can resolve for this geometry, from the fizzle-and-transition territory natural cookoff produces to the idealized CJ-class ceiling the boost-imposed initial state admits.

8.1. Configuration

The bed is a three-layer Cartesian slab spanning $x \in [0, 12]$ cm: 4 cm of inert steel at $T = 575$ K on each end (representing two hot fragments), with 4 cm of cold porous HMX at $T = 300$ K in the middle ($x \in [4, 8]$ cm). The HMX is at 70% theoretical maximum density ($\phi_{s,0} = 0.70$, $\rho_{s,0} = 1900$ kg/m³), particle radius $a_p = 50$ μ m, ignition temperature $T_{\text{ign}} = 800$ K. The HMX layers are assumed vented in the XCHEM cookoff phase (player='v': pressure throughout the explosive remains at the initial value, with pressure-dependent reaction-rate terms reducing to unity); the steel layers act as inert thermal reservoirs and as rigid boundaries on the XDDT side.

No outer-wall thermal ramp is applied — the steel fragments are the entire heat source, conducting energy into the HMX bulk by simple thermal contact.

The HMX bed and its boundary conditions are deliberately symmetric; no preferential side is built into the geometry. As shown in §8.2, the XCHEM precursor stage nevertheless develops a small natural asymmetry between the two steel/HMX interfaces, with the right-hand-side (RHS) ignition spike becoming somewhat more intense than the left-hand-side (LHS) spike before XCHEM termination. That asymmetry is preserved in the XTIV handoff and propagates into the XDDT integration as a slight LHS-of-mid bias in the collision location.

The headline coupled-pipeline output for this configuration is shown in Fig. 8-1, which combines the XCHEM thermal profiles in the pre-ignition phase (panel a) with the XDDT post-handoff wave dynamics (panel b) on a common geometry axis. Panel (b) is the working visual the rest of this chapter refers back to: the converging twin-front structure, the collision frame, and the four bed-state regions (steel jacket, unreacted HMX, burning HMX, and HMX residue).

8.2. XCHEM precursor stage: pre-runaway handoff

The XCHEM cookoff phase resolves heat conduction from the hot steel fragments inward through the HMX bed until just before chemical runaway. XCHEM is run with the maximum temperature ($t_{\max}$) set to the steel initial temperature plus a small margin (~ 600 K, just above the 575 K fragment temperature), which terminates the integration before any chemistry-driven temperature spike develops. Typical cookoff-coupled practice sets $t_{\max}$ slightly above the maximum boundary temperature — often a few hundred K higher, but rarely above ~ 800 K — so the integration halts at the onset of runaway with a modest overshoot (the SCO cylinder of Chapter 7 uses $t_{\max} \approx 800$ K for that reason). Setting $t_{\max}$ far above the maximum boundary temperature is problematic: near runaway the integrator can step several hundred to several thousand K in a single step, so a permissive $t_{\max}$ produces a large overshoot and an artificially high ignition spike in the handoff state. The choice here is more conservative still: at 600 K the integration halts well before chemistry begins to run away, delivering a clean pre-runaway handoff on which the XDDT-side hot-spot boost owns the entire ignition-to-DDT phase, as described in §8.4.

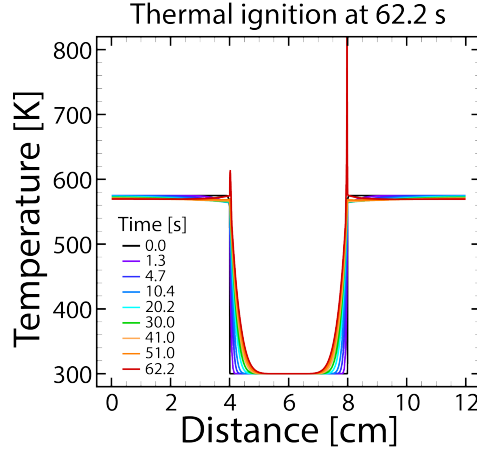
The handoff state is a single-zone `.xtiv` transfer file written by the `xtiv_extract.py` tool against the `schema-2` format. It carries a near-symmetric $T(x)$ profile across the bed: bulk HMX still cool ($T \sim 320\text{--}400$ K in the central 1.5 cm), thermally graded outer slabs on both sides peaking just inside the steel/HMX interfaces, and a small natural asymmetry favoring the RHS interface. This is the clean pre-runaway state that the asymmetric twin hot-spot boost described in §8.4 layers on top of: XDDT itself owns the entire ignition-to-DDT phase.

8.3. Boosted XDDT integration of the handoff state

The XDDT integration is run with the `IBC=3` transfer-file mode (§3.3), the dual-edge smoothing knob (`xtiv_smooth_l_in` and `xtiv_smooth_outer_l_in`) set symmetrically to 0.2 mm on both edges, and an asymmetric twin hot-spot boost (`&xtiv_boost`): inner-edge patch at $(\ell, T, P) = (20$

Twin assymmetric cookoff in HMX sandwiched between two steel fragments at 575 K

(a) Pre-thermal ignition, XCHEM



(b) Post-thermal ignition, XDDT

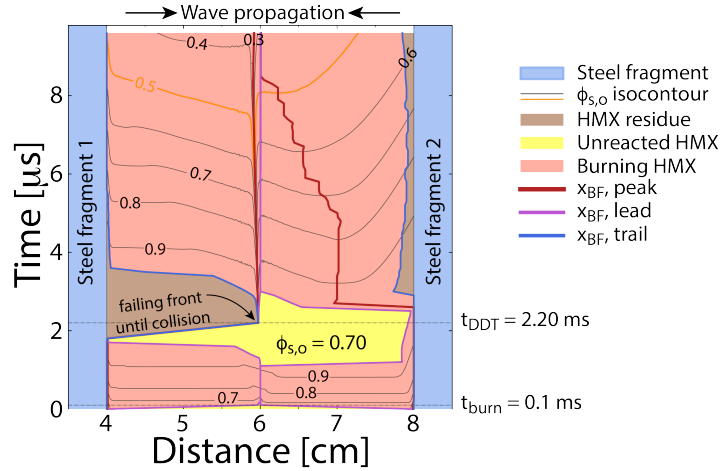


Figure 8-1. Twin-fragment-impact wave dynamics for the asymmetric cookoff configuration of §8.1. **(a)** Pre-ignition XCHEM phase: temperature profiles $T(r)$ at multiple time slices during the slow heating of the cold HMX core by the two flanking 575 K steel fragments; runaway at the steel/HMX interfaces (red curves). **(b)** Post-ignition XDDT phase: wave dynamics over $t \in [0, 9.7] \mu\text{s}$, with time advancing upward and position horizontal. Bed-state regions are colored by physical state: cornflowerblue for the inert steel jacket, yellow for unreacted HMX, tomato for the active reaction (burning) zone, and saddlebrown for the post-reaction residue. The inward-propagating LHS and RHS reaction bands collide near mid-bed at $t \approx 2 \mu\text{s}$; the BN-DDT criterion fires at $t_{\text{DDT}} = 2.20 \mu\text{s}$ at $x_{\text{DDT}} = 5.98 \text{ cm}$, slightly LHS-of-center because of the deliberate boost-intensity asymmetry that breaks the symmetric-stall lock-in (§8.4). $\phi_{s,o}$ contour lines (gray) trace the solid-volume-fraction structure across the three reaction-state regimes (§8.3.3).

mm, 900 K, 50 MPa), outer-edge patch at $(\ell, T, P) = (20 \text{ mm}, 1100 \text{ K}, 100 \text{ MPa})$, the asymmetry matching the natural-cookoff intensity profile carried in the handoff state. The mesh is adaptive (`imesh_in` = 1, `npt_in` = 4500); tolerances are the tight defaults of `rel_in` = 10^{-6} , `ae_in` = 10^{-10} . Four Eulerian gauges instrument the bed at $x = 1.0, 1.8, 2.2$, and 3.0 cm in HMX-internal coordinates — quarter-bed, mid-bed-LHS, mid-bed-RHS, three-quarter-bed — to capture both inward-propagating fronts and their collision.

8.3.1. *Primary 1.40 μs run — propagation phase*

The 1.40 μs run terminates when the legacy BURNING HAS REACHED END OF BED diagnostic fires at the LHS bed boundary. Within this propagation-phase window the diagnostics are well-behaved and the headline numbers can be quoted directly:

- $t_{\text{ign}} = t_{\text{burn}} = 0.10 \mu\text{s}$ — compaction-driven and surface-burn-driven kinetics engage simultaneously at both interfaces, an artefact of the pre-deposited hot-spot boost rather than natural ignition.
- Burn-front lead and peak both propagate inward from each interface into the bed. Fourteen burn-front samples are captured before the bed-end check fires.
- Peak mixture pressure $P_{\text{max}} = 2.01 \text{ GPa}$, measured at the burn front.
- Fitted detonation velocity $D = +8.05 \text{ km/s}$, recovered by least-squares fit on the last half of the burn-front trajectory in the LHS-bed frame. This is well above the cold-granular CJ velocity $D_{\text{CJ}} = 6.36 \text{ km/s}$ (HMX at $T_0 = 300 \text{ K}$, $\rho_0 = 1330 \text{ kg/m}^3$), but the cold-granular CJ is the wrong baseline for a bed driven from the boost-augmented initial state. Benchmarking against a TIGER hot-CJ calculation at the matching internal-energy condition (§8.5) places the observed $D = 8.05 \text{ km/s}$ on the hot-CJ trend at $\rho \approx 1.54 \text{ g/cc}$ ($\phi_s \approx 0.81$ of TMD), consistent with a CJ-class wave running in the hot, partially compacted bed that the boost establishes ahead of the front — not an overdriven detonation in the classical sense.
- The mesh adapts thirteen times during the run, with the point count rising from `NPT` = 163 to 671 as refinement tracks the wave front.

The 1.40 μs run is the publication value to cite for D . Its bed-end-driven termination is the natural cutoff before the two inward fronts have time to collide — a feature, not a limitation, for the propagation-phase analysis.

8.3.2. *Extended 9.70 μs run — collision and DDT closure*

To capture the collision and DDT closure, a companion run is extended past the bed-end check by setting the namelist flag `bed_end_disable_in` = `.true.` in `&numerical` (further discussed in §8.4). The extended run uses the same handoff and the same boost but integrates to $t = 9.70 \mu\text{s}$, where the chemistry Jacobian becomes ill-conditioned and DDEBDF returns `IDID=-7` (a known limitation; see §10). Within the extended window the wave-position structure is:

- 0–1.5 μs : outer-edge front propagates inward through the LHS half of the bed, with its peak migrating from $x \approx 6.65$ to $x \approx 6.81$ cm in absolute coordinates (i.e., from the inner steel/HMX interface inward). An inner-edge front develops in parallel on the RHS half but is not picked up by the burn-front tracker until the wave intensifies enough to clear the noise floor of the active-reaction-band tracker (§4).
- ~ 2.0 μs : the two inward fronts collide near mid-bed. The burn-front peak jumps to $x \approx 5.98$ cm — mid-bed biased slightly LHS-of-center, consistent with the LHS-of-mid geometric prediction of §8.1. The compaction-front tracker direction reverses through the collision frame.
- $t_{\text{DDT}} = 2.20$ μs at $x_{\text{DDT}} = 5.98$ cm: the burn front overtakes the compaction front position at ignition. The classical BN-DDT criterion fires at the collision, not via individual-front intensification.
- 2.5–7.0 μs : a post-collision reaction zone develops and slowly retreats inward (peak at $x \approx 5.98$ to ≈ 4.35 cm) as reaction products vent through the permeable steel/HMX interfaces. The peak mixture pressure over the full window is $P_{\text{max}} = 2.87$ GPa, achieved during the collision frame.
- 9.70 μs : DDEBDF fails with IDID=–7 on chemistry-Jacobian conditioning (further discussion in §10). The integration is terminated.

The fitted D on the extended run is -1.18 km/s, but this value is not a velocity: the least-squares fit averages the outward-then-inward trajectory of the burn-front peak (propagation toward mid-bed, the collision jump, and the post-collision retreat) and the negative sign here is a fitter artefact rather than a physically meaningful inward propagation. The D figure of merit for this scenario is the 1.40 μs run value of $+8.05$ km/s. The diagnostic exposed by the extended run — that the BF-peak-following D fitter does not correctly handle non-monotone front trajectories — is an open issue (§10).

8.3.3. Solid-fraction structure across the bed

The solid volume fraction ϕ_s field provides a complementary view of the wave structure to the BF-tracker trajectories discussed above. At handoff $\phi_s = 0.70$ throughout the HMX layer (the deck’s `poro_in` initial packing fraction). Post-handoff, ϕ_s departs from this initial value in two opposite directions, marking two distinct physical regimes:

Consumption regime ($\phi_s < 0.70$). Behind each BF trail, where reaction has completed, ϕ_s drops to 0.10–0.30 as solid HMX is converted to gas-phase products. The remaining solid is condensed-carbon residue.

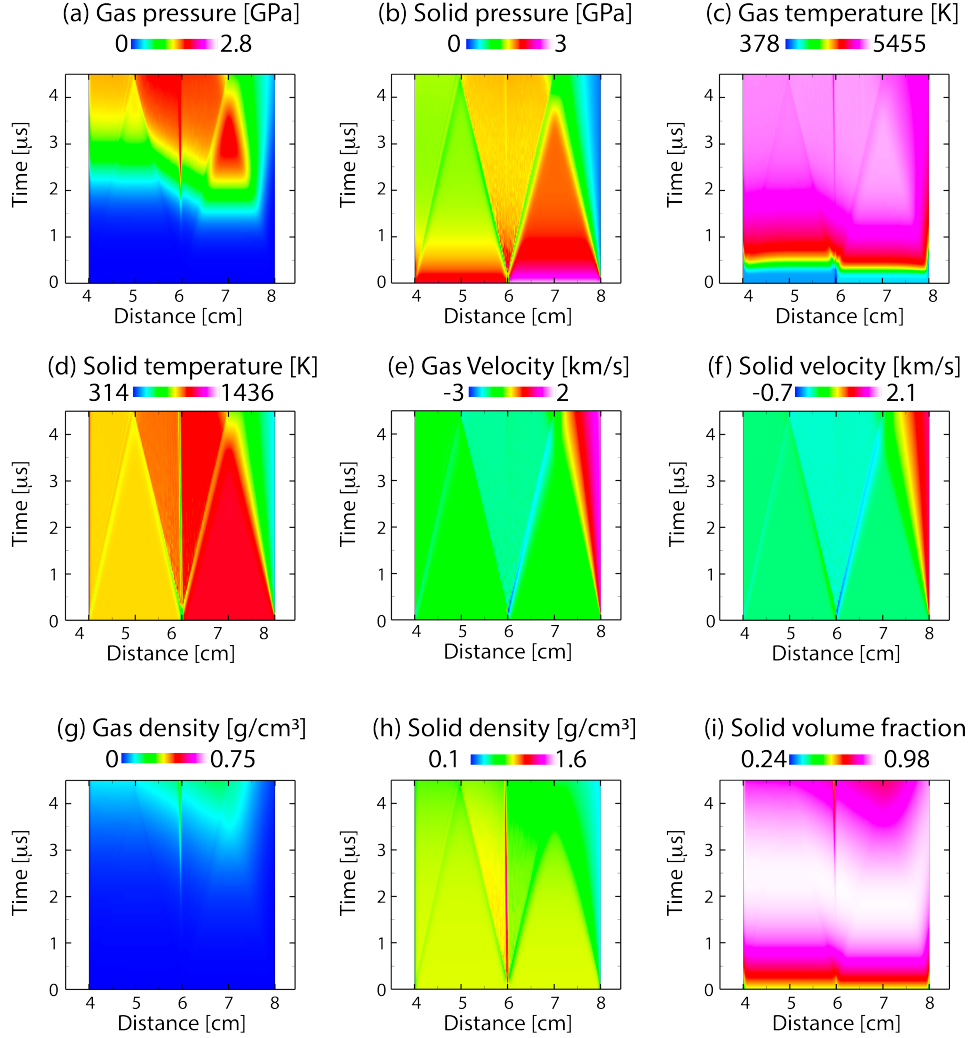


Figure 8-2. State-variable contour fields for the extended 2frag XDDT run (companion to Fig. 8-1(b)). Each panel shows one BN-model field as a function of time (vertical) and bed position $x \in [4, 8]$ cm (horizontal): **(a)** gas pressure; **(b)** solid pressure; **(c)** gas temperature; **(d)** solid temperature; **(e)** gas velocity (sign reflects propagation direction of each front); **(f)** solid velocity; **(g)** gas density; **(h)** solid density; **(i)** solid volume fraction. The two converging fronts and their collision near $t \approx 2 \mu\text{s}$ are visible in every field, with the strongest collision-frame signatures in gas pressure (a) and gas temperature (c). Panel (i) shows the post-collision compaction structure that drives ϕ_s above the initial 0.70 packing toward TMD at the rigid steel walls (§8.3.3).

Compaction regime ($\phi_s > 0.70$). Ahead of each burn front the BN compaction wave squeezes the granular bed above the initial 0.70 packing toward theoretical maximum density ($\phi_s = 1$), driven by the BN compaction relaxation source. Post-collision the merged reaction zone vents products outward through the permeable steel/HMX interfaces, pressing the residual solid against the rigid steel-bounded walls; ϕ_s peaks at 0.96 near the inner steel/HMX interface around $t = 2.7 \mu\text{s}$ and remains above 0.85 along both walls throughout the post-collision phase.

The unreacted-HMX core — the central region between the two inward-propagating fronts, before the converging waves engulf it — remains at the initial $\phi_s = 0.70$ throughout its existence. This three-regime structure (depleted residue at low ϕ_s , initial unreacted bed at $\phi_s = 0.70$, initial compaction at high ϕ_s) governs the ϕ_s -contour topology visible across the bed in the wave-dynamics figure: closely spaced contour lines below 0.70 mark the residue zones; the absence of internal contour lines marks the unreacted core (a single uniform value); and closely spaced contour lines above 0.70 mark the compaction-driven squeeze ahead of each front and the post-collision compression at the rigid walls.

8.4. Methodology development: getting the run to integrate

The twin-fragment-impact case is the headline result of this report, but it is also the case that exposed the largest number of methodology limitations in the coupled XTIV pipeline. Several configurations failed before the boosted deck of §8.3 converged on a usable result; a brief development record is preserved here both to substantiate the methodology choices in the deck and to flag the specific failure modes for future coupled-pipeline work. Each prerequisite is reported in the SAND-2 *Bug / How found / Fix* style [2].

Symmetric boost configurations stalled the DDT diagnostic. **Bug.** Earlier attempts placed equal-intensity hot-spot patches on both steel/HMX interfaces (configurations `sym_1` and `sym_3` in the development tree), reasoning from the geometric symmetry of the problem. Physically, two inward fronts meeting symmetrically should produce a strong collision-frame pressure spike — and they do in the BN field data — but the BN-DDT *criterion* as implemented could not fire on the event. The criterion was originally coded for single-wave propagation (it watches a designated burn front overtake the compaction-front-at-ignition position from one side), so when two mirror-image waves converge with equal amplitude and zero net velocity at mid-bed, neither has a “leading” role for the criterion to track. The tracker latched its burn-front peak onto the geometric center, the single-side overtaking test never evaluated true, and the run returned without a recorded DDT marker even though the field data showed the violent collision the symmetric configuration should produce. **How found.** Burn-front trajectory plots from the symmetric-boost runs showed both fronts arriving at mid-bed with comparable amplitudes and zero net velocity at the BF-peak location, while the underlying 3D-field data showed strong pressure and temperature peaks at the collision frame — the violent event was happening, the diagnostic just wasn’t recognizing it. **Fix.** The boosted deck uses an *asymmetric* twin hot-spot boost matching the natural-cookoff intensity profile carried in the XCHEM handoff: the RHS patch is hotter and at higher pressure than the LHS patch, giving the RHS-driven front a slight head start. The asymmetry gives the single-direction

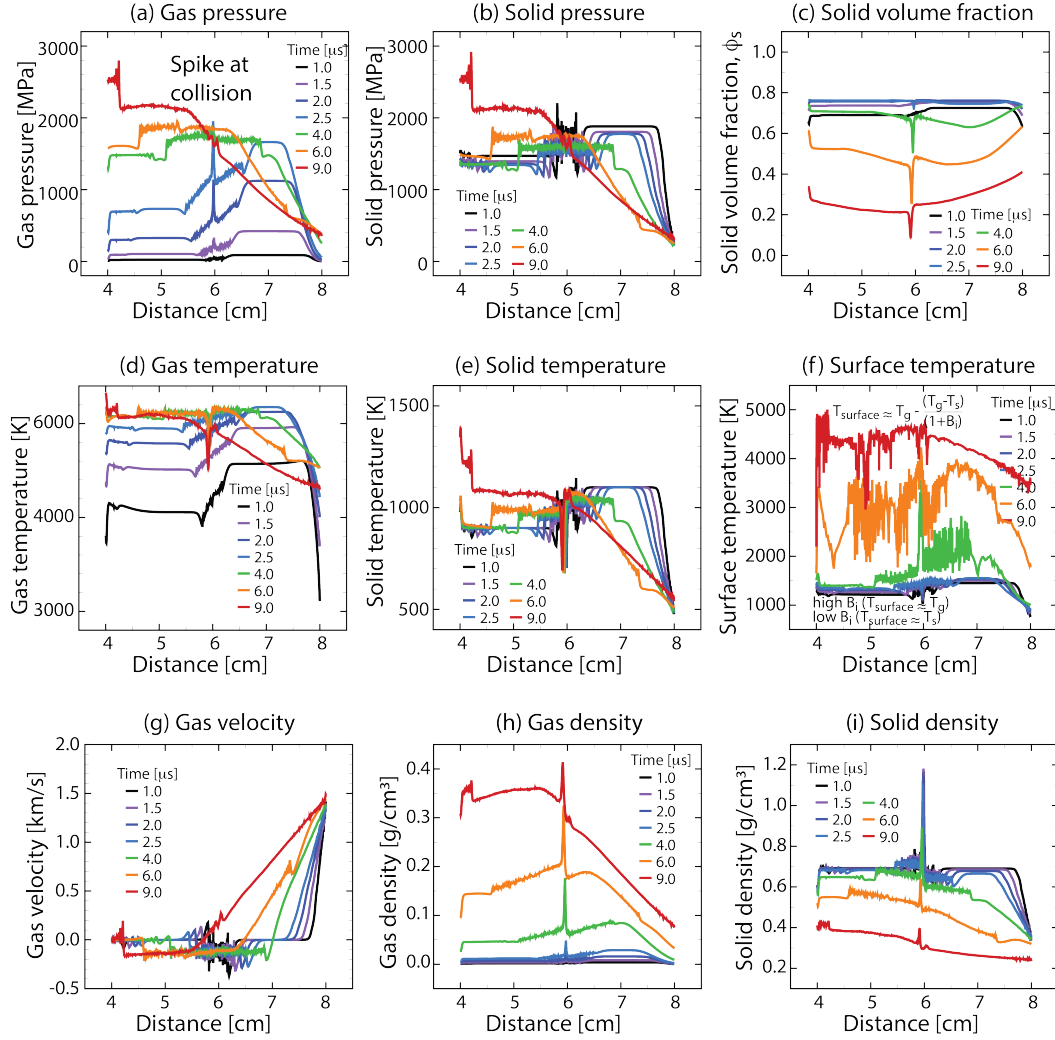


Figure 8-3. Radial profiles of the BN state variables at multiple time slices for the 2frag extended XDDT run, with each panel showing one field as a function of bed position $x \in [4, 8]$ cm. The time slices are color-coded across the pre-collision approach phase, the collision frame at $t \approx 2 \mu\text{s}$, and the post-collision retreat. Pre-collision profiles show the LHS and RHS active reaction bands as separate spatial features in the pressure and temperature panels; the collision frame produces a single merged peak near $x = 5.98$ cm. Post-collision, the merged reaction zone retreats inward while ϕ_s at the rigid steel boundaries climbs toward TMD as residual solid is pressed into the walls.

DDT criterion a clear “lead” front to track; the collision then occurs slightly LHS-of-mid and the criterion fires cleanly. The asymmetry is therefore both a physically reasonable choice (it matches the natural-cockoff handoff state, in which the two steel/HMX interfaces are never exactly identical) and a diagnostic workaround for the single-direction DDT criterion. Implementing the asymmetric specification required extending the boost machinery to accept independent inner- and outer-edge hot-spot parameters. Generalizing the DDT criterion to handle twin converging waves natively is a proposed enhancement in §10.

Fixed-mesh integration fails before any usable result. Bug. With `imesh_in = 0` (fixed-mesh path), the run dies with DDEBDF returning `IDID=-7` at $t \approx 1.04 \mu\text{s}$ — before D has had a chance to stabilize. The two-front collision regime is not captured; the published $1.40 \mu\text{s}$ value cannot be recovered. **How found.** A bisection on `imesh_in` between adaptive (1) and fixed (0) cleanly separated runs that did and did not survive past $1.04 \mu\text{s}$. No tolerance combination on the fixed-mesh side improved the outcome. **Fix.** Adaptive XDDT (`imesh_in = 1`) is required. This introduced a new dependency on the XDDT mesh-integrity diagnostics, which themselves had to be repaired before adaptive XDDT would track the converging fronts correctly, as detailed below.

Adaptive mesh refinement defects. Bug. The converging twin-front geometry needs adaptive mesh refinement to track both fronts at adequate resolution through the collision frame; the unidirectional cases of Chapter 6 did not exercise the refinement machinery comparably. Enabling adaptive refinement on this case surfaced three latent defects in the refinement code: an indexing inconsistency in how refined cells were associated back to the handoff state, a boundary-clamping defect that allowed refined nodes to drift past the bed boundary during refinement, and an overly tight cell-width-match tolerance that caused refinement to abort spuriously where the two converging fronts gathered. **How found.** The three defects surfaced in sequence: each fix exposed the next as the integrator progressed further into the run before failing again. The failure mode shifted from a hard abort, to a refinement loop that never converged, to integrator tolerance failure right at the converging-wave collision. **Fix.** Each defect was corrected as it surfaced. Most consequential was rephrasing the cell-width tolerance as a percentage of cell width rather than an absolute floor, which preserves mesh integrity in the standard cases while letting refinement complete in the converging-wave region. With those repairs in place the adaptive XDDT path correctly tracks both inward-propagating fronts through the collision frame.

Dual-edge smoothing (both inner and outer steel/HMX interfaces). Bug. The original handoff-state smoothing knob (§5) applied a tanh ramp on T , P , ϕ_s , ρ_g over the inner HMX edge only. For the twin-fragment geometry, both steel/HMX interfaces carry ignition spikes in the handoff, and the unsmoothed outer edge caused integrator failures within the first few hundred nanoseconds. **How found.** Comparison runs with and without inner-edge smoothing showed the outer-edge gradient as the load-bearing failure mode: inner-only smoothing did not improve survival. **Fix.** A companion smoothing parameter was added that applies the tanh ramp at the outer bed edge in addition to the inner edge. With both edges smoothed at 0.2 mm , the integrator clears the handoff-state stiffness without erasing the boost-driven ignition spikes that follow. The 0.2 mm

value is the smallest that lets the integrator clear the first nanosecond; larger values (≥ 5 mm) wash out the boost and degrade ignition.

XCHEM termination tuning. Bug. The conventional XCHEM termination criterion — run until $T_{\max} = 4000$ K is reached anywhere in the bed — proved incompatible with the twin-fragment geometry. The XCHEM integration thrashed on grid refinement of the converging near-runaway region, accumulating excessive cell counts long before any cell crossed the $T_{\max}$ threshold. **How found.** A sweep over $t_{\max} \in \{585, 600, 625, 675, 825\}$ K (directories preserved under `xchem/tests/runs/2frag/`) showed cleanly converging runs only at the lowest values, with the run wall-clock time and grid refinement count diverging beyond ~ 700 K. **Fix.** The boosted deck uses $t_{\max} = T_{\text{steel,initial}} + \varepsilon \approx 600$ K, which terminates XCHEM well before any chemistry-driven runaway and delivers a clean pre-runaway state to XDDT. The trade-off is methodological: XDDT now owns the entire ignition-to-DDT phase, with the asymmetric twin hot-spot boost standing in for the sub-grid-scale shock-driven ignition physics that XCHEM’s 1D thermal model cannot resolve. A more principled XCHEM termination criterion remains future work (§10).

Premature bed-end termination of the 1.40 μs run. Bug. XDDT’s legacy bed-end check terminates the run as soon as a burn-front trail reaches either bed boundary. In the twin-fragment geometry the inward-propagating left-side front reaches the left bed-end at $t \approx 1.40$ μs — before the two fronts have a chance to collide — and the check fires, terminating the integration mid-physics. **How found.** Establishing that DDT fires at the front collision (rather than at one of the individual inward-propagation events) required running well past 1.40 μs . The boosted deck alone could not support that analysis. **Fix.** A namelist flag was added that lets the user disable the bed-end check when the physics calls for it. With the flag set, the integration continues through the front-on-front collision and the post-collision retreat until the solver fails for other reasons at $t \approx 9.70$ μs . The extended run is therefore reproducible from the boosted deck plus that single flag, with no source modifications required. Direction-aware bed-end checking — terminating only when a front reaches the boundary it is actually propagating toward — is the more principled long-term fix and is folded into the broader direction-aware diagnostic redesign in §10.

D fitter behavior on non-monotone trajectories. Bug. The least-squares D fitter operates on the burn-front peak position as a function of time and assumes a monotone trajectory. On the extended run, the burn-front peak migrates inward from each interface, jumps at the collision, and then retreats inward; the resulting non-monotone trajectory yields $D = -1.18$ km/s, which has neither sign nor magnitude that corresponds to a physically meaningful detonation velocity. **How found.** The extended run’s `coupled.out` reported the fitter result verbatim; the burn-front trajectory plot made the non-monotone structure obvious. **Fix.** Until the fitter is generalized to handle non-monotone trajectories, the recommended convention is to cite D from the 1.40 μs run and to treat the extended-run D output as a fitter artefact. The 1.40 μs cap is, in retrospect, well-suited to keeping the fitter inside the monotone-propagation window, even though the cap was not designed for that purpose.

8.5. Hot-CJ benchmark for the boosted initial state

The 1.40 μ s detonation velocity $D = +8.05$ km/s (§8.3.1) was originally reported as “above D_{CJ} ” against the TIGER cold-granular CJ value of 6.36 km/s for HMX at $T_0 = 300$ K, $\rho_0 = 1.33$ g/cc (70% TMD). That framing is incorrect: the cold-granular CJ applies to a bed that the boost-augmented configuration is no longer in. The asymmetric twin hot-spot boost layers ~ 900 – 1100 K initial temperature on top of the XCHEM pre-runaway handoff, and the BN compaction wave running ahead of the burn front squeezes the granular bed above its initial 0.70 packing toward TMD (§8.3.3). The wave is therefore running into hot, partially compacted HMX, whose CJ ceiling sits well above the cold-granular reference. Benchmarking the observed D against the appropriate hot-bed CJ velocity is the correct comparison; this section provides it.

8.5.1. TIGER hot-CJ calculation

The reference CJ velocities are computed with TIGER, the thermochemical equilibrium code, using the JCZS3 product database and the JCZ3 equation of state for the detonation products. The hot-bed condition is imposed at the level of the unreacted-HMX initial state: the internal energy is set to $e_{\text{initial}} = 1.38$ kJ/g ($= 329.7$ cal/g), the value HMX attains at ~ 1000 K, matching the elevated initial temperature established by the asymmetric twin hot-spot boost ahead of the inward-propagating reactive fronts. At each of three densities — $\rho = 1.20, 1.33, 1.90$ g/cc, spanning the granular initial state through to TMD — TIGER returns:

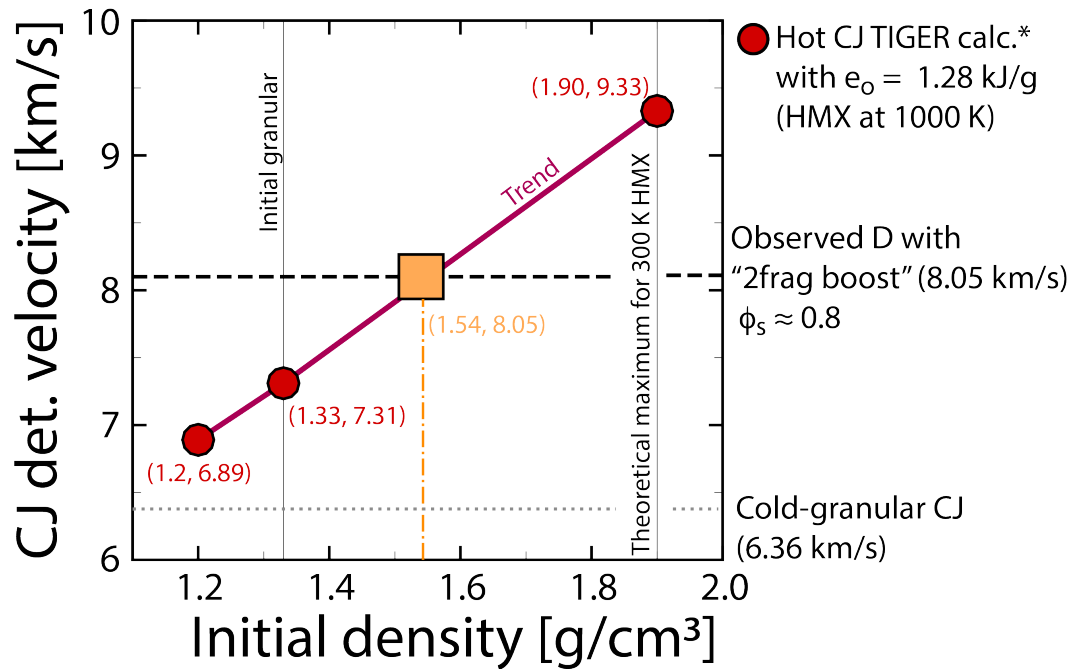
ρ (g/cc)	D_{CJ} (km/s)
1.20	6.89
1.33	7.31
1.90	9.33

A linear fit through the three points gives $D_{\text{CJ}}(\rho) = 3.50\rho + 2.67$ km/s on the hot-bed trend. The TIGER calculation does not resolve transport or shock dynamics — it is a Chapman-Jouguet steady-wave calculation against an idealized hot-bed initial state — and the result is a thermodynamic ceiling rather than a prediction of the BN model’s transient behavior. Its value here is as a benchmark for the BN model’s recovered D : agreement between the two establishes thermodynamic consistency between the boosted-deck wave and the CJ steady state for the same initial-energy condition.

8.5.2. Bracketing the BN result

Fig. 8-4 plots the TIGER hot-CJ points and their linear fit alongside the observed BN $D = 8.05$ km/s. The observed velocity intersects the hot-CJ trend at $\rho \approx 1.54$ g/cc, corresponding to a solid volume fraction $\phi_s \approx 0.81$ of TMD ($\rho_{\text{TMD}} = 1.90$ g/cc). This implied burn-front density is physically consistent with the BN compaction-wave structure documented in §8.3.3: ahead of each front the granular bed is squeezed from its initial $\phi_s = 0.70$ packing into the 0.80–0.95 range, depending

on position, before the reaction completes. The intersection at $\phi_s \approx 0.81$ therefore lands in the physically expected window of partial compaction immediately ahead of the burn front, not at the initial granular density and not at TMD.



*TIGER calculations with JCZS3 database using JCZ3 EOS.

Figure 8-4. Hot-bed CJ velocity vs density: bracketing the twin-fragment-impact BN result. Red filled circles are TIGER CJ calculations using the JCZS3 product database with the JCZ3 equation of state, at unreacted-HMX initial energy $e_{\text{initial}} = 1.38 \text{ kJ/g}$ ($= 329.7 \text{ cal/g}$), corresponding to HMX at $\sim 1000 \text{ K}$. The linear fit through the three points (red line) is the hot-CJ trend. The observed twin-fragment-impact $D = 8.05 \text{ km/s}$ (black dashed) intersects the hot-CJ trend at $\rho \approx 1.54 \text{ g/cc}$ ($\phi_s \approx 0.81$ of TMD, orange square), consistent with a CJ-class wave running in the partially compacted bed ahead of the burn front. The cold-granular CJ reference ($D = 6.36 \text{ km/s}$ at $T_0 = 300 \text{ K}$, $\rho_0 = 1.33 \text{ g/cc}$, gray dotted) shows that the original “overdriven by $\sim 27\%$ ” framing in §8.3.1 compared the wrong baseline.

8.5.3. What the bracket establishes, and what it does not

The CJ bracket establishes a single, narrow claim: that the BN model’s boosted-DDT velocity of $D = 8.05 \text{ km/s}$ is thermodynamically consistent with the CJ ceiling for the boost-imposed initial state, at the partial-compaction density the BN compaction wave produces ahead of the front. The wave is running at its CJ ceiling for the bed it sees, not above it; the original “overdriven” label was an artefact of comparing against the wrong baseline.

The bracket does not establish field realism. Both the TIGER hot-CJ calculation and the BN boosted-DDT integration take an idealized initial state as input. HMX in a real cookoff scenario does not pre-equilibrate to $e_{\text{initial}} = 1.38 \text{ kJ/g}$ uniformly, and the 900/1100 K boost layered onto the XDDT handoff exceeds what natural cookoff of vented HMX produces (Chapter 6 demonstrates the natural- cookoff outcome for the matched single-fragment slab geometry: fizzle, not detonation).

The TIGER and BN calculations are comparative model outcomes against an idealized hot-bed initial state, not predictions of a physically realizable detonation following a realistic cookoff handoff.

What the bracket contributes, then, is methodological rather than predictive. It shows that the BN model’s response to a sufficiently energetic preheated and compacted initial state is internally consistent with CJ thermodynamics in the same initial condition — a property the model should have, and the bracket confirms it does. Combined with the natural- cookoff fizzle and transition-territory cases of Chapters 6 and 7, the boosted twin-fragment-impact result establishes the upper end of the regime spectrum the coupled pipeline can resolve, with the hot-CJ benchmark providing the thermodynamic anchor for that upper bound.

8.6. Natural-cookoff baseline: fragment-spacing sensitivity

Having presented the boost-augmented run — the idealized upper-bound probe and its CJ-class wave — the chapter now turns to what the same laminate steel-HMX-steel geometry produces under *natural* cookoff, without any imposed boost. The natural-cookoff cases are the physically realizable companion to the boosted result, and they are themselves illuminating: the fragment-spacing sweep reported here spans three distinct positions in the cookoff-regime spectrum, with the outcome depending non-monotonically on the fragment spacing. The sweep also grounds the boost choice in the boosted deck: the boost is the methodology lever that lifts the accessible regime from the fizzle-and-transition territory natural cookoff produces in this geometry up to the CJ-class ceiling benchmarked in §8.5, not an arbitrary parameter probe.

8.6.1. Configuration and parameter sweep

The natural-cookoff sensitivity study sweeps the distance between the two hot fragments — 1, 2, and 4 cm — while holding the surrounding steel dimension at 4 cm on each side of the porous HMX. The BN model parameters are held identical across all three cases. The natural-cookoff configurations differ from the boosted deck of §8.3 in three respects.

First, the steel temperatures are lower than the boosted deck’s 575 K: $T_{\text{steel}} = 530$ K for the 1 cm bed, 540 K for the 2 and 4 cm beds. The reduction in fragment temperature was made to maximize the thermal soak between the two hot steel fragments.

Second, XCHEM is run *through* natural ignition rather than terminated at low t_{max} as the boosted deck does. The natural-cookoff XCHEM trajectories run for ~12, ~25, and ~67 minutes respectively before bed-state runaway is captured. The XCHEM-to-XDDT handoff is then taken at the final XCHEM state, with the natural-cookoff ignition spikes preserved into the XDDT initial condition.

Third, the BN ignition temperature T_{ign} is reduced from its default 800 K value to just above the smaller of the two natural-cookoff handoff spikes per the XTIV T_{ign} calibration convention, so that the surface burning switch arms cleanly on the higher-temperature side at handoff and engages with modest compaction work on the cooler side. The 1 cm case uses $T_{\text{ign}} = 600$ K; 2 cm uses 586 K; 4 cm uses 600 K.

None of the three natural-cockoff cases apply the asymmetric twin hot-spot boost. All other BN parameters — particle radius, surface burn rate coefficients, compaction terms, smoothing length (0.2 mm at each edge), adaptive-mesh settings — are held identical to the boosted deck. The study is therefore a controlled probe of fragment-spacing sensitivity in the natural-cockoff path, with the steel-temperature variation as a secondary, lightly varied parameter.

8.6.2. Results: pre-ignition character and XDDT outcome

Fig. 8-5 shows the matched XCHEM-XDDT results for the three fragment spacings. Panel (a) gives the XCHEM temperature profiles and includes the final time slice (just before chemistry-driven runaway); panel (b) gives the matching XDDT bed-state-vs-time wave dynamics with state-colored regions and burn-front-tracker traces.

The three cases occupy three distinct positions in the natural-cockoff regime spectrum, summarized in Table 8-1.

Table 8-1. Natural-cockoff outcomes across the fragment-spacing sweep of §8.6. Survival time is the time XDDT integrates before solver failure or bed-end termination; $|D|$ is the magnitude of the fitted detonation velocity on the last half of the burn-front trajectory.

Gap	XCHEM ignition character	Survival	$ D $ (km/s)	$P_{\max}$ (MPa)	Regime
1 cm	merged central, >1000 K	0.24 μ s	—	—	solver fail, no BF
2 cm	twin interface spikes	1.44 μ s	1.68	601	transition
4 cm	asymmetric single-side spike	7.90 μ s	1.43	502	transition

1 cm bed: pathological pre-runaway state. The laminate steel-HMX-steel geometry at 1 cm separation brings the two thermal penetrations into full overlap: the XCHEM temperature trajectory shows the two-sided preheating merge into a single central spike that reaches $T > 1000$ K at the final time slice (Fig. 8-5(a), leftmost panel). This is the most intense XCHEM ignition character of the three cases — and counterintuitively, the case that XDDT can integrate the shortest. The post-handoff XDDT state inherits the central sharply peaked temperature feature in the bed interior; the integrator chokes on the steep gradients within 0.24 μ s, before the BN kinetic switch can arm, and returns without any burn-front detection. No reaction wave develops; no $|D|$ was calculated for this case.

2 cm bed: compaction-then-collision in three phases. At 2 cm separation the XCHEM temperature trajectory keeps the two thermal sides distinct: two ignition spikes peak at the steel/HMX interfaces with the bulk between them preheated to 450–500 K but not merged into a single peak (Fig. 8-5(a), middle panel). The XDDT integration captures a striking three-phase structure (Fig. 8-5(b), middle panel). An early compaction-driven reaction band forms near the LHS steel/HMX interface and persists through $t \approx 0.4$ μ s. The band then drops below the BF tracker’s noise floor for ~ 200 ns ($t \in [0.5, 0.6]$ μ s) — a real physical pause as the compaction-driven activity tails off below the threshold for surface-burn engagement. At $t \approx 0.7$ μ s the two-front collision triggers volumetric

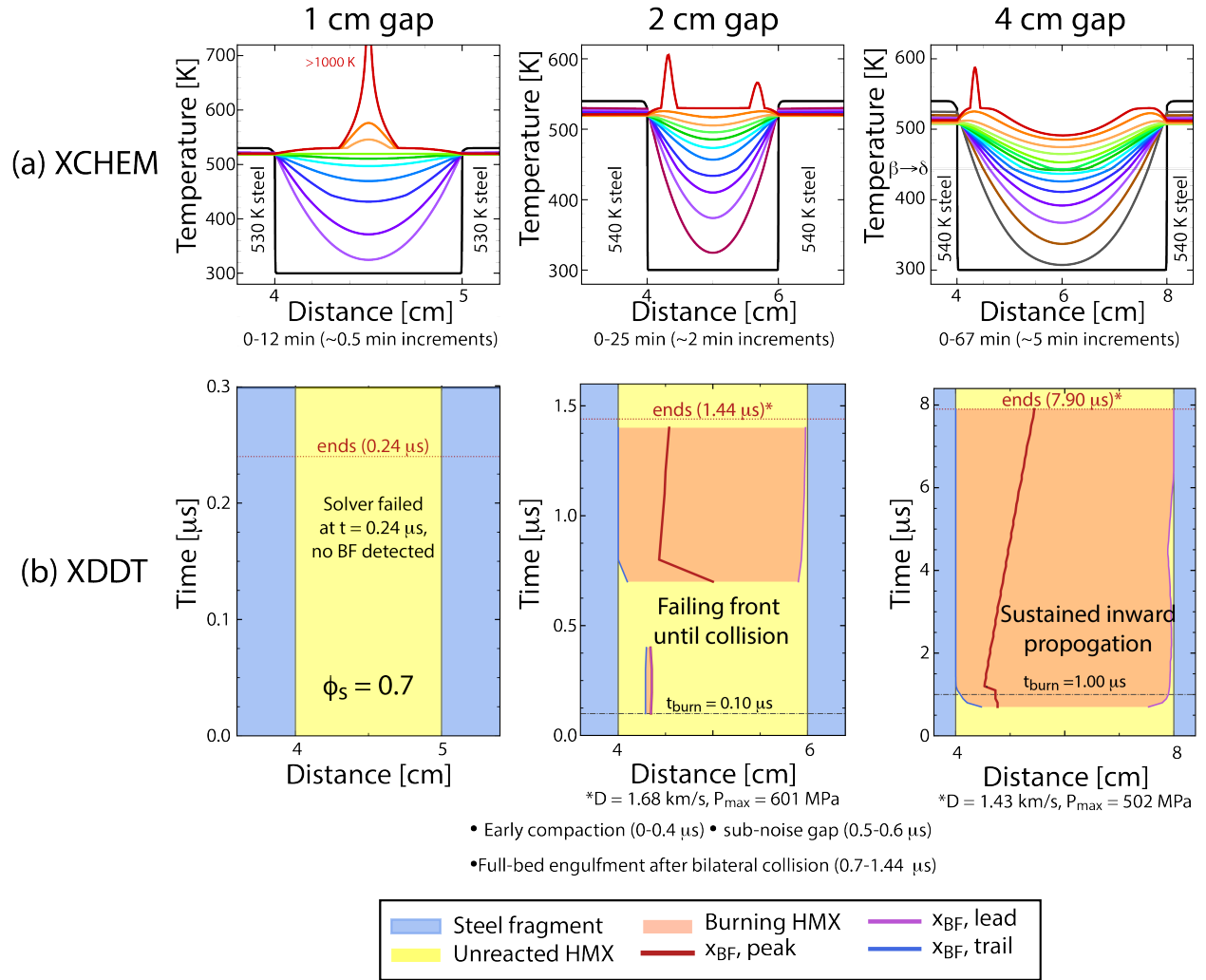


Figure 8-5. Fragment-spacing sensitivity of natural cookoff in the laminate steel-HMX-steel geometry. **(a)** XCHEM pre-ignition temperature profiles including the final time slice (showing chemistry-driven runaway) for HMX beds of 1, 2, and 4 cm sandwiched between 530–540 K steel fragments. The 1 cm panel shows a merged central temperature-spike at $T > 1000$ K; the 2 cm panel shows two distinct interface spikes with bulk between them preheated to 450–500 K; the 4 cm panel shows an asymmetric single-side spike with the $\beta \rightarrow \delta$ HMX phase transition shoulder visible near 432 K. **(b)** Matching XDDT bed-state-vs-time plots showing the post-handoff wave dynamics for each case. Colors: cornflowerblue inert steel jacket; yellow unreacted HMX; lightsalmon active reaction band; firebrick line $x_{\text{BF, peak}}$; mediumorchid $x_{\text{BF, lead}}$; royalblue $x_{\text{BF, trail}}$. The 1 cm panel fails before any burn front develops; the 2 cm panel shows a three-phase structure (early LHS-pinned compaction $\rightarrow$ sub-noise gap $\rightarrow$ full-bed engulfment after two-front collision); the 4 cm panel shows sustained inward propagation with surface-burn engagement. Asterisked outcome scalars underneath each XDDT panel: fitted $|D|$ and P_{max} .

engulfment of the bed: a full-bed active-reaction band develops that extends from $x = 4.0$ cm (LHS interface) to $x = 5.98$ cm (RHS interface) for the remaining ~ 0.7 μs of integration. The fitted velocity is $|D| = 1.68$ km/s — transition class, well below the CJ ceiling.

4 cm bed: sustained inward propagation. At 4 cm separation the XCHEM temperature trajectory keeps the two thermal sides far enough apart that the two-sided preheating only weakly overlaps in the interior (Fig. 8-5(a), right panel). An asymmetric single-side ignition spike develops at the LHS steel/HMX interface; the RHS side reaches a smaller shoulder but no comparable spike. The XDDT integration captures the inward propagation cleanly (Fig. 8-5(b), right panel): the burn-front lead reaches the RHS interface by $t \approx 1$ μs (stuck at the outer boundary thereafter), while the burn-front peak migrates slowly inward from $x \approx 4.5$ cm at $t = 0.7$ μs to $x \approx 5.5$ cm at $t = 7.9$ μs , with the burn front first reaching $x = 5$ cm at $t = 4.70$ μs . The fitted velocity is $|D| = 1.43$ km/s — transition class again, similar to the 2 cm case but with a much longer-surviving integration (7.90 μs vs 1.44 μs).

8.6.3. *Fragment spacing as the regime parameter*

The fragment-spacing sweep exposes a non-monotonic relationship between two-sided preheating overlap and XDDT survival. Counter to a naive “more preheating means more reaction” expectation, the 1 cm bed — which has the most intense two-sided preheating in the XCHEM phase — produces the shortest XDDT integration of any of the three cases. The mechanism is straightforward: full two-sided overlap produces a localized spike whose gradients exceed what the BN integrator can resolve at the handoff seam. Spreading the two heat sources further apart relaxes those gradients. The 2 cm case admits two distinct interface spikes and a clean compaction-then-collision evolution; the 4 cm case admits the longest-surviving sustained-propagation evolution.

None of the three natural-cockoff cases approach the detonation regime. The 2 and 4 cm cases produce transition propagation at $|D| \sim 1.4$ – 1.7 km/s, well below the cold-granular CJ velocity ($D_{\text{CJ}} = 6.36$ km/s) and below the cylinder slow-cockoff case’s transition velocity ($|D| = 5.386$ km/s, Chapter 7). The 4 cm case here is particularly informative: it matches the boosted twin-fragment-impact deck’s fragment spacing (4 cm) but with the natural-cockoff handoff state in place of the boost-augmented initial condition. Its outcome — transition propagation at $|D| = 1.43$ km/s — establishes the natural-cockoff baseline for this fragment spacing.

The natural-cockoff baseline puts the boost choice of the boosted deck in context. The asymmetric twin hot-spot boost of §8.3 is what extends the accessible regime from natural-cockoff transition propagation ($|D| \sim 1.4$ km/s, this section) up through the CJ-class detonation regime ($|D| = 8.05$ km/s, §8.3.1) for the artificially elevated initial state benchmarked in §8.5. The boost was necessary to let the boosted deck exercise the BN model’s upper-bound detonation regime against an idealized hot, partially compacted initial state. Natural cockoff of the same geometry, without boost, cannot reach that regime — at least under the current surface-burn model, which uses a Heaviside switch on T_{ign} . Replacing that switch with Arrhenius kinetics is identified as future work.

8.7. Interpretation: hybrid initiation regime

The twin-fragment-impact case occupies a hybrid position in the DDT/SDT initiation-regime space introduced in §1.3.1. The heat conducting inward from the steel fragments is a thermal component that XCHEM resolves correctly, producing a graded preconditioning of the HMX bulk and the small natural asymmetry between interfaces. The compaction-driven hot-spot formation at each steel/HMX interface — which a 1D thermal code cannot resolve — is a shock-like component captured here through the asymmetric twin hot-spot boost in the XDDT phase. Neither XCHEM-only (which would miss the compaction-driven hot-spot seeding) nor XDDT-only (which would miss the thermal preconditioning of the bulk) is sufficient.

The DDT outcome differs qualitatively from the single-fragment slab case of Chapter 6 and from the converging-wave cylinder of Chapter 7. Where the cylinder DDT emerges from a single inward-propagating reactive wave amplifying through geometric focusing, and where the single-fragment slab fails to engage the surface-burn kinetics at all (contact-face ignition without bulk preheating), the twin-fragment-impact case fires the BN-DDT criterion at the collision of two opposing reactive waves. The classical intensity-amplification picture of converging-wave detonation transition is replaced by an interaction picture in which the DDT criterion fires at the moment two compaction fronts meet — a third regime, beyond the cookoff and SDT regimes that XTIV is built to span, but produced naturally as a consequence of the two-sided initial condition.

The headline detonation velocity — $D = +8.05$ km/s — is the strongest DDT outcome reported in this report's demonstration matrix. Its relationship to the standard CJ benchmarks deserves care. Comparing against the cold-granular CJ ($D_{\text{CJ}} = 6.36$ km/s at $T_0 = 300$ K, $\rho_0 = 1.33$ g/cc) would label the result as overdriven by $\sim 27\%$, but that comparison crosses initial states: the cold-CJ baseline applies to a bed that the boost-augmented configuration is no longer in. The TIGER hot-CJ bracketing of §8.5 places the observed $D = 8.05$ km/s squarely on the hot-CJ trend at the partial- compaction density the BN compaction wave produces ahead of the front, supporting the interpretation that the wave is running at the CJ ceiling for the bed it actually sees — not above it. The result is thermodynamically consistent rather than anomalous, but the initial state on which the consistency holds is itself an idealization.

The numerical fragility of the boosted configuration, documented in §8.4, suggests that the boost-augmented result should be interpreted with care: small changes in the boost intensity, the smoothing length, or the XCHEM termination criterion materially affect the outcome, and the boosted deck as written is best understood as a demonstration of the qualitative DDT mechanism rather than as a calibrated quantitative prediction. The natural-cookoff fragment-spacing sweep of §8.6 sharpens that reading: across fragment spacings of 1, 2, and 4 cm the same geometry produces solver failure, transition-class compaction-then-collision propagation, and sustained inward propagation respectively, with no case reaching the detonation regime the boosted deck accesses. Together with the natural-cookoff demonstrations of Chapter 6 (single-fragment fizzle) and Chapter 7 (slow cook-off transition-territory propagation), the fragment-spacing sweep provides the bracketing physically realizable cases for this geometry; the boosted twin-fragment configuration here probes the upper-bound CJ-class regime that none of those natural-cookoff cases access. Future work directions for tightening the configuration are outlined in §10.

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9. METHODOLOGY FINDINGS

9.1. Regime structure across the demonstration matrix

The demonstration configurations of this report — the single-fragment natural-cockoff slab (§6.1), the slow cook-off cylinder (Chapter 7), and the twin-fragment impact case (Chapter 8) — establish a regime structure governed by the extent of bulk preheating in the bed at the moment of ignition.

Table 9-1. Coupled XCHEM–XDDT regime structure across four demonstration cases. Bulk preheating extent governs whether the BN model produces a CJ-class detonation, transition-territory propagation, or fizzle.

Case	Preheat extent	Geometry	Surface burn	D (km/s)	$P_{\max}$ (MPa)
Slab, single-fragment natural	none, contact-face only	flat	no	−2.10	143
Cylinder slow cook-off	partial, outer wall	$1/r$	yes	−5.386	15619
Twin-fragment, 4 cm gap, natural	two-sided, weak (natural)	flat	yes	+1.43	502
Twin-fragment impact (boost)	two-sided, twin (imposed)	flat	yes	+8.05	2014

The four cases span the regime structure from a stalled, essentially non-propagating response (single-fragment fizzle, no surface burn) through transition propagation (~ 22 –85% of cold-granular D_{CJ} , spanned by the cylinder slow-cockoff and the natural-cockoff twin-fragment case) to a CJ-class detonation at the hot-and-compacted initial state produced by the imposed twin-fragment boost. The single-fragment slab demonstrates the lower-bound regime directly: contact-face ignition in a cold bulk produces only compaction-driven activity pinned at the steel/HMX interface; the BN surface-burn kinetic switch never arms, the bed beyond the contact face never sees the reaction, and the integration terminates after $\sim 4 \mu\text{s}$ with $D \approx -2 \text{ km/s}$ and $P_{\max}$ two orders of magnitude below CJ. The slow cookoff cylinder produces an intermediate state: the ignition is still single-cell-class, but the surrounding HMX bed has been preheated by the AOP-4382 wall ramp (outer wall at $\sim 500 \text{ K}$ at the moment of ignition), supporting surface-burn engagement once the wave traverses the preheated bed. The natural-cockoff twin-fragment case probes the laminate counterpart of the slab geometry without boost: laminate steel-HMX-steel preheating at 540 K , 4 cm HMX bed, no imposed hot-spot patch. The result lands in the same transition band as the cylinder ($|D| = 1.43 \text{ km/s}$, $P_{\max} = 502 \text{ MPa}$, §8.6), establishing that even two-sided cookoff in a thin reactive bed produces transition propagation rather than detonation when the initial state is the natural-cockoff handoff. The twin-fragment impact case with boost extends the regime spectrum further: imposed twin hot-spot patches preheat the bed from both sides above what natural cookoff produces, the resulting reaction bands collide near mid-bed, and the BN-DDT criterion fires at the front-on-front collision, producing a CJ-class detonation in the artificially preheated initial state (see §8.5 for the hot-bed CJ benchmark analysis).

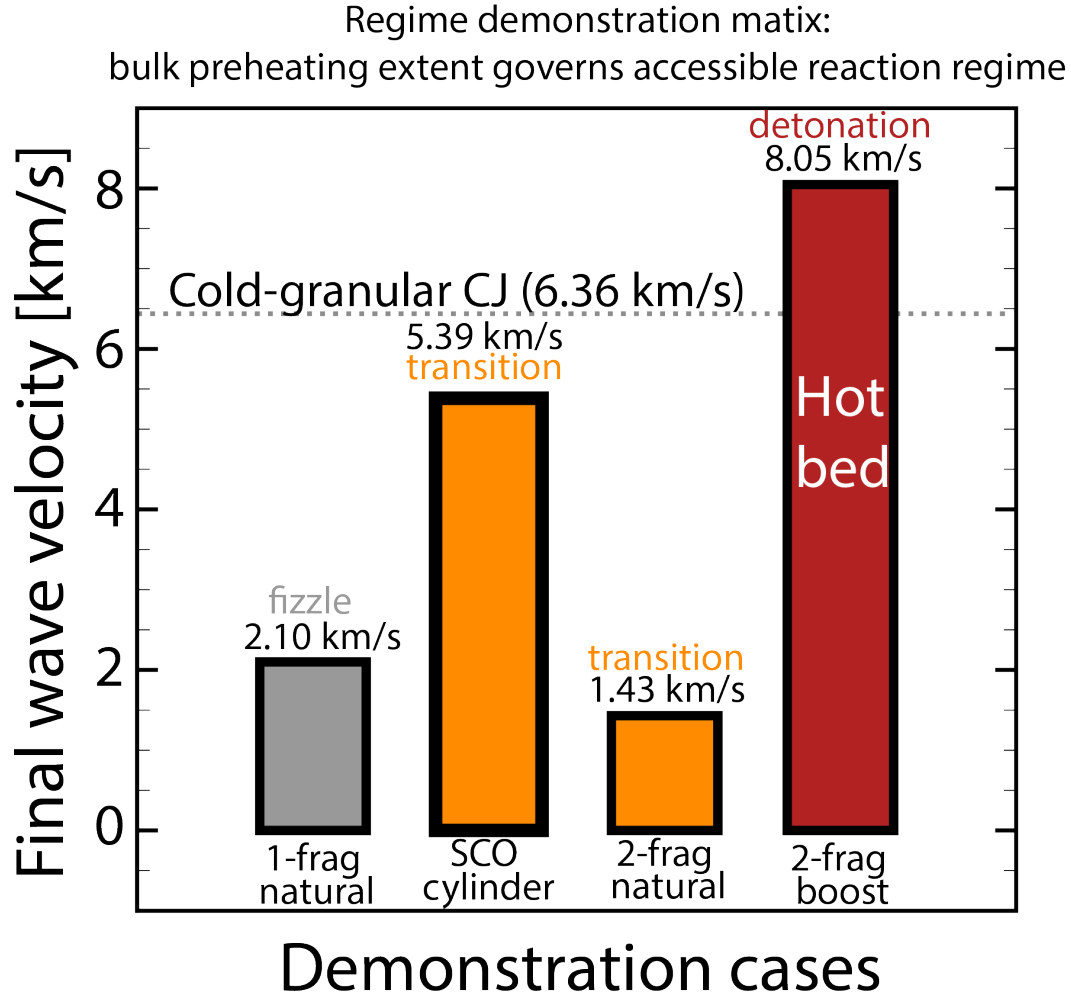


Figure 9-1. Regime sweep across the four demonstration cases of Table 9-1, showing the predicted detonation (burn) velocity $|D|$ as a function of regime classification. Bars are colored by regime: fizzle (gray, slab single-fragment natural), transition (orange, cylinder slow cook-off), and CJ-class for the boost-imposed initial state (firebrick, twin-fragment impact). The cold-granular CJ velocity ($D_{CJ} = 6.36$ km/s at $T_0 = 300$ K, $\rho_0 = 1.33$ g/cc, dashed) is the TIGER reference for the natural-cookoff cases (slab and cylinder) and sits above both transition outcomes. The hot-CJ velocity for the twin-fragment-impact boosted initial state ($\rho \approx 1.54$ g/cc, $\phi_s \approx 0.81$ of TMD; firebrick segment) corresponds with the boosted result determined by imposing an elevated initial condition. This is not an overdriven detonation (see §8.5). The only case that did not engage surface burn was the 1-frag natural case.

9.2. Bulk-preheating extent as the regime parameter

Two competing reaction mechanisms govern wave fate after ignition. The compaction-driven burn term (a1star) sustains reaction at low velocity in cold bed but cannot reach DDT alone. The pressure-dependent surface-burn term ($a_2 (P/P_{\text{ref}})^{bn} H$) requires the local mixture temperature to exceed T_{ign} , which typically obtains only in preheated bed — or, in pure SDT scenarios, behind a shock front strong enough that mechanical heating drives T_{mix} above T_{ign} . Reaching DDT requires surface-burn engagement; the question for any given coupled scenario is whether the bed configuration already supplies the threshold temperature ahead of the wave, or whether the wave must generate it through its own mechanical work.

In natural cookoff handoffs the latter path is closed. The ignition-region pressures produced by realistic cookoff are not strong enough to mechanically drive T_{mix} above T_{ign} in cold downstream bed. A cold-bed natural- cookoff wave therefore propagates only as long as compaction- driven burn sustains it, which is to say: as a slow deflagration that decays once the source-region energy depletes. This is the mechanism behind the fizzle outcome of the single-fragment slab in §6.1. In preheated-bed scenarios, by contrast, the bed itself supplies the threshold temperature; the surface-burn term engages in the preheated band, the wave amplifies via compaction-burn feedback (and via geometric focusing in cylindrical or spherical geometries), and the transition-territory or detonation-class regimes become accessible.

9.2.1. Fragment count and bulk preheating extent

The bulk-preheating-extent regime parameter has a direct implication for the fragment-impact problem. A single hot fragment in contact with cold reactive material lacks the thermal mass to preheat the bulk through T_{ign} over more than a millimeter-scale skin (§6.4); the local ignition that results is single-cell-class, the surrounding bulk is too cold to support sustained surface-burn kinetics, and the wave never establishes outward propagation at all. This is the regime exemplified by the single-fragment slab demonstration of §6.1, which produces $D \approx -2$ km/s (burn-front activity pinned at the steel/HMX interface, no propagating wave, no surface-burn engagement) — the fizzle outcome.

Multiple fragments, by contrast, can preheat the bulk from both sides. Their thermal influence overlaps in the bed interior and provides the multi-centimeter preheated extent that the BN model's compaction-burn coupling requires to engage the surface-burn kinetics across enough bed length for sustained DDT-class propagation. The twin-fragment-impact case (§8, $D \approx 8$ km/s) demonstrates this directly: two converging hot fragments produce sufficient two-sided preheating that the BN model fires the DDT criterion at the front-on-front collision near mid-bed, yielding detonation-class velocity at or above D_{CJ} .

The contrast between the two cases reveals a regime parameter that connects fragment-impact analysis to the bulk-preheating- extent regime structure: *fragment count and coordination control bulk preheating extent*, which in turn controls the accessible reaction regime in the BN model. Isolated fragments deliver localized ignition without bulk preheating and produce benign deflagration outcomes; coordinated fragment groups delivering simultaneous two-sided or multi-source thermal

loading can preheat the bed sufficiently to support DDT-class propagation. The slow-cockoff cylinder (§7) provides a third path to bulk preheating — via slow thermal diffusion from a heated outer boundary over hours — producing transition-territory propagation in the same regime structure.

Methodology contribution. Characterizing the BN model’s response to realistic ignition scenarios requires resolving bulk preheating extent, not just the local ignition spike. Coupled cookoff-to-DDT modeling that handles only the ignition site and not the surrounding bed thermal state cannot distinguish weak-deflagration cases from transition-territory or detonation-class cases, because the distinguishing physics lives in the preheated HMX bed (or absence thereof) rather than at the ignition kernel.

9.3. Implications for IM safety analysis

A scope reminder before drawing implications. The four demonstrations are deliberately exercised against a *sensitive HMX* fill, not an IM-qualified composition. This is a methodology choice: the pipeline cannot be tested against the violent-response regime if the fill never reaches that regime under the imposed conditions. Sensitive HMX in realistic geometries under standardized AOP-4382 heating guarantees that the XCHEM–XDDT chain is exercised end-to-end against post-ignition wave dynamics, where the modeling capabilities reported here are needed. The implications below are about the *modeling capabilities* the pipeline brings to IM safety analysis, not about IM compliance of the specific demonstration configurations.

The regime structure of §9.1 carries direct implications for IM safety analysis. Two are worth calling out explicitly.

First, the natural-cockoff result for the 18-inch sensitive-HMX cylinder under NATO AOP-4382 conditions shows transition-territory propagation rather than full detonation, despite the deliberately sensitive fill. That outcome — transition-class response on a configuration chosen to favor the violent-response regime — demonstrates that the pipeline can resolve the margin between transition-territory and detonation-class outcomes without conflating them, which is the property an IM-design model needs. An IM-qualified fill in the same geometry would be expected to land further down the regime spectrum (deflagration or weak transition), and the pipeline’s ability to track that shift quantitatively is the capability IM design depends on. The design lever the regime parameter exposes is the preheat penetration ratio (the fraction of bed depth preheated through T_{ign} at the moment of ignition): reducing bed size relative to cookoff timescale, or extending cookoff timescale through case-design choices that slow heat conduction inward, both shift the response toward DDT. Conversely, design choices that bound the preheated bed extent (insulating layers, rapid case rupture, smaller charges) preserve cold-bed bulk and therefore the deflagration-class margin.

Second, the fragment-count regime structure of §9.2.1 is directly applicable to fragment-impact threat scenarios. Incident fragments arriving isolated and asynchronously to a reactive bed deliver localized ignition without bulk preheating, producing benign deflagration-class outcomes. Coherent fragment groups arriving simultaneously at multiple bed surfaces can preheat the bed sufficiently to support DDT-class propagation. This distinction connects fragment-impact analysis to the bulk-preheating-extent regime parameter and provides a principled basis for evaluating

fragment-threat severity in IM design analysis: the relevant figure of merit is not the peak fragment temperature in isolation but the *spatial coherence and simultaneity* of multiple incident fragments and their resulting cumulative preheating extent in the bed at the moment of ignition.

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10. LIMITATIONS AND FUTURE WORK

The pipeline as documented in this report exercises the XCHEM–XDDT methodology end-to-end on the demonstration matrix of Chapters 6 through 8, but several issues surfaced during the current development that the present work cannot close. Each is described below: its current state, its impact on the present results, and the path to resolution. The list is enumerated rather than exhaustive — these are the items that emerged with enough clarity during demonstration runs to be worth recording as bounded future work, not a comprehensive audit of every known shortcoming.

10.1. Outer-boundary-condition implementation

Current state. XDDT’s outer boundary condition on the IBC=3 path currently falls back to a Neumann (zero-gradient, transmissive) treatment for all non-trivial confinement strings carried in the `.xtiv` transfer file. The `free`, `wall`, and `layer:X` options that the schema allows are all reduced to the same Neumann implementation in the present build; only one outer-BC type is genuinely available. The same dispatcher serves the inner edge for hollow cylindrical and spherical configurations (§2.3), so the `layer:X` Neumann fallback applies identically there: hollow annular or shell beds run today with free (Neumann) inner and outer surfaces, with finite-impedance case material deferred to the same impedance-match work described below.

Impact on present results. The Neumann fallback lets reaction products vent through the outer boundary in every demonstration. The effect is most visible in the slow cook-off cylinder of Chapter 7, where the inward-propagating wave’s late-time deceleration is partially attributable to the open outer boundary: a true confining case would maintain pressure behind the wave longer and likely extend transition-territory behavior further toward detonation.

Path to resolution. Add a second outer-BC type — an *impedance-matched* condition that couples the reactive bed to a finite-impedance confining layer through reflection and transmission coefficients computed from the confining material’s acoustic properties. This is the boundary condition needed for steel-jacketed warhead-class geometries where the case is neither rigid (perfect reflector) nor transparent (the Neumann fallback’s effective behavior); it would also subsume the rigid-wall case as the infinite-impedance limit. The work is bounded scope — adding the impedance-match branch to the existing BC dispatcher — but requires care in the dispatch logic to preserve backward compatibility with the Neumann path exercised by the current demonstrations.

10.2. One-dimensional cylindrical and spherical symmetry as a modeling idealization

Current state. The XTIV pipeline implements one-dimensional cookoff-to-DDT analysis in three geometry options matching the pre-ignition XCHEM solver: slab, cylinder, and sphere. In the cylindrical and spherical cases the 1D treatment implicitly assumes ignition propagates uniformly along the symmetry coordinate — i.e., a ring of finite radius for the cylindrical case or a shell of finite radius for the spherical case ignites simultaneously, then drives a 1D-radial reactive wave inward toward the geometric center. This is the standard idealization underlying 1D cookoff modeling generally, but it is worth acknowledging explicitly that simultaneous ring/shell ignition over a multi-centimeter radius is physically unrealizable for natural-cookoff scenarios in practice; in the real world, ignition occurs at a point (or a small cluster of points), not over an extended ring or shell of perfect simultaneity.

Impact on present results. The 1D symmetry assumption affects what claims the cylinder demonstration of Chapter 7 can support. The inward-propagating converging-wave amplification observed there is a consequence of the imposed cylindrical symmetry, not a prediction of what a physically asymmetric cookoff event would produce. In a real cookoff scenario where ignition is localized to a single point or a small cluster of points on or near the outer wall, the early-time wave structure would be predominantly outward-expansion-driven (point-source giving an expanding-shock geometry) rather than converging-inward, and the $1/r$ geometric-focusing amplification mechanism the 1D analysis exposes would be attenuated to whatever extent the real event departs from the symmetric ring/shell idealization. Asymmetry is intrinsic to the physical problem. Even in the nominally symmetric slab geometries of this report the natural-cookoff XCHEM phase produces non-trivial asymmetry between the two steel/HMX interfaces (§8.2), so the symmetric configurations enforced by the 1D cylindrical and spherical solvers are stricter idealizations still.

Where the idealization is closer to reality. Two cases worth identifying narrow the gap between the 1D idealization and the physical problem. First, in external-heating cookoff tests such as the One-Dimensional Time-to-eXplosion (ODTX) sphere (§10.3), where the outer-wall temperature is held uniform around the entire surface, uniform external heating delivers near-uniform shell-surface temperatures and shell-simultaneous ignition is approximately what the test is designed to produce; the 1D shell idealization is, in this case, close to the physical reality the test embodies. Second, point-source ignition events — such as a compaction-driven hot spot in the interior of the bed — propagate *outward* in approximately spherical fashion, which is captured cleanly by the 1D spherical solver if the hot spot is treated as a centered initial condition. The configurations where 1D symmetry is most idealized are those where ignition is at the outer boundary *and* the natural physical scenario involves localized rather than distributed initiation; the slow cook-off cylinder of Chapter 7 falls in this category.

Why the idealization is retained. Two practical considerations support the 1D approach in the present work. First, the full Baer-Nunziato two-phase model with the BN compaction-burn coupling has no published 3D detonation implementation; even 2D axisymmetric integration of the full BN model in the production-grade hydrocode CTH was attempted during this work and could not be brought to a converged result. The 1D analysis is therefore the practical state of the art for end-to-end demonstration of the BN-coupled cookoff-DDT methodology, not a deliberate restriction. Second,

the quantitative effects the 1D demonstrations expose — geometric acceleration, inward-front amplification, collision-frame pressure spikes — are real physical mechanisms that are present and observable even in geometries that depart from strict 1D symmetry. The 1D analysis identifies the mechanism; multi-dimensional work would calibrate the magnitude. Hot fragments themselves are not spherical either, so the geometric idealization in the present work is one approximation among many that the broader cookoff modeling program accepts in service of end-to-end demonstration.

Transverse confinement is absent by construction. A second consequence of the 1D treatment, related to but distinct from the ring/shell ignition idealization above, is the implicit treatment of confinement. In physical 2D or 3D charges, whether a deflagration grows into a steady detonation depends on confinement against *lateral* pressure relief: case-wall strength, hoop stress, and the geometry of any vent or unconfined surface perpendicular to the wave direction. A charge with weak lateral confinement can fail to support DDT even when bulk kinetics and porosity are otherwise favorable. In the 1D treatment used here there is no transverse coordinate along which products can expand, so lateral confinement is implicitly perfect; the only confinement degree of freedom is the outer (or inner) boundary condition discussed in §10.1, which represents wall reflection along the symmetry direction only. The 1D model therefore tends to favor wave growth and propagation relative to a real charge with a finite-strength case and finite lateral inertia. Quantitative differences between simulated wave speeds, transition times, or peak pressures and a physical analog experiment can be due in part to this missing lateral-relief channel rather than to errors in the bulk kinetics or constitutive parameters; the 1D demonstrations in this report should therefore be read as upper-bound estimates of geometric-focusing and converging-wave amplification under idealized lateral confinement, not as predictions for laterally-vented configurations.

Path to resolution. A multi-dimensional implementation of the full BN compaction-burn coupling, with the equation-stiffness and grid-resolution constraints of the 1D solver carried into a 2D or 3D framework, is the proper resolution to the 1D symmetry idealization. The present work establishes that the 1D path is itself sufficiently constrained by equation stiffness and grid resolution to leave clear room for methodology development before multi-dimensional work becomes tractable, but the multi-dimensional generalization remains the right long-term direction for the BN-coupled cookoff-DDT methodology. Until that work is available, the 1D cylindrical and spherical results in this report should be read as identifying the geometric-focusing and converging-wave mechanisms in the model rather than as quantitative predictions of physical events with locally varying ignition.

10.3. Spherical-geometry stability and the ODTX handoff window

A natural extension of the demonstration matrix in this report would be a spherical-geometry case covering the One-Dimensional Time-to-eXplosion (ODTX) test—a standard IM-relevant scenario consisting of a small (~0.25-inch radius) HMX core inside an aluminum jacket held at a fixed boundary temperature, with thermal runaway expected on a timescale of seconds to minutes. Sphere geometry is also the strongest available test of geometric DDT amplification: the $1/r^2$ source term concentrates inward-propagating waves more aggressively than the $1/r$ source of the slow-cook-off cylinder demonstrated in Chapter 7. The ODTX test was attempted during the current

development and produced three distinct, instructive limitations of the present XTIV pipeline that together prevent a clean sphere demonstration in this report.

Sub-zone XCHEM runaway. HMX thermal runaway in ODTX is intrinsically explosive: the kinetic chain reaction has its own timescale (set by the chemistry, not the heating rate), and once the kinetic threshold is crossed locally, the temperature transitions from the soak plateau (~ 550 K) to the post-runaway state (~ 3000 K) in tens of milliseconds. Three XCHEM cookoff runs were attempted at outer-wall boundary temperatures of 555.55 K (standard 540°F), 500 K, and 475 K, with progressively denser zone-output cadences (0.1 s intervals at the slowest case). In every case the transition through the SAND-2 / SCO-equivalent handoff window (~ 800 – 1200 K) compressed into a *single* consecutive zone-output snapshot, leaving no intermediate state available for extraction. The runaway is genuinely sub-zone at any practical XCHEM print cadence—only the slow-soak phase ($T < 600$ K) and the post-runaway state ($T > 3000$ K) are resolvable, with no zones in between.

Stop-at- T workaround. The runaway-window problem can be sidestepped by lowering the XCHEM solver’s runaway threshold (the `tmax` parameter in `tchem.in`), causing XCHEM to halt as soon as any cell reaches a chosen handoff temperature. Setting `tmax` = 800 K produced a clean handoff state at $T_{\text{max}} = 957$ K (overshoot is set by the integrator catching the threshold within a step) on the 475 K outer-wall ODTX run, with the hot region distributed across cells 1–6 of the HMX bed. This is a usable methodology recipe for any future fast-cookoff scenario: *set XCHEM tmax to the desired handoff temperature rather than the kinetic-chain-completion threshold*, and XCHEM will halt at the onset of runaway.

Smoothing-convention mismatch for cold-bulk handoffs. The handoff produced by the stop-at- T workaround is the problematic kind of state for the XDDT integrator. Unlike the SCO cylinder—where slow soak warms the entire bed roughly uniformly and the handoff peak is just a single-cell spike above an already warm bulk—the ODTX sphere has a sharp *multi-cell* hot shell ($T = 957$ K at cell 1, decreasing through 951, 929, 871, 787, 706 K at cells 2–6) abutting a still-cold bulk ($T \approx 530$ K in the deep interior). The current handoff-state smoothing knob (§5) was designed for the SCO-cylinder regime, where the inner-edge “feature” is one cell wide and the smoothing target is a warm bulk. Applied to the ODTX configuration, the tanh ramp pulls the entire multi-cell hot shell toward the cold-bulk target $T \approx 537$ K, erasing the only reactive feature: with `xtiv_smooth_l_in` = 0.0001 m (0.1 mm, ~ 2 cells wide), the post-smoothing state has $T \leq 686$ K throughout the bed—below any reasonable T_{ign} , so the surface-burn switch fires nowhere and no reaction develops.

Integrator failure on the unsmoothed handoff. Setting the smoothing length `xtiv_smooth_l_in` to zero (smoothing disabled) preserves the multi-cell hot shell but exposes the underlying issue: the gradient at the hot-face / cold-bulk transition is too sharp for the BN integrator in spherical geometry. Tracing the $t = 0.1$ μs output snapshot shows clean integration through the cold portion of the bed ($x = 0$ to $x \approx 0.62$ cm, with $T_g \approx 475$ – 680 K and burn rate identically

zero), then numerical chaos at the hot face ($x \approx 0.63$ cm onward): gas temperature pinned at the TIGER table cap of 6000 K, solid temperature exceeding 10^4 K, mixture pressure overflowing the format width, and the local burn rate growing from zero through -1.3×10^4 to -2.1×10^{25} kg/m³/s in three consecutive grid cells. The integrator catches the divergence and halts with IDID = -6 at $t = 61$ ns. Schema 2 handoff (§3.3.1) and adaptive mesh refinement (`imesh_in` = 1) are both enabled in this run; neither mitigates the failure mode.

Path forward. The sphere demonstration is therefore deferred to future work, with three concrete enhancement directions identified by the present exploration:

1. *Smoothing-convention generalization for cold-bulk handoffs.* The smoothing routine should detect the multi-cell extent of the handoff hot region (rather than assuming a single-cell spike) and choose its smoothing target from a node *just inside* the hot region’s trailing edge, not from the deep-bed bulk. Bounded scope; replaces a few lines in `xtiv_glue.f90`.
2. *Event-driven XCHEM zone output.* Augment the time-cadence-based zone print with an event trigger that forces a zone whenever local T crosses configurable thresholds (e.g., 600 K, 800 K, 1200 K, 2000 K). The `tmax` stop-at- T recipe described above is a special case of this; full event-driven output would capture the full runaway transition for diagnostic purposes even when stop-at- T isn’t desired.
3. *Distributed activation-energy (DAE) rate replacement (§10.5).* A smooth Arrhenius/DAE rate in place of the empirical $rP^n H$ form would soften the kinetic chain reaction itself, making runaway less explosive in XCHEM and the post-handoff dynamics less susceptible to integrator instabilities at the hot face in XDDT. This is the most fundamental of the three paths because it addresses the root cause—a binary burn rate that turns on at a threshold—rather than its downstream symptoms.

The combination of these three enhancements is expected to make the ODTX sphere case a feasible demonstration in the next report of the DDT series.

10.4. Direction-aware diagnostics

Current state. Three diagnostic subsystems — the compaction-front tracker, the BN-DDT criterion, and the bed-end halt check — were originally implemented under an outward-propagation assumption inherited from the piston-driven cases of SAND-1 and SAND-2. That assumption fails for cookoff-driven cases, where ignition is at the outer edge and waves propagate inward. The current work added a propagation-direction signal computed from the recent burn-front history, and rewired all three diagnostics to consult it. The direction-aware redesign was prerequisite to interpreting the slow cook-off cylinder of Chapter 7 cleanly.

Remaining limitation. The direction-aware diagnostics handle single-direction propagation (outward or inward) correctly, but they do not natively handle the two-front converging-wave case the twin-fragment demonstration of Chapter 8 exercises. In a symmetric two-front collision, neither front has a clear “leading” role, and the single-direction DDT criterion (“burn front overtakes the

compaction-front-at-ignition position from one side”) cannot fire on the collision event even when the field data shows the violent pressure spike that defines DDT. Chapter 8 works around this with an asymmetric twin hot-spot boost that gives the criterion a designated leading front, but this is a diagnostic workaround rather than a clean physical generalization.

Path to resolution. Generalize the DDT criterion to handle two-front converging-wave geometries natively: detect that two propagation-direction signals coexist (one positive, one negative), and reformulate the overtaking test as a collision-frame test rather than a single-side test. The bed-end halt check would benefit from the same generalization, terminating only when a front reaches the boundary it is actually propagating toward (rather than firing on the originating-end reaction of either side). Bounded scope; the propagation-direction infrastructure already exists, only the comparator logic in the three diagnostics needs to be extended.

10.5. True Arrhenius and distributed activation energy kinetics

The pressure-dependent surface-burn rate $a_2(P/P_{\text{ref}})^{b_n} H$ used in the BN compaction equation (eq. 1.1 of §1.3.2) is an empirical $r = aP^n$ form commonly used to fit solid-propellant strand-burn data [15], turned on by a Heaviside step at a single ignition threshold T_{ign} . Although it is sometimes referred to as the “Arrhenius rate” in BN literature, it is not a true Arrhenius rate in the chemical-kinetics sense—it lacks the explicit activation energy E_a and pre-exponential factor A that a chemistry-derived rate would carry, and the temperature dependence is reduced to a hard threshold rather than a smooth $\exp(-E_a/RT)$ activation.

The path forward, motivated by the Universal Cookoff Model (UCM) [4, 19, 20] that already drives the XCHEM prethermal-ignition phase of the present pipeline, is to replace the empirical $rP^n H$ *surface-burn* form with the distributed activation-energy (DAE) rate expression used in UCM:

$$r_x = \frac{d[X]}{dt} = A \left(\frac{P}{P_o} \right)^n \lambda(T) T^m \exp \left[-\frac{E + \xi \sigma}{T} \right] [X], \quad (10.1)$$

where $[X]/[X_o]$ is the remaining-reactant fraction, $\xi = \text{NORM.INV}([X]/[X_o], 0, 1)$ is the inverse standard-normal cumulative distribution function of the reaction progress (running from $\xi \approx +3$ at the pristine start to $\xi \approx -3$ at full consumption), E is the mean activation energy in K, σ sets the breadth of the activation-energy distribution, T^m is a temperature pre-multiplier, $\lambda(T)$ is a melting-acceleration factor (a cosine ramp over the melt mush zone, typically peaking at $10\text{--}20 \times$ the unmolten rate), and $(P/P_o)^n$ is a pressure multiplier of the same Saint-Robert form used in the current BN surface-burn term.

The structural reach of this single rate expression is what makes it the natural replacement. The activation-energy modulation $E + \xi \sigma$ shifts the effective barrier with reaction progress: $\sigma > 0$ gives a higher barrier at the start that decreases with consumption (autocatalytic kinetics, rate accelerating as products accumulate; characteristic of HMX and other ideal explosives), while $\sigma < 0$ gives a lower barrier at the start that rises with consumption (diffusion-limited kinetics, rate decelerating as a barrier layer builds; characteristic of ammonium nitrate and other non-ideal explosives). A single calibrated expression therefore spans the full kinetic regime structure from diffusion-limited

through single-Arrhenius (recovered when $\sigma = 0$) to fully autocatalytic, with the smooth thermal dependence of $\exp(-(E + \xi\sigma)/T)$ providing the orders-of-magnitude temperature range the BN $rP^n H$ form lacks entirely above T_{ign} .

The compaction-burn term in equation (1.1) is a separate physical mechanism—Lee–Tarver hot-spot generation under compaction work [13], calibrated against the Sandusky–Bernecker compaction-induced-combustion experiments [17]—that need not be replaced with a thermal-kinetic form. A unified DAE framework could treat both the thermal-driven and mechanical-driven pathways with separate activation-energy distributions, but the surface-burn replacement described here is the first and most urgent step.

Two motivations support this enhancement. First, it unifies the kinetics representation between the cookoff (XCHEM, DAE-based) and the post-ignition (XDDT, currently rP^n with Heaviside gate) phases of the XTIV pipeline, eliminating the present discontinuity at the handoff frame where the kinetic representation changes mid-simulation. Second, the DAE parameter space resolves the calibration overload identified in §1.3.2—where the present two-parameter a_2, b_n surface-burn form must absorb physics it does not structurally represent, leading to parameters that seldom transfer between scenarios. The DAE form can match multiple regimes simultaneously (strand burn, piston-driven DDT, slow cookoff, fragment impact) within a single calibrated rate expression, and it generalizes naturally across explosive material classes from ideal (HMX-like) to non-ideal (AN-like).

Path: Implementation of the DAE surface-burn rate as a drop-in replacement for the $rP^n H$ term in the XDDT compaction equation; recalibration against the existing SAND-1/SAND-2 validation matrix (McAfee piston-driven DDT [11], Price–Bernecker thermal ignition [12], Sandusky compaction [17]) to verify the new form matches established quantitative results before extending to broader data including non-ideal materials. The Los Alamos Gapstick Test [21] provides recent rate-stick measurements that would be an interesting additional calibration target for the DAE rate, complementary to the legacy validation matrix above.

Calibrating against the Gapstick Test would, however, require two prerequisite code-development tasks beyond the rate replacement itself. First, the present single-reactive-layer XDDT geometry would have to be extended to a multilayer configuration with alternating reactive and inert layers. Second, the inert layers would need to carry a propagating pressure wave that transmits detonation from one reactive segment across the gap into the next; the steel layers of Chapter 8 were treated as thermal reservoirs and rigid boundaries rather than as hydrodynamic media carrying a shock, so the XDDT solver’s ability to propagate a shock through an inert layer would have to be verified or developed. Both extensions are non-trivial but doable, and naturally fall within the scope of a successor SAND report.

The development, calibration, and validation of the DAE-based surface-burn rate—and the broader question of unifying the BN post-ignition kinetics with UCM’s prethermal-ignition framework—will be the subject of a future SAND report in this series.

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11. CONCLUSIONS

The XTIV pipeline couples the XCHEM prethermal-ignition code to the XDDT post-ignition compressible solver, enabling end-to-end simulation of cook-off-driven violent reaction in granular explosives. The pipeline architecture (§3), diagnostic redesigns (§4), and handoff-state smoothing knob (§5) collectively make natural cook-off ignition states tractable for XDDT integration without the imposed hot-spot patches that earlier work would have required.

11.1. Outlook for the next report in the series

The methodology development reported here suggests several lines of follow-on work. Foremost among them is the replacement of the empirical pressure-dependent surface-burn term $a_2(P_g/P_{\text{ref}})^{b_n}H$ in the BN compaction equation with the distributed activation-energy (DAE) rate of the Universal Cookoff Model [4, 19, 20], as proposed in §10.5. This enhancement would unify the kinetics representation across the cookoff (XCHEM, DAE-based) and the post-ignition (XDDT, currently rP^n with Heaviside gate) phases of the XTIV pipeline, eliminate the calibration-overload limitation identified in §1.3.2, and—perhaps most consequentially—extend the BN methodology beyond pure HMX in two directions. First, to HMX-based composite formulations such as EDC-37, LX-07, LX-10, LX-14, PBX-9501, and PBXN-9 [20], where the present single-rate-with-binary-thermal-switch form is poorly suited to the multi-stage decomposition kinetics of multi-component reactive systems and where calibration of a_2 and b_n against pure-HMX strand data does not transfer cleanly. Second, to non-ideal energetic materials such as ammonium nitrate (AN) and AN-based explosives, where diffusion-limited kinetics ($\sigma < 0$ in the DAE form) dominate the reaction progression and the BN rP^nH form—which has no mechanism for rate deceleration with reaction progress—cannot represent the regime at all.

Companion items for the same future report include direction-aware diagnostics for inward-propagating reactive waves (§10.4), implementation of an impedance-matched outer-boundary-condition in place of the present Neumann fallback (§10.1), and multi-microsecond stability of the spherical-geometry integration (§10.3). The DAE-based kinetics extension would position the BN methodology to address the broader insensitive munitions, transportation, and storage scenarios encountered in real-world safety analysis, beyond the pure-HMX cookoff regime that motivates the present report. Together these items constitute the natural scope of the next report in the DDT series.

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APPENDIX A. XTIV TRANSFER-FILE SCHEMA

This appendix reproduces the file-format details needed to write a custom extractor or to read the transfer file from another program. Section 3.1 gave the high-level shape; the tables and formulas below complete the specification.

A.1. File conventions

- File extension: `.xtiv`.
- ASCII text, UTF-8, LF line endings.
- Sections delimited by `[section_name]` headers in column 1. Section order is fixed (Table A-1); reorderings are a parse error.
- Within a section, lines are either `key = value` pairs or a single data block (a header line of column names followed by N rows of whitespace-separated values).
- `!` starts a comment; everything from `!` to end-of-line is ignored.
- Blank lines are ignored.
- Numeric values use Fortran-style scientific notation (`1.234e-5`, `1.234E-5`, `0.234d-5` are all accepted).
- String values are unquoted unless they contain whitespace; quoted values use single quotes (e.g., `'layer:ALUM'`).
- All values are SI.

Table A-1. File section order.

Section	Contains
<code>[meta]</code>	Schema version, problem identity, handoff timing, build provenance
<code>[geometry]</code>	1D coordinate system and reactive-layer extent
<code>[material]</code>	Reactive-material constants for ρ_c reconstruction
<code>[confinement]</code>	Inner- and outer-edge boundary-condition selectors
<code>[grid]</code>	Per-node coordinates and refinement-rank metadata
<code>[fields]</code>	Per-node primitive state
<code>[end]</code>	Sentinel; XDDT verifies this is the last section

A.2. Section keys

Table A-2. Keys in [meta].

Key	Type	Meaning
schema_version	integer	1 or 2
problem_title	string	human-readable problem identifier
xchem_build	string	XCHEM git SHA at write time
xchem_input	string	basename of <code>tchem.in</code> used
extractor_build	string	extractor git SHA at write time
write_time	string	ISO-8601 UTC timestamp
t_src	real (s)	XCHEM time at the extracted zone
zone_index	integer	1-based index into <code>ofield.plt</code> zones
ignition_flag	boolean	T if XCHEM hit <code>tmaxx</code> by <code>t_src</code>
ignition_cause	string	one of 'tmaxx', 'solver_id_neg', 'final_time', 'time_trigger'
tmax_reached	real (K)	maximum T in reactive layer at <code>t_src</code>

Table A-3. Keys in [geometry].

Key	Type	Meaning
geom	integer	0 = slab, 1 = cylinder, 2 = sphere
pgeom	real	slab area (m^2), cylinder height (m), or 0 (sphere)
kexp_name	string	XCHEM layer name of the reactive layer
x0_exp	real (m)	inner edge of reactive layer
x1_exp	real (m)	outer edge of reactive layer
evol	real (m^3)	excess volume (tubing + ullage)
maxstrain	real	upper clip on total strain (run-derived)

Table A-4. Keys in [material]. Per-ramp triplets (`start_e_j`, `end_e_j`, `del_e_j`) appear only for $j = 1 \dots \text{ndv}$.

Key	Type	Meaning
name	string	reactive material identifier
rho_co	real (kg/m^3)	condensed density at <code>tlay</code> , no strain
bvol	real ($1/\text{K}$)	base volumetric thermal expansion coefficient
bslope	real ($1/\text{K}^2$)	linear T -slope of the expansion coefficient
tlay	real (K)	reference (layer initial) temperature
phi_o	real	initial gas volume fraction
ndv	integer	number of phase-change ramps (0, 1, or 2)
start_e_j	real (K)	start temperature of ramp j
end_e_j	real (K)	end temperature of ramp j
del_e_j	real	strain magnitude across ramp j

Table A-5. Keys in [confinement].

Key	Type	Meaning
inner_side	string	'centerline', 'free', or 'layer:<name>'
outer_side	string	'free' or 'layer:<name>'

A.3. Per-node blocks

Grid. The [grid] block begins with the key `n_nodes = N` followed by a header line `columns = i r_m ilev` and N data rows.

Table A-6. Columns in the [grid] data block.

Column	Type	Meaning
i	integer	1-based node index, monotonic
r_m	real (m)	physical position
ilev	integer	refinement rank (1 = coarsest)

Fields. The [fields] block begins with a header line `columns = ...` and one row per node, in the same order as [grid]. Columns differ by schema version.

Table A-7. Columns in the [fields] data block. Schema v1 carries `p_Pa`; schema v2 replaces it with `mass_fraction`. All other columns are identical between versions.

Schema	Column	Unit	Meaning
v1, v2	T_K	K	temperature
v1, v2	rho_g	kg/m ³	gas partial density (strained frame)
v1, v2	alpha_s	–	solid volume fraction (strained)
v1	p_Pa	Pa	gas pressure (cross-check only on v1)
v1, v2	S_f	–	unreacted-mass fraction
v2	mass_fraction	–	mass-balance regime indicator
v1, v2	rho_m_check	kg/m ³	extractor's ρ_c recomputation

The mass-balance closure formula for `mass_fraction` (schema v2 only) is given in equation (3.1) of §3.1.

A.4. Condensed-phase density reconstruction

XDDT does not consume a per-node ρ_m from the file. Instead, it reconstructs ρ_c at each node from the transferred temperature T , the [material] constants, and the run-derived `maxstrain` from [geometry]. The same formula is used by the extractor (writing `rho_m_check`) and by XDDT (reading and verifying):

$$\text{strain}(T) = (\beta_v + \beta_s T) (T - T_{\text{lay}}) + \sum_{j=1}^{n_{\text{dv}}} \Delta\epsilon_j f_j(T), \quad (\text{A.1})$$

$$f_j(T) = \begin{cases} 0 & T \leq T_{\text{start},j}, \\ \frac{1}{2} \left[1 - \cos\left(\frac{\pi(T - T_{\text{start},j})}{T_{\text{end},j} - T_{\text{start},j}}\right) \right] & T_{\text{start},j} < T < T_{\text{end},j}, \\ 1 & T \geq T_{\text{end},j}. \end{cases} \quad (\text{A.2})$$

The total strain is then clipped against `maxstrain`,

$$\text{strain}(T) \leftarrow \min(\text{strain}(T), \text{maxstrain}), \quad (\text{A.3})$$

and the condensed-phase density is reconstructed as

$$\rho_c(T) = \frac{\rho_{co}}{1 + \text{strain}(T)}. \quad (\text{A.4})$$

The mapping between symbols above and `[material]` keys is: $\beta_v \leftrightarrow \text{bvol}$, $\beta_s \leftrightarrow \text{bslope}$, $T_{\text{lay}} \leftrightarrow \text{tlay}$, $\rho_{co} \leftrightarrow \text{rho_co}$, $n_{\text{dv}} \leftrightarrow \text{ndv}$, $\Delta\epsilon_j \leftrightarrow \text{del_e_j}$, $T_{\text{start},j} \leftrightarrow \text{start_e_j}$, $T_{\text{end},j} \leftrightarrow \text{end_e_j}$. `maxstrain` comes from `[geometry]` and is derived inside the extractor as `evol/vem/(1-phi_o)`, where `vem` is the explosive-layer volume implied by `geom`, `pgeom`, `x0_exp`, and `x1_exp`.

For $T < T_{\text{lay}}$, the base term is negative (condensed phase compressed below ρ_{co}); no lower clamp is applied, matching XCHEM's behavior.

A.5. Schema versioning and error conditions

Versioning rules.

- `schema_version` is the first key in `[meta]`. XDDT checks this before parsing anything else.
- XDDT accepts `schema_version` $\in \{1, 2\}$. Any other value is fatal and is reported with a message naming both the file's version and the accepted set.
- Adding a new field to an existing section, adding a new section, or changing the meaning of an existing field all require a version bump. Removing or renaming a field requires a version bump.
- `v1` and `v2` are both locked once committed. Future schema work happens on `v3+`; both extractor and XDDT must be updated together.

Error conditions. XDDT aborts with a diagnostic message naming the file and the violated rule on:

- Missing or unreadable file.
- `schema_version` absent or not in {1, 2}.
- Section out of order or missing.
- `n_nodes` exceeds XDDT's compile-time maximum (NPT0, currently 4001).
- `[material]` name does not match XDDT's `mat.dat` reactive entry.
- `rho_m_check` mismatch greater than 1% at any node.
- `[fields]` row count differs from `n_nodes`.
- `[end]` marker missing.
- `geom` not in {0, 1, 2}.
- `inner_side` or `outer_side` malformed (unknown selector).
- (schema v2 only) `mass_fraction` mismatch greater than 1% at any non-pristine node.

XDDT logs (but does not abort on) soft warnings, including `ilev` inconsistencies and large extractor-vs-XDDT pressure deviations on the `v1` path.

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APPENDIX B. REPRODUCIBILITY, USER EXPERTISE, AND SHIPPED TEMPLATES

This appendix records the level of user expertise assumed by the XTIV pipeline as documented in this report, the principal numerical controls a user must exercise to reproduce the demonstrations of Chapters 6–8, and the template set shipped in the source-tree `templates/` directory that constitutes the present path toward a more “push-button” workflow. It was added in response to reviewer feedback noting that the detailed model and numerical insight required to run the present build would be unrealistic for a non-expert user — a fair point that deserves an explicit statement in the report.

B.1. Assumed user expertise

The pipeline as documented here is an expert tool. Producing stable, physically meaningful coupled XCHEMA–XDDT results currently requires working familiarity with: (i) finite-volume reactive-flow solvers and their failure modes (stiff source terms, integrator return codes, mesh–time-step coupling under adaptive refinement); (ii) the Baer–Nunziato two-phase closure and its calibration parameters (`a1star`, `a2`, `bn`, T_{ign}); (iii) the conventions of the underlying XCHEMA cookoff code (`tchem.in` namelists, layer/material strings, zone-output cadence, the `tmax` termination criterion); and (iv) the handoff-state smoothing convention introduced in §5 and its interaction with the chosen geometry and confinement. A user without this background is unlikely to diagnose the failure modes documented in Chapter 10—sub-zone XCHEMA runaway, smoothing-convention mismatches at cold-bulk handoffs, integrator divergence at sharp hot-face gradients—when they occur on a new scenario. This is a real reproducibility constraint, not a documentation gap: the underlying physics spans many orders of magnitude in spatial and temporal scale, and the numerics needed to follow it across that range do not yet collapse to a small set of defaults.

B.2. Key numerical controls and sensitivity steps

The headline results of this report depend on a small set of numerical controls whose appropriate settings are scenario-dependent and were established through the sensitivity exploration documented in the demonstration chapters. A user reproducing the demonstrations should expect to revisit, at minimum, the controls listed in Table B-1; deviations from the tabulated values should be made deliberately and verified against the reported diagnostics.

Table B-1. Principal numerical controls exercised to produce the demonstrations in Chapters 6–8. Each control has a scenario-dependent setting; the cross-reference column points to the section that documents the sensitivity exploration behind the chosen value.

Control	Role	See
XCHEM tmax (K)	termination temperature for the cookoff phase	§10.3, §3
XCHEM zone-output cadence	resolves the handoff window	§10.3
imesh_in (0/1)	fixed vs. adaptive XDDT mesh	§7, §8
xtiv_smooth_l_in	inner-edge handoff-state smoothing length	§5
xtiv_smooth_outer_l_in	outer-edge handoff-state smoothing length	§5
rel_in, ae_in	DDEBDF relative / absolute tolerances	§3
nstep_in	integration-step cap (run length)	§6, §8
ibc_in (=3 coupled)	outer boundary-condition selector	§10.1
Hot-spot boost (hot_spot_l/T/P_in)	imposed-ignition seed parameters	§8

The minimum sensitivity-check protocol used during the current development is to (i) confirm that halving `xtiv_smooth_l_in` does not shift the primary diagnostics (t_{ddt} , D_2 , P_{max}) outside their reported precision, (ii) repeat with `imesh_in` toggled where geometry permits, and (iii) verify that XCHEM tmax is set close to T_{ign} and below the kinetic-runaway threshold for the chosen scenario (§10.3). As a rule of thumb, $t_{\text{max}} \approx 800$ K — or alternatively a value slightly above the highest applied boundary temperature — is a good choice: either indicates that thermal runaway is imminent or close, and produces a clean pre-runaway handoff state without the integrator having to chase the kinetic-chain transition through the sub-zone-runaway window (§10.3). XDDT results that fail any of these checks should not be accepted without accompanying details describing the diagnostic shift and the physical or numerical reasons for it.

A caveat applies to the run-length diagnostic specifically. The present BN surface-burn term uses a pseudo-Arrhenius rate gated by a Heaviside step at T_{ign} (§10.5); small numerical perturbations near that switch can shift the moment the surface-burn kinetics first engage, which propagates into the run-length values (t_{ign} , t_{ddt} , and x_{ddt}) reported across otherwise-equivalent runs. The detonation-class diagnostics D_2 and P_{max} are governed by the post-engagement wave dynamics rather than the engagement instant and remain fairly consistent across the same perturbations. Replacing the Heaviside-gated surface-burn rate with a smooth chemistry model — the distributed activation-energy (DAE) form proposed in §10.5 is the natural candidate — is expected to improve run-length consistency by removing the threshold-switch sensitivity at the source.

B.3. Shipped templates and the push-button gap

The source-tree `templates/` directory ships scenario decks for the four problem classes exercised in this report: `cookoff_slab/` (single-edge thermal cookoff; §6), `cookoff_cylinder/` (slow cookoff cylinder per NATO AOP-4382 [5]; Chapter 7), `fragment_impact/` (twin-fragment colliding-wave; Chapter 8), and `shocked_slab/` (piston-driven shock initiation; under construction, see §10). Each subdirectory ships a complete, runnable pair of input decks (`tchem.in` for

XCHEM and `coupled.inp` for XDDT) together with a small driver script that executes the full pipeline end-to-end.

The templates close part of the reproducibility gap but not all of it. They give a user a known-good starting point for each of the four regimes, but they cannot remove the requirement to re-exercise the sensitivity controls of §B.2 when the new scenario departs materially from the seed case (different bed length, different confinement, different fill material). A genuinely “push-button” workflow would additionally require: (i) a broader outer-boundary-condition implementation (§10.1) so that a user is not restricted to the Neumann-fallback case; (ii) the smoothing-convention generalization for cold-bulk handoffs (§10.3) so that spherical and other multi-cell-hot-shell scenarios run without bespoke smoothing choices; (iii) the direction-aware diagnostic generalization (§10.4) so that two-front converging-wave cases do not require asymmetric hot-spot workarounds; and (iv) the distributed-activation-energy surface-burn replacement (§10.5) to remove the threshold-driven integrator stiffness that motivates much of the present sensitivity exploration. These four developments, already itemized in Chapter 10 as bounded future work, are also the principal items separating the current expert-only workflow from a template-driven one usable by a non-expert.

The two listings that follow reproduce, verbatim, the decks shipped in `templates/fragment_impact/` — the actual runnable XCHEM and XDDT input files that produce the twin-fragment colliding-wave result of Chapter 8 ($D_2 = 8.05$ km/s, $t_{\text{ddt}} = 2.20$ μ s, $P_{\text{max}} = 2.87$ GPa). The remaining three scenario subdirectories ship analogous deck pairs and are not reproduced. Non-ASCII characters present in the source files (the micro sign, em-dashes) have been transliterated to ASCII for the verbatim listings; consult the source-tree files for the originals.

B.4. XCHEM input deck (`fragment_impact/tchem.in`)

```
!-----!
! TCHEM input to solve the energy equation with a source term for      !
!   chemistry for 1-D multilayered geometries using grid refinement.  !
!   Up to 20 layers are allowed: |1|2|...|19|20|. Units are SI.       !
!-----!
! BASIC INFORMATION                                                    !
!-----!
Two Frag Ignition ! Title
slab ! geometry (options are slab, cylinder, sphere)
3      ! number of layers (20 maximum)
cm      ! units for interface and pgeom (m, cm, mm, inch)
0 4 8 12 ! interface locations (units above)
1 !1.0    ! pgeom: slab area, cylinder height, 0 for sphere (1 entry)
!-----!
! MATERIAL LAYERS                                                    !
!-----!
steel      575.  0.5 n ! Layer name, tlay, emiss,
ict_HMX_1150 300.  0.5 v ! Layer name, tlay, emiss,
steel      575.  0.5 n ! Layer name, tlay, emiss, n v p i HMX_ucm_1150
!-----!
! PRESSURIZATION (based on ideal gas, P=nRT/V                        !
!   n [kgmol] = initial + generated (in specified layers)           !
!-----!
```

```

!   R = 8314 m^/Users/mlhobbs/tchem2/input/propellant/25/2883 Pa/(K kgmol)
!   T [K]      = ideal gas ave = integral(phi dV)/integral(phi/T dV) !
!   V [m^3]    = integral(phi dV) for each specifid layer          !
!-----!
psi          ! Pressure units (options are pa, psi, atm, bar)
12.09        ! Initial pressure
12.09! 1e6    ! Maximum pressure
ideal        ! gas eos (bkws, covol) or (ideal)
0            ! Excess volume, m^3 (tubing and ullage)
!-----!
! CONTACT RESISTANCE BETWEEN LAYERS                                !
!   nlay-1 entries required for area, gap, and kgap                !
!-----!
F            ! cont      flag for contact: F=off, T=on
0.0          ! area      fraction in contact (= 0 gap =1. no gap)
0.0001       ! gap, m     gap dimension
0.1          ! kgap, W/mK thermal cond. of fluid in gap
!-----!
! INNER BC (esb = emmissivity*Stephan-Boltzmann is already specified) !
! Flux BC = fin + hin(Tin-T0) + radin*esb(Tin^4-T0^4); T BC = Tin    !
!-----!
0.           ! bcin      0.= flux boundary; 1.=temperature boundary
0.           ! fin       inner flux, W/m^2
0.           ! hin       convection coefficient, W/(m^2K)
0.           ! radin     0.= radiation BC on, 1.= radiation BC
0            ! ttin      number of time,temp points to follow
!-----!
! OUTER BC (esb = emmissivity*Stephan-Boltzmann is already specified) !
! Flux BC = fex + hex(Tex-Tnpt) + rad*esb(Tex^4-Tnpt^4); T BC = Tex  !
!-----!
0.           ! bcex      0.= flux boundary; 1.=temperature boundary
0.           ! fex       flux
0.           ! hex       convection coefficient
0.           ! radex     0.= radiation BC off, 1.= radiation BC on
0            ! ttex      number of time,temp points to follow
!-----!
! GRID AND PRINT CONTROL                                           !
!-----!
100          ! pnts      number of nodes to add to small layer
T            ! refit     T=refine grid, F=use initial grid
T F F T      ! ltemp; lspe; lsiti; ltrad (T=print)
!-----!
! TIME AND NUMERICAL CONTROL                                       !
!-----!
0.           ! tstart    Initial starting time
100          ! tend      Stop time
1.0          ! dt        Time step (refinement only activated at this dt)
1            ! iprint    Print every iprint*dt seconds
1.E-08       ! re        Relative error for stiff solver
1.E-12       ! ae        Absolute error for stiff solver
600.0        ! tmax      Max. T before leaving solver
100          ! itempt    # of ID=-1 from ODE solver before stopping
0.0002       ! tolrf     Error tolerance for adding nodes
0.00002      ! tolrc     Error tolerance for deleting nodes

```

B.5. XDDT input deck (fragment_impact/coupled.inp)

```
=====
! DDT Input File - 2frag (coupled) -- CANONICAL configuration
!
! SAND-3 result: twin-fragment colliding-wave DDT in HMX.
! Publication numbers from this deck:
!   D2      = +8.05 km/s (HMX CJ regime)
!   t_ddt   = 2.20 us at x_ddt = 5.98 cm (mid-bed converging-wave collision)
!   P_max   = 2.87 GPa
!   t_ign   = 0.10 us
!
! Methodology that produced these:
!   XCHEM 'v' (vented) emission to ofield.plt for the cold deck
!   ilev indexing + boundary-node clamp (handoff mesh integrity)
!   REFINED1D width-match tolerance (relative 1%, was abs 1e-10)
!   dual-edge smoother (xtiv_smooth_outer_l_in companion to inner)
!   outer-edge hot-spot patch in xtiv_apply_boost (this deck uses it)
!
! XCHEM handoff: tmax = 600 K (= T_steel_initial + 25, i.e. catch the
! pre-runaway state before XCHEM's print cadence loses spike resolution
! to a localized hotspot -- see the tmax sweep in
! xchem/tests/runs/2frag/v_tmax{585,600,625,675,825}/).
!
! Asymmetric inner+outer boost reflects the physical asymmetry of the
! two ignition spikes: the outer (RHS) interface is more intense, so it
! gets stronger boost (1100K/100MPa vs inner 900K/50MPa). Symmetric
! boost configurations (sym_1, sym_3 in coupled_boost_sym_*/) produce
! mirror-image stalled fronts that meet at mid-bed without forming DDT;
! asymmetry breaks the lock-in and yields a propagating wave.
!
! Smoothing is 0.2 mm (not 5 mm): boost-driven cases need narrow ramps
! to preserve the seeded hot-spot temperature; 5 mm washes the boost
! out and degrades ignition.
!
! Adaptive XDDT (imesh_in=1) gives the integrator headroom -- without it
! the run dies at IDID=-7 around 1.04 us before D2 stabilizes; with it
! the run reaches the t_ddt-population time of 2.20 us cleanly. The
! 1.40 us nstep_in cap below stops the run BEFORE the BF tracker's
! "peak-following" D2 fit gets confused by the post-collision retreat.
! Within 0-1.40 us the BF peak migrates monotonically
! outward and D2 = +8.05 km/s is well-defined.
=====

&material_props
  poro_in  = 0.70d0      ! Initial solid volume fraction (70% TMD)
  gamo_in  = 1.90d3      ! True solid density [kg/m^3]
  tmo_in   = 300.0d0     ! Initial solid temperature [K]
  tgo_in   = 300.0d0     ! Initial gas temperature [K]
  cpm_in   = 1.5d3       ! Solid specific heat [J/(kg K)]
  hpyr_in  = 2.3d6       ! Heat of pyrolysis [J/kg]
  a1_in    = 5.0d19      ! Pre-exponential factor (not exercised)
  a2_in    = 0.1227d0    ! Burn rate coeff, high-p branch [m/(s*100 MPa)]
```

```

a3_in      = 0.0d0      ! Burn law coefficient (temperature)
ta_in      = 26520.0d0   ! Activation temperature [K]
bn_in      = 1.0d0      ! Burn law exponent, high-p branch
ap_in      = 5.0d-5      ! Particle radius [m] (100 um diameter)
tign_in    = 800.0d0     ! Ignition temperature [K]
dreact_coeff_in = 2.0d-12 ! Hot-spot nucleation rate coefficient
alstar_coeff_in = 100.0d0 ! Compaction-driven burn acceleration coefficient
htfact_in  = 10.0d0     ! Biot number multiplier for surface temperature
alstar_pexp_in = 2.0d0
dreact_pexp_in = 2.0d0
/

&GAS_EOS
  IEOS_GAS_IN = 1                      ! TIGER tabular
  EOS_TABLE_FILE_IN = '/Users/mlhobbs/xddt/tbl/hmx/hmx.tbl'
/

&jwl_eos
  r1q_in    = 4.2d0
  r2q_in    = 1.0d0
  omg_in    = 0.25d0
  aq_in     = 2.544820d11 ! A_J [Pa] (HMX at DMO=1.33 kg/m^3 * 1000)
  bq_in     = -1.156722d8 ! B_J [Pa]
  cq_in     = 0.0d0       ! C_J [Pa]
  cpq_in    = 2.391600d3  ! c_v,g [J/(kg K)]
  hdet_in   = 7.878083d6  ! Q_det [J/kg]
  ! HMX polynomials evaluated at DMO = phi_s0*gamma_s0 = 0.70*1900 = 1330 kg/m^3
/

&solid_eos
  gov_in    = 2.1d0
  akto_in   = 1.35d10      ! Pa
  ans_in    = 9.8d0
  const1_in = 33.0d0
  const2_in = 0.01d0
/

&compaction
  rjotm_in  = 1.0d-3      ! 1/(Pa*s) [SI]
  tau1_in   = 1.2d7       ! Pa
  pa1_in    = 1.0d0
  pa2_in    = 0.70d0      ! Match initial porosity
  pa3_in    = 0.0d0
  pa4_in    = 1.0d0
/

&transport
  visr_in   = 5.0d-5      ! Pa*s
  akr_in    = 7.0d-2      ! W/(m*K)
  prn_in    = 1.71d0
  psdr_in   = 358.0d0     ! kg/m^3
  resv_in   = 3.88d-6     ! Pa*s
  resk_in   = 5.36d-2     ! W/(m*K)
  akm_in    = 4.0d-1      ! W/(m*K)

```

```

    betar_in = 5.0d2          ! 1/m
/

&geometry
    bl_in    = 0.04d0        ! Bed length [m] = 4 cm HMX bed (overridden by .xtiv)
/

&piston_bc
    ! Not used for IBC=3, but must be present
    x0_in    = 0.0d0
    v0_in    = 0.0d0
    tat0_in  = 1.0d0
    x1_in    = 0.04d0        ! Right boundary = bed length [m]
    v1_in    = 0.0d0
    tat1_in  = 0.0d0
/

&ignition_bc
    ! IBC=3: XTIV transfer-file mode -- initial condition comes from the
    ! .xtiv handoff (see &xtiv_initialize below).  pg_ign/tg_ign/l1_ign
    ! values are legacy IBC=1 parameters; read but unused on IBC=3 path.
    ibc_in    = 3
    pg_ign_in = 1.0d7        ! unused at IBC=3
    tg_ign_in = 3000.0d0     ! unused at IBC=3
    l1_ign_in = 0.005d0     ! unused at IBC=3
/

&numerical
    npt_in    = 4500        ! must be >= n_nodes in .xtiv (RWORK sizing).
                           ! 2frag is a 3-layer cookoff; HMX layer is
                           ! ~100 base + AMR refinement, expect ~600--1500
                           ! handoff nodes; 4500 leaves comfortable margin.
    nstep_in  = 14          ! cap at t=1.40 us (= 14 * dt_in) -- see header
                           ! for D2-fit rationale.  Increase to study
                           ! post-collision dynamics.

    dt_in     = 0.1d-6      ! Output time step [s]
    arv_in    = 0.75d0
    rel_in    = 1.0d-6
    ae_in     = 1.0d-10
    rtolref_in = 5d-4
    rtolcom_in = 5d-5
    nadapt_in = 1
    imesh_in  = 1          ! ADAPTIVE -- required for clean t_ddt
/

&monitoring
    npstat_in = 4
    ! Gauge layout for the 2frag colliding-wave geometry.  HMX bed spans
    ! [0.04, 0.08] m in absolute coords; the .xtiv handoff puts the bed at
    ! [0, 0.04] m in XDDT-internal coords (HMX-only frame).  Gauges placed
    ! at quarter-bed, mid-bed-LHS, mid-bed-RHS, three-quarter-bed to
    ! capture both inward fronts and their collision regime:
    !   1: 0.010 m -- 1 cm into bed from LHS interface (LHS-front path)

```

```

!   2: 0.018 m -- mid-bed, biased LHS (collision likely to land here
!           given RHS>LHS asymmetry)
!   3: 0.022 m -- mid-bed, biased RHS (collision regime, other side)
!   4: 0.030 m -- 1 cm into bed from RHS interface (RHS-front path)
xpstat_in = 0.010d0, 0.018d0, 0.022d0, 0.030d0, 0.0d0,
            0.0d0, 0.0d0, 0.0d0, 0.0d0, 0.0d0
/

&output_control
  iplot_in      = 0
  ispmx_in      = 2
  ilistmx_in    = 25
  max_stiff_in  = 10000
  lwrite_bf_in  = .true.
  lwrite_piston_in = .true.
  lwrite_compact_in = .true.
  lwrite_3d_in  = .true.
  vm_thresh_in  = 5.0d0      ! m/s
  dphi_thresh_in = 0.01d0
/

&xtiv_initialize
  xtiv_file_in = 'handoff.xtiv'           ! relative to cwd; written by ./run script
  xtiv_expected_material_in = 'ict_HMX_1150' ! matches 2frag/tchem.in layer 2 (index 1)
  xtiv_outer_bc_in = 'auto'               ! resolves from .xtiv [confinement] block.
                                           ! 2frag has steel-bounded HMX on both sides,
                                           ! neither vented to atmosphere -- likely
                                           ! resolves to 'wall'/'free' on outer edge.
                                           ! Adjust if 'auto' picks the wrong default.

  xtiv_smooth_l_in      = 0.0002d0        ! m -- 5 mm inner-edge tanh ramp
  xtiv_smooth_outer_l_in = 0.0002d0        ! m -- 5 mm outer-edge tanh ramp
                                           ! 2frag has ignition spikes at both
                                           ! steel/HMX interfaces; symmetric 5 mm
                                           ! ramps on each edge.
/

&xtiv_boost
! sym_2: asymmetric matching natural physics -- inner canonical, outer stronger
! (RHS is more intense per the natural-cookoff observation in deck header).
  t_scale_in      = 1.0d0
  hot_spot_l_in   = 0.020d0      ! m inner-edge: 20 mm patch
  hot_spot_T_in   = 900.0d0      ! K
  hot_spot_P_in   = 5.0d7        ! Pa = 50 MPa
  hot_spot_outer_l_in = 0.020d0  ! m outer-edge: 20 mm patch (stronger T, P)
  hot_spot_outer_T_in = 1100.0d0 ! K
  hot_spot_outer_P_in = 1.0d8    ! Pa = 100 MPa
/

```

APPENDIX C. POSTLOG

A novel aspect of this report’s preparation is the substantive use of large language model (LLM) tools to assist with the modeling, simulation, and writing tasks documented in §1 through §11. This postlog documents that workflow transparently and addresses three questions that may arise on review: how the human–AI collaboration was structured and what each party contributed, what reproducibility and code-protection considerations apply, and what export-control and distribution status is appropriate.

C.1. Workflow and division of labor

The report was prepared through a human project lead (the author) and a chat-based AI assistant used primarily for interactive prose drafting and figure-script authorship.

The author retained full authority and responsibility for all technical decisions, methodology choices, calibration and parameter selections, scenario design, deck configuration, validation, and final review of generated content. The report’s direction, technical judgement, physical insight, and accountability rest with the author. No AI-produced content was incorporated without the author’s review.

For the benefit of others preparing complex technical reports with LLM assistance, the following examples illustrate the categories of prompts that proved most useful during the preparation of this report. They are presented in generic form, applicable to any technical report, and span structural editing, consistency cleanup, citation management, formatting fixes, and verification.

Structure and organization.

- “I think Section X.Y would flow better in a different place. What do you think, and if you agree, make the move and add transition prose so it doesn’t read as a non sequitur.”
- “Read this chapter end-to-end and tell me where the structure interrupts the narrative arc.”

Editorial review of a draft.

- “I just rewrote this chapter. Give me honest feedback on what works and what doesn’t.”
- “Read the introduction and tell me where the prose is too dense for a non-specialist reader (e.g., a program manager or funding reviewer), with concrete examples.”

Consistency cleanup.

- “Find every place this document uses these terms interchangeably and standardize on one. Suggest which one before changing anything.”
- “The report self-references itself by its publication number throughout. Replace those self-references with neutral phrasing like ‘this report’ or ‘the current work’, but leave external citations alone.”
- “Scan this section for typos, broken sentences, or grammar issues that look like the result of mid-edit collisions.”
- “Find places where I’ve used editorial or opinion language (e.g., ‘iconic’, ‘striking’, ‘revolutionary’) and flag them so I can decide whether to remove.”

Citations and bibliography.

- “Verify the bibliography is in order of first citation in the document, and tell me which entries are out of place.”
- “Add this reference. Put the bibitem in the right position and insert a citation sentence in this paragraph.”
- “Verify that every cross-reference in the document resolves to a defined label.”

Formatting and layout.

- “There is a paragraph where a long identifier is overflowing the right margin. Find it and make it breakable while keeping the visual style.”
- “Add a bit more space between Figure X and its caption.”

Generation.

- “Generate a nomenclature section from the symbols and abbreviations actually used in the document.”
- “Draft a paragraph that makes explicit the dependency chain between this proposed future work and the prerequisites it sits behind.”

Verification after edits.

- “After these edits, verify that braces and environments balance and that cross-references still resolve.”

A pattern that proved consistently useful across the categories above is to ask for a plan and confirmation before changes, make the changes, then ask for a verification pass afterward. That three-beat workflow most reliably catches reflexive edits and missed line-break edge cases.

C.2. Reproducibility and code protection

A reasonable question from review concerns the reproducibility of this work by external readers — specifically, whether uploading this report (or its predecessors [1, 2]) to an LLM tool would allow a third party to reproduce the work in some compressed timeframe.

Scientific publication is designed to enable methodology dissemination. The physics models in this work are published in the open literature: the Baer–Nunziato two-phase formulation in Refs. [8, 9]; the McAfee–Asay–Campbell–Ramsay DDT validation experiments in Ref. [11]; the Price–Bernecker thermal-ignition experiments in Ref. [12]; the Sandusky–Glancy–Campbell–Krall ball-propellant compaction experiments in Ref. [17]; the TIGER thermochemical-equilibrium formulation in Ref. [7]; and the JCZS3 intermolecular potential database in Ref. [22]. An external party with access to this report and the cited literature could in principle reconstruct the methodology — but doing so requires sustained domain expertise in two-phase reactive-flow modeling, Fortran software engineering, IM-relevant validation practice, and the experimental literature that calibrates the constitutive choices. An LLM tool can summarize published material, draft surrounding prose, and accelerate implementation, but it does not substitute for the domain expertise and physical judgement that distinguish a working coupled simulation from a methodology summary.

The XDDT and XCHEM source codes are not part of this publication. The deck files, output summaries, and validation scenarios that appear in the figures of this report describe *what was computed*, not how the codes implement the two-phase equations or solve them. The codes themselves remain in Sandia-internal repositories.

C.3. Export-control assessment

This SAND report deals with the export-controlled material HMX. The technical information included herein is entirely captured by previously published open-publication information. The experimental data upon which the model was validated and compared (Refs. [11, 12, 17]) are published in publicly available International Detonation Symposia and JANNAF proceedings. The Baer–Nunziato two-phase model is published in the open literature [8, 9]. The contribution of this report and its predecessors [1, 2] is the modernization of the original Baer Fortran-77 source to current Fortran standards, the correction of historical software bugs, the introduction of the coupled XCHEM–XDDT pipeline, and the demonstration of that pipeline on representative test configurations. The software itself is not part of this publication.

All coupled simulation demonstrations in this report use HMX as the reactive material. Cross-material gas-equation-of-state sensitivity work involving additional explosives was conducted in the predecessor report [2], where only the *reacted* (product-gas) EOS data were applied for relative sensitivity comparisons; that work is cited here but not extended. This material does not rise to the level of Export Controlled Controlled Unclassified Information and is appropriate for unlimited distribution.

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APPENDIX D. NOMENCLATURE

Symbol	Units	Description
<i>Acronyms</i>		
AMR	–	Adaptive Mesh Refinement
AOP	–	Allied Ordnance Publication (NATO)
BN	–	Baer–Nunziato (two-phase mixture-theory framework)
C-J	–	Chapman–Jouguet
DDT	–	Deflagration to Detonation Transition
EOS	–	Equation of state
HMX	–	Cyclotetramethylene-tetranitramine (the explosive used here)
IBC	–	Initial-boundary condition (XDDT initialization mode selector)
IM	–	Insensitive munitions
MMP	–	MicroMechanics Pressurization model
SAND	–	Sandia (National Laboratories) Report
SAND-1	–	First report of the present series [1] (XDDT code modularization)
SAND-2	–	Second report of the present series [2] (SI units, adaptive gridding, TIGER EOS)
SHA	–	Secure Hash Algorithm; used as a unique identifier for a specific source-code version
SI	–	Système International (units)
TIGER	–	Thermochemical-equilibrium code / EOS database
TMD	–	Theoretical Maximum Density
XCHEM	–	eXplosive CHEM istry (prethermal-ignition code)
XDDT	–	eXplosive D eflagration to D etonation T ransition (the code)
XTIV	–	eXplosive T hermal I gnition to V iolence (the coupling pipeline)
<i>XTIV pipeline</i>		
$ibc = 3$	–	XDDT initialization mode for XTIV transfer file
m_f	–	Mass-balance diagnostic, $\rho_g(1 - \phi_s)/[\rho_{co}(1 - \phi_o)(1 - S_f)]$
P	Pa	Pressure in handoff state
S_f	–	Surviving solid mass fraction, m_c/m_{co} (MMP)
T	K	Solid temperature in handoff state
$xtiv_smooth_1$	m	Inner-edge smoothing length parameter
π_y	kg/m ³	XDDT pyrolysis-progress variable (F_sol slot 8)
ρ_{co}	kg/m ³	Initial (pristine) condensed-phase density
ρ_g	kg/m ³	Gas density (partial) in handoff state
ϕ_o	–	Initial gas volume fraction (pre-cockoff)
ϕ_s	–	Solid volume fraction in handoff state
<i>Diagnostics</i>		
ipk_bf	–	Node index of band peak

Symbol	Units	Description
r_{burn}	m/s	Pressure-dependent surface-burn velocity ($a_2(P/P_{\text{ref}})^{b_n}H$)
$x_{\text{BF,lead}}$	m	Leading edge of active-reaction band
$x_{\text{BF,peak}}$	m	Position of peak burn-rate node
$x_{\text{BF,trail}}$	m	Trailing edge of active-reaction band
$\dot{\phi}_s$	1/s	Solid volume fraction time derivative
<i>BN model parameters (inherited from SAND-1, SAND-2)</i>		
a_2	m/s	Surface-burn coefficient at $P = P_{\text{ref}}$
b_n	–	Burn-law exponent
D	m/s	Steady detonation-wave speed (the D_2 of SAND-1's D_1/D_2 convention; subscript dropped throughout, see §1.3.6)
D_1	m/s	Transient detonation-wave speed through an inert plug ahead of the reaction zone (SAND-1 convention; not observed in any demonstration in this report)
D_{CJ}	m/s	Ideal (Chapman–Jouguet, steady self-sustaining) detonation velocity, computed from TIGER thermochemistry; 6.36 km/s for HMX at $\rho_0 = 1330 \text{ kg/m}^3$
$H(\cdot)$	–	Heaviside step function ($H(x) = 1$ if $x > 0$, else 0)
P_g	Pa	Gas pressure (used in surface-burn term; <code>pg_nod</code>)
P_{mix}	Pa	Mixture pressure $\phi_s P_s + \phi_g P_g - P_{\text{intergranular}}$ (used in compaction-burn term; <code>pmix</code> in source)
P_{ref}	–	Reference pressure for HMX strand-burn calibration; $P_{\text{ref}} = 100 \text{ MPa} = 10^8 \text{ Pa} = 10^9 \text{ dyn/cm}^2 = 1 \text{ kbar} = 0.1 \text{ GPa} \approx 986.92 \text{ atm}$
T_{ign}	K	Ignition temperature threshold
T_{surface}	K	Granular surface temperature (interface heat-transfer model); distinct from gas T_g and bulk solid T_s ; <code>tinf</code> in source
α_0	1/s	Compaction-burn rate coefficient (<code>alstar_coeff_in</code>); full coefficient is $\alpha^* = \alpha_0(P_{\text{mix}}/P_{\text{ref}})^2$
$\gamma_{s,0}$	kg/m^3	True solid (crystal) density
ζ	K	Surface-temperature integral function (BN state variable); running total of past gas–grain heating delivered to the grain surface; stored as $\rho_s \zeta$ (<code>fh_nod</code> in source, BN state slot 9); see §1.3.3
$\phi_{s,0}$	–	Initial solid volume fraction (= TMD fraction)
<i>IM testing</i>		
<i>(NATO AOP-4382, MIL-STD-2105)</i>		
<i>Slow cook-off ramp rate: 15 °C/hr</i>		

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